CBC data analysis

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First time run order: Spleen → CBC → Spleen

Requires clean_meta_data.csv from the spleen analysis markdown

Libraries

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(writexl)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
library(skimr)
library(naniar)
```

```
##
## Attaching package: 'naniar'
##
## The following object is masked from 'package:skimr':
##
##     n_complete
```

```r
library(ggplot2)
library(ggpubr)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```r
library(DHARMa)
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```r
library(scales)
```

```
##
## Attaching package: 'scales'
```

```
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
library(haven)
library(broom.mixed)
library(readr)
```

Setting the colors (Okabe-Ito palette and Paul Tol's Bright)

```r
color_black <- "#000000"
color_mustard <- "#E69F00"
color_light_blue <- "#56B4E9"
color_green <- "#009E73"
color_yellow <- "#F0E442"
color_dark_blue <- "#0072B2"
color_orange <- "#D55E00"
color_purple <- "#CC79A7"

color_grey <- "#BBBBBB"
color_dark_purple <- "#AA3377"
color_light_purple <- "#EE6677"
color_sand <- "#CCBB44"
color_cyan <- "#66CCEE"
color_olive <- "#228833"
color_navy <- "#4477AA"
```

Setting up paths

```r
base_path <- "/Users/zehao/Documents/Research Projects/Graham Group Thesis Research (Summer 2025)"
data_path <- file.path(base_path, "Metadata")
clean_data_path <- file.path(base_path, "Cleaned data")
figure_path <- file.path(base_path, "Figures")
cbc_figure_path <- file.path(figure_path, "CBC Figures")

dir.create(cbc_figure_path,
           showWarnings = FALSE,
           recursive = TRUE)
```

Setting up vectors

```
cbc_number <- c("neu_number",
                "lym_number",
                "mon_number",
                "eos_number",
                "bas_number")

cbc_percent <- c("neu_percent",
                 "lym_percent",
                 "mon_percent",
                 "eos_percent",
                 "bas_percent")
```

Defining functions

```
data_saver <- function(data_name,
                       path = "") {
  write_csv(x = data_name,
            file = path)
}

plot_saver <- function(filename,
                       plot,
                       path = cbc_figure_path,
                       width = 8,
                       height = 6,
                       dpi = 300) {
  ggsave(filename = filename,
         plot = plot,
         path = path,
         width = width,
         height = height,
         units = "in",
         dpi = dpi)
}

cbc_summarizer <- function(data,
                           group_name) {
  data %>%
    summarise(neu_percent = mean(neu_percent,
                                 na.rm = TRUE) / 100,
              lym_percent = mean(lym_percent,
                                 na.rm = TRUE) / 100,
              mon_percent = mean(mon_percent,
                                 na.rm = TRUE) / 100,
              eos_percent = mean(eos_percent,
                                 na.rm = TRUE) / 100,
              bas_percent = mean(bas_percent,
                                 na.rm = TRUE) / 100) %>%
```

```r
    mutate(group = group_name) %>%
    pivot_longer(cols = neu_percent:bas_percent,
                 names_to = "cell_type",
                 values_to = "cbc_percent")
}

emm_plotter <- function(emm_df,
                        cbc_input,
                        y_col,
                        y_label,
                        title = NULL,
                        floor_zero = FALSE) {
  if (floor_zero) {
    emm_df <- emm_df %>% mutate(lower.CL = pmax(lower.CL, 0))
  }

  ggplot(emm_df, aes(x = phase,
                     y = response,
                     color = t_muris_dose,
                     fill = t_muris_dose,
                     group = t_muris_dose)) +
    geom_point(data = cbc_input,
               aes(x = phase,
                   y = .data[[y_col]],
                   fill = t_muris_dose,
                   color = t_muris_dose),
               position = position_jitterdodge(jitter.width = 0.1,
                                               dodge.width = 0.5),
               alpha = 0.3,
               size = 1.2,
               inherit.aes = FALSE) +
    geom_line(position = position_dodge(width = 0.5),
              linewidth = 0.7,
              alpha = 0.7) +
    geom_errorbar(aes(ymin = lower.CL,
                      ymax = upper.CL),
                  width = 0.2,
                  alpha = 0.7,
                  position = position_dodge(width = 0.5)) +
    geom_point(size = 3,
               shape = 21,
               color = "black",
               position = position_dodge(width = 0.5)) +
    geom_vline(xintercept = 2.5,
               linetype = "dashed",
               color = "grey40",
               alpha = 0.6) +
    annotate("text",
             x = 2.5,
             y = Inf,
             label = "Infection",
             vjust = 1.5,
             hjust = -0.1,
```

```
            size = 3,
            fontface = "italic",
            alpha = 0.6) +
    facet_grid(lab_or_rw ~ strain,
               labeller = labeller(lab_or_rw = c("Lab" = "Laboratory",
                                                  "RW" = "Rewilded"))) +
    scale_fill_manual(values = c("Zero" = color_dark_blue,
                                 "Low"  = color_yellow,
                                 "High" = color_orange),
                      name = expression(bolditalic("T. muris") ~ bold("Dose"))) +
    scale_color_manual(values = c("Zero" = color_dark_blue,
                                  "Low"  = color_yellow,
                                  "High" = color_orange),
                       name = expression(bolditalic("T. muris") ~ bold("Dose"))) +
    scale_y_continuous(name = y_label) +
    # add expand = expansion(mult = c(0.05, 0.3)) if adding sig astrix.
    theme_minimal() +
    theme(strip.background = element_rect(fill = "white",
                                          color = "black",
                                          linewidth = 0.8),
          strip.text = element_text(size = 10,
                                    face = "bold"),
          panel.border = element_rect(color = "black",
                                      fill = NA,
                                      linewidth = 0.8),
          panel.grid = element_blank(),
          plot.title = element_text(hjust = 0.5,
                                    face = "bold"),
          axis.title.x = element_text(face = "bold"),
          axis.title.y = element_text(face = "bold")) +
    xlab("Phase") +
    ggtitle(title)
}

cbc_plotter <- function(cbc_data,
                        cbc_error,
                        y_label,
                        title = NULL) {
  ggplot(cbc_data, aes(x = phase,
                       y = number,
                       fill = t_muris_dose,
                       color = t_muris_dose,
                       group = t_muris_dose)) +
  geom_point(position = position_dodge(width = 0.5),
             size = 1.5,
             alpha = 0.3) +
  geom_line(data = cbc_error,
            aes(y = mean, group = t_muris_dose),
            linewidth = 0.5,
            alpha = 0.6,
            position = position_dodge(width = 0.5)) +
  geom_point(data = cbc_error,
             aes(y = mean),
```

```r
            position = position_dodge(width = 0.5),
            size = 2.5,
            alpha = 1,
            shape = 21,
            color = "black") +
geom_errorbar(data = cbc_error,
              aes(y = mean,
                  ymin = mean - se,
                  ymax = mean + se),
              position = position_dodge(width = 0.5),
              width = 0.15,
              alpha = 0.6) +
geom_vline(xintercept = 2.5,
           linetype = "dashed",
           color = "grey40",
           alpha = 0.6) +
annotate("text",
         x = 2.5,
         y = Inf,
         label = "Infection",
         vjust = 1.5,
         hjust = -0.1,
         size = 3,
         fontface = "italic",
         alpha = 0.6) +
facet_grid(lab_or_rw ~ strain,
           labeller = labeller(lab_or_rw = c("Lab" = "Laboratory",
                                             "RW" = "Rewilded"))) +
scale_fill_manual(values = c("Zero" = color_dark_blue,
                             "Low"  = color_yellow,
                             "High" = color_orange),
                  name = expression(bolditalic("T. muris") ~ bold("Dose"))) +
scale_color_manual(values = c("Zero" = color_dark_blue,
                              "Low"  = color_yellow,
                              "High" = color_orange),
                   name = expression(bolditalic("T. muris") ~ bold("Dose"))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                      color = "black",
                                      linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5,
                                face = "bold"),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold")) +
xlab("Phase") +
ylab(y_label) +
ggtitle(title)
```

```
}
```

Loading Data

```
# Mouse CBC data
cbc_data_1 <- read_excel(file.path(data_path,
                                   "DCV_25SF.W-3thruNSF1trapout.XLSX"))
cbc_data_2 <- read_excel(file.path(data_path,
                                   "DCV_25SF.W3_NSF2.XLSX"))
meta_data   <- read_csv(file.path(clean_data_path,
                                  "clean_meta_data.csv"))
```

```
## Rows: 110 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): strain, experiment, t_muris_dose, lab_or_rw, cohort, source
## dbl (1): mouse_id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# CDC NHANES human data
nhanes_data_2017_18 <- read_xpt(file.path(data_path,
                                          "CBC_J_NHANES_2017_2018.xpt"))
nhanes_data_2021_23 <- read_xpt(file.path(data_path,
                                          "CBC_L_NHANES_2021_2023.xpt"))
```

Cleaning Mouse CBC Data

```
clean_cbc_data_1 <- cbc_data_1 %>%
  clean_names() %>%
  filter(id != "BG",
         !id %in% c("25SF.W5.1330.0.1",
                    "25SF.W3.1311.0",
                    "25SF.17.1380.-3",
                    "25SF.W5.1330.0",
                    "25SF.W5.1330.2")) %>%
  separate(col = id,
           into = c("experiment",
                    "cage_wedge",
                    "mouse_id",
                    "week",
                    "repeat"),
           sep = "\\.",
           fill = "right",
           extra = "merge",
           remove = FALSE) %>%
```

```r
  select(mouse_id,
         all_of(cbc_number),
         neu_percent_2,
         lym_percent_2,
         mon_percent,
         eos_percent_2,
         bas_percent_2,
         cage_wedge,
         week,
         `repeat`) %>%
  rename(neu_percent = neu_percent_2,
         lym_percent = lym_percent_2,
         eos_percent = eos_percent_2,
         bas_percent = bas_percent_2) %>%
  mutate(week = as.integer(week),
         cage_wedge = factor(cage_wedge),
         mouse_id = as.character(mouse_id),
         across(all_of(c(cbc_number,
                          cbc_percent)), ~ as.numeric(.)))

clean_cbc_data_2 <- cbc_data_2 %>%
  clean_names() %>%
  filter(id != "BG") %>%
  separate(col = id,
           into = c("experiment",
                    "cage_wedge",
                    "mouse_id",
                    "week",
                    "repeat"),
           sep = "\\.",
           fill = "right",
           extra = "merge",
           remove = FALSE) %>%
  select(mouse_id,
         all_of(cbc_number),
         neu_percent_2,
         lym_percent_2,
         mon_percent,
         eos_percent_2,
         bas_percent_2,
         cage_wedge,
         week,
         `repeat`) %>%
  rename(neu_percent = neu_percent_2,
         lym_percent = lym_percent_2,
         eos_percent = eos_percent_2,
         bas_percent = bas_percent_2) %>%
  mutate(week = as.integer(week),
         cage_wedge = factor(cage_wedge),
         mouse_id = as.character(mouse_id),
         across(all_of(c(cbc_number, cbc_percent)), ~ as.numeric(.)))
```

Combining Mouse CBC Data and Adding Metadata

```
cleaned_meta_data <- meta_data %>%
  mutate(mouse_id = as.character(mouse_id))

cleaned_combined_cbc_data <- bind_rows(clean_cbc_data_1,
                                       clean_cbc_data_2) %>%
  select(-`repeat`) %>%
  left_join(cleaned_meta_data, by = "mouse_id") %>%
  mutate(cage_wedge = tolower(cage_wedge)) %>%
  mutate(week = as.integer(week),
         strain = factor(strain),
         lab_or_rw = factor(lab_or_rw),
         t_muris_dose = factor(t_muris_dose),
         mouse_id = factor(mouse_id)) %>%
  mutate(strain = factor(strain,
                         levels = c("C57BL/6J",
                                    "B10.BR")),
         lab_or_rw = factor(lab_or_rw,
                            levels = c("Lab",
                                       "RW")),
         t_muris_dose = factor(t_muris_dose,
                               levels = c("Zero",
                                          "Low",
                                          "High"))) %>%
  filter(!is.na(lab_or_rw)) %>%
  mutate(phase = case_when(
    week %in% c(-4, -3) ~ "Pre-RW",
    week == 0           ~ "During RW",
    week %in% c(3, 4)   ~ "Post-RW"
  ) %>% factor(levels = c("Pre-RW", "During RW", "Post-RW")))

cbc_endpoint <- cleaned_combined_cbc_data %>%
  filter(phase == "Post-RW")

# [132, 193] was removed because it has different naming. Check how to fix this issue soon.
```

```
data_saver(data_name = cleaned_combined_cbc_data,
           path = file.path(clean_data_path,
                            "cleaned_combined_cbc_data.csv"))
```

Cleaning NHANES Human Data

```
cleaned_nhanes_data_2017_18 <- nhanes_data_2017_18 %>%
  clean_names() %>%
  rename(respondent_id = seqn,
         lym_percent = lbxlypct,
         mon_percent = lbxmopct,
         neu_percent = lbxnepct,
```

```r
         bas_percent = lbxbapct,
         eos_percent = lbxeopct,
         lym_number  = lbdlymno,
         neu_number  = lbdneno,
         mon_number  = lbdmono,
         bas_number  = lbdbano,
         eos_number  = lbdeono) %>%
  select(respondent_id,
         all_of(cbc_number),
         all_of(cbc_percent)) %>%
  drop_na() %>%
  mutate(respondent_id = as.character(respondent_id),
         across(all_of(c(cbc_number, cbc_percent)), ~ as.numeric(.)))

cleaned_nhanes_data_2021_23 <- nhanes_data_2021_23 %>%
  clean_names() %>%
  rename(respondent_id = seqn,
         lym_percent = lbxlypct,
         mon_percent = lbxmopct,
         neu_percent = lbxnepct,
         bas_percent = lbxbapct,
         eos_percent = lbxeopct,
         lym_number  = lbdlymno,
         neu_number  = lbdneno,
         mon_number  = lbdmono,
         bas_number  = lbdbano,
         eos_number  = lbdeono) %>%
  select(respondent_id,
         all_of(cbc_number),
         all_of(cbc_percent)) %>%
  drop_na() %>%
  mutate(respondent_id = as.character(respondent_id),
         across(all_of(c(cbc_number, cbc_percent)), ~ as.numeric(.)))

data_saver(data_name = cleaned_nhanes_data_2017_18,
           path = file.path(clean_data_path, "cleaned_nhanes_data_2017_18.csv"))

data_saver(data_name = cleaned_nhanes_data_2021_23,
           path = file.path(clean_data_path, "cleaned_nhanes_data_2021_23.csv"))
```

Making long-form mouse CBC data to be plotted

```r
cleaned_combined_cbc_data_long <- cleaned_combined_cbc_data %>%
  pivot_longer(cols = c(all_of(cbc_percent),
                        all_of(cbc_number)),
               names_to = c("cell_type",
                            ".value"),
               names_pattern = "(.*)_(number|percent)") %>%
  filter(!is.na(lab_or_rw))

data_saver(data_name = cleaned_combined_cbc_data_long,
```

```
           path = file.path(clean_data_path,
                            "cleaned_combined_cbc_data_long.csv"))
```

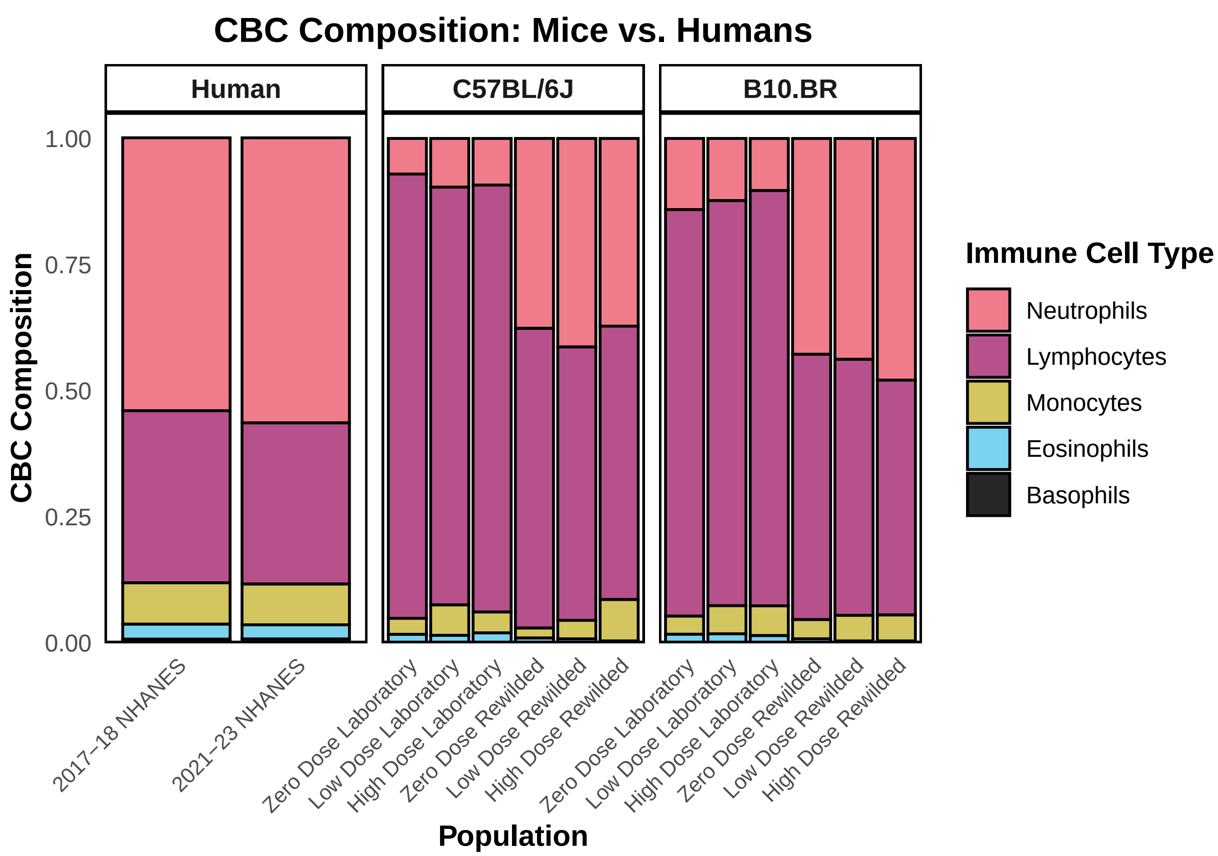
Composition Plot for Human vs. Mouse understanding

```
human_17_18 <- cbc_summarizer(cleaned_nhanes_data_2017_18,
                              "Human_2017_18") %>%
  mutate(strain = "Human")

human_21_23 <- cbc_summarizer(cleaned_nhanes_data_2021_23,
                              "Human_2021_23") %>%
  mutate(strain = "Human")

mean_mouse <- cbc_endpoint %>%
  group_by(strain, t_muris_dose, lab_or_rw) %>%
  summarise(neu_percent = mean(neu_percent,
                               na.rm = TRUE) / 100,
            lym_percent = mean(lym_percent,
                               na.rm = TRUE) / 100,
            mon_percent = mean(mon_percent,
                               na.rm = TRUE) / 100,
            eos_percent = mean(eos_percent,
                               na.rm = TRUE) / 100,
            bas_percent = mean(bas_percent,
                               na.rm = TRUE) / 100,
            .groups = "drop") %>%
  mutate(group = paste(strain, t_muris_dose, lab_or_rw)) %>%
  pivot_longer(cols = neu_percent:bas_percent,
               names_to = "cell_type",
               values_to = "cbc_percent")

bar_plot_input <- bind_rows(human_17_18,
                            human_21_23,
                            mean_mouse) %>%
  mutate(group = factor(group,
                        levels = c(
                          "Human_2017_18",
                          "Human_2021_23",
                          "C57BL/6J Zero Lab",
                          "B10.BR Zero Lab",
                          "C57BL/6J Low Lab",
                          "B10.BR Low Lab",
                          "C57BL/6J High Lab",
                          "B10.BR High Lab",
                          "C57BL/6J Zero RW",
                          "B10.BR Zero RW",
                          "C57BL/6J Low RW",
                          "B10.BR Low RW",
                          "C57BL/6J High RW",
                          "B10.BR High RW")),
         cell_type = factor(cell_type,
```

```r
                              levels = c("neu_percent",
                                         "lym_percent",
                                         "mon_percent",
                                         "eos_percent",
                                         "bas_percent")),
         strain = factor(strain, levels = c("Human", "C57BL/6J", "B10.BR")),
         facet_label = case_when(
           strain == "Human" ~ gsub("Human_", "", group),
           TRUE ~ gsub("C57BL/6J |B10.BR ", "", group)
         ),
         facet_label = factor(facet_label,
                                levels = c("2017_18", "2021_23",
                                           "Zero Lab", "Low Lab", "High Lab",
                                           "Zero RW", "Low RW", "High RW")))

data_saver(data_name = bar_plot_input,
           path = file.path(clean_data_path,
                            "input_data_cbc_composition.csv"))
```

```r
comp_plot <- ggplot(bar_plot_input,
                    aes(x = facet_label,
                        y = cbc_percent,
                        fill = cell_type)) +
  geom_col(position = "stack",
           color = "black",
           linewidth = 0.5,
           alpha = 0.85) +
  scale_fill_manual(values = c("neu_percent" = color_light_purple,
                               "lym_percent" = color_dark_purple,
                               "mon_percent" = color_sand,
                               "eos_percent" = color_cyan,
                               "bas_percent" = color_black),
                    labels = c("neu_percent" = "Neutrophils",
                               "lym_percent" = "Lymphocytes",
                               "mon_percent" = "Monocytes",
                               "eos_percent" = "Eosinophils",
                               "bas_percent" = "Basophils"),
                    name = expression(bold("Immune Cell Type"))) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.05))) +
  scale_x_discrete(labels = c("2017_18" = "2017-18 NHANES",
                               "2021_23" = "2021-23 NHANES",
                               "Zero Lab" = "Zero Dose Laboratory",
                               "Low Lab" = "Low Dose Laboratory",
                               "High Lab" = "High Dose Laboratory",
                               "Zero RW" = "Zero Dose Rewilded",
                               "Low RW" = "Low Dose Rewilded",
                               "High RW" = "High Dose Rewilded")) +
  facet_wrap(~ strain, scales = "free_x") +
  theme_minimal() +
  theme(strip.background = element_rect(fill = "white",
                                        color = "black",
                                        linewidth = 0.8),
        strip.text = element_text(size = 10,
```

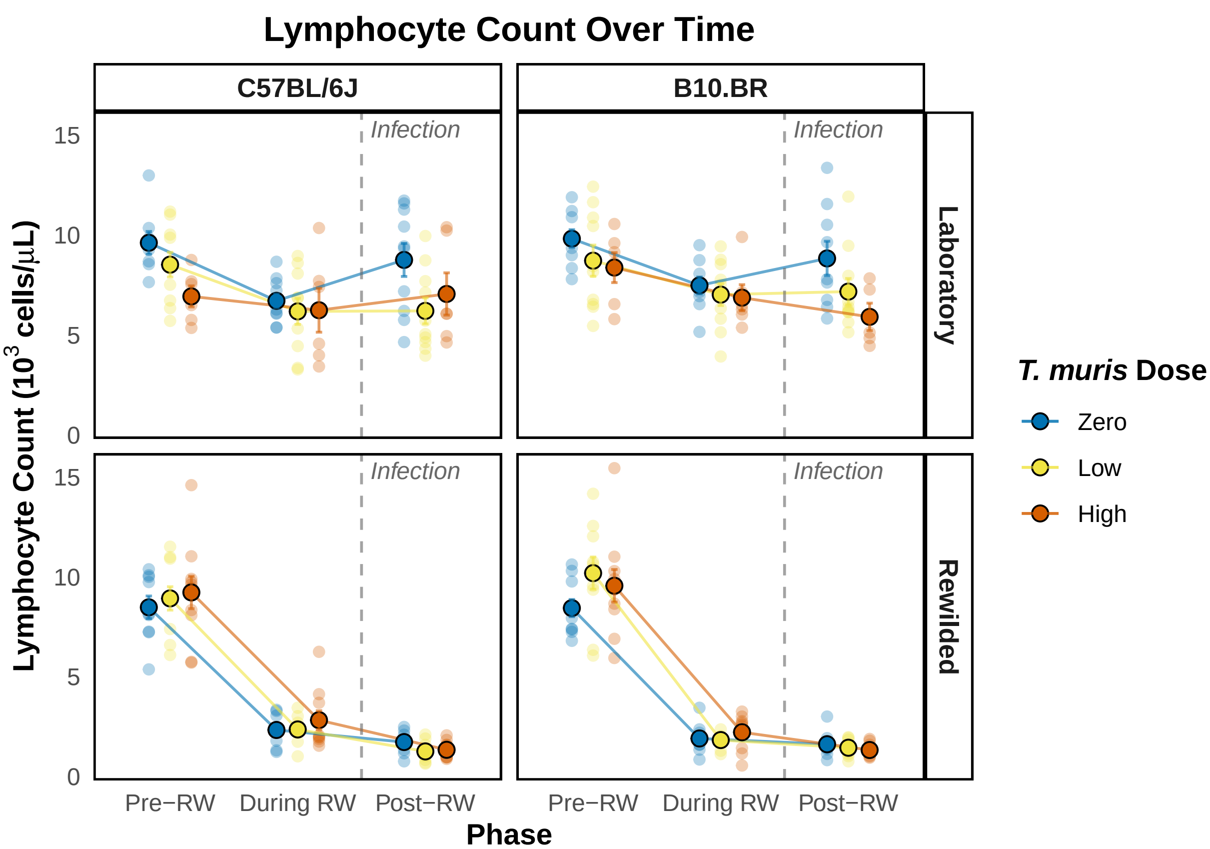
```
                                   face = "bold"),
        panel.border = element_rect(color = "black",
                                     fill = NA,
                                     linewidth = 0.8),
        panel.grid = element_blank(),
        axis.title.x = element_text(face = "bold"),
        axis.title.y = element_text(face = "bold"),
        plot.title = element_text(hjust = 0.5,
                                   face = "bold"),
        axis.text.x = element_text(angle = 45,
                                    hjust = 1,
                                    size = 8)) +
  xlab("Population") +
  ylab("CBC Composition") +
  ggtitle("CBC Composition: Mice vs. Humans")

comp_plot
```




```
plot_saver("cbc_composition_plot.png",
           plot = comp_plot)
```

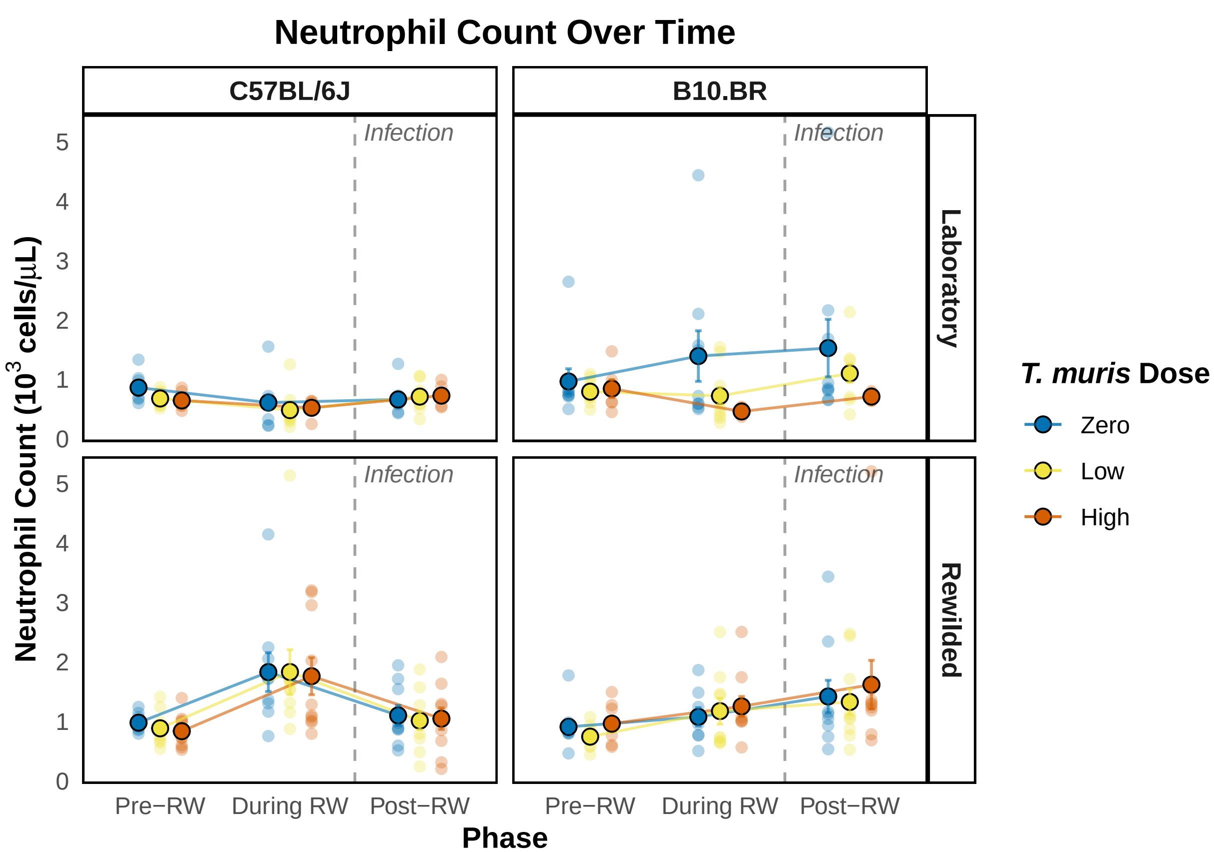
Plotting the data as it is, looking at over time changes.

```
lym_plot_data <- cleaned_combined_cbc_data_long %>%
  filter(cell_type == "lym")

standard_error_lym <- lym_plot_data %>%
  group_by(phase, t_muris_dose, strain, lab_or_rw) %>%
  summarise(mean = mean(number),
            sd = sd(number),
            n = n(),
            se = sd / sqrt(n),
            .groups = "drop")

lym_plot <- cbc_plotter(lym_plot_data,
                        standard_error_lym,
                        expression(bold("Lymphocyte Count (10"^3~"cells/"*mu*"L)")),
                        "Lymphocyte Count Over Time")

lym_plot
```




```
plot_saver("lym_number.png", plot = lym_plot)
```

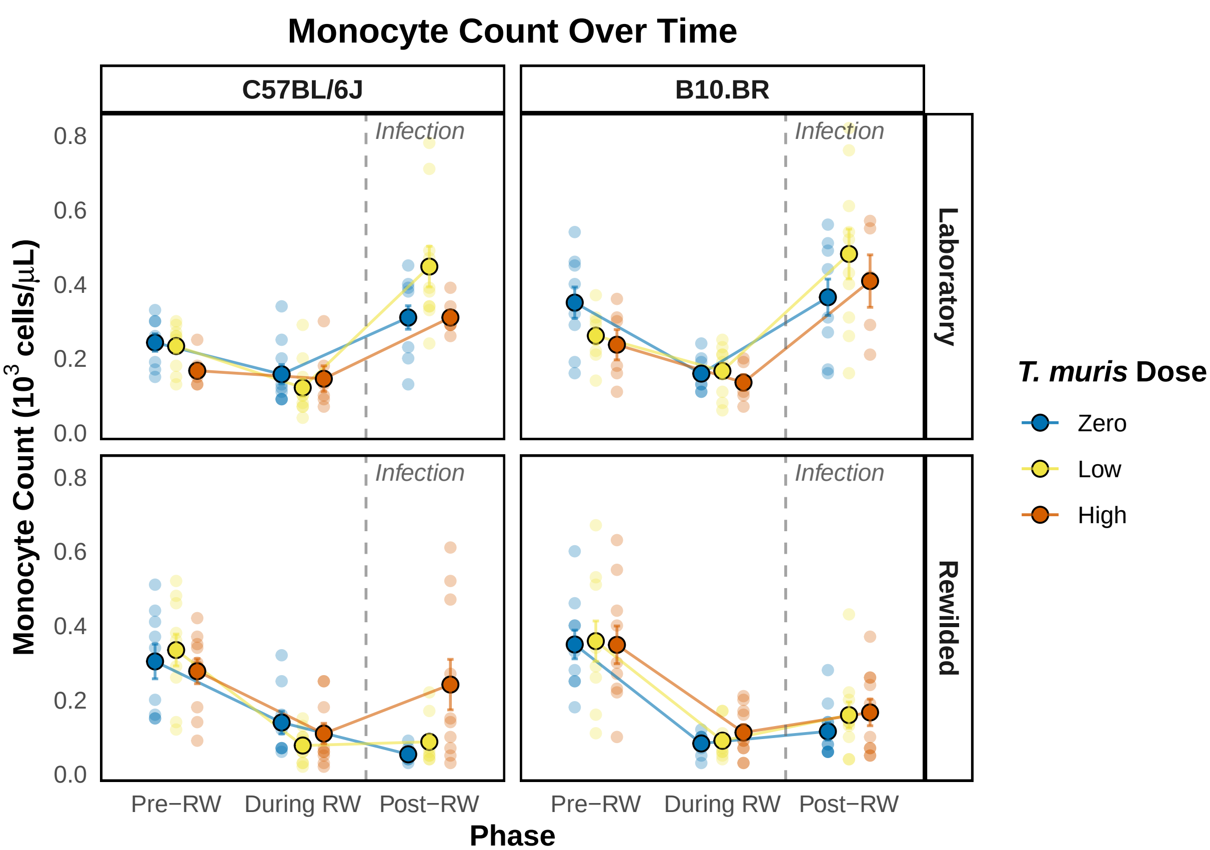
```r
neu_plot_data <- cleaned_combined_cbc_data_long %>%
  filter(cell_type == "neu")

standard_error_neu <- neu_plot_data %>%
  group_by(phase, t_muris_dose, strain, lab_or_rw) %>%
  summarise(mean = mean(number),
            sd = sd(number),
            n = n(),
            se = sd / sqrt(n),
            .groups = "drop")

neu_plot <- cbc_plotter(neu_plot_data,
                        standard_error_neu,
                        expression(bold("Neutrophil Count (10"^3~"cells/"*mu*"L)")),
                        "Neutrophil Count Over Time")

neu_plot
```




```r
plot_saver("neu_number.png", plot = neu_plot)
```

```r
mon_plot_data <- cleaned_combined_cbc_data_long %>%
  filter(cell_type == "mon")

standard_error_mon <- mon_plot_data %>%
  group_by(phase, t_muris_dose, strain, lab_or_rw) %>%
```

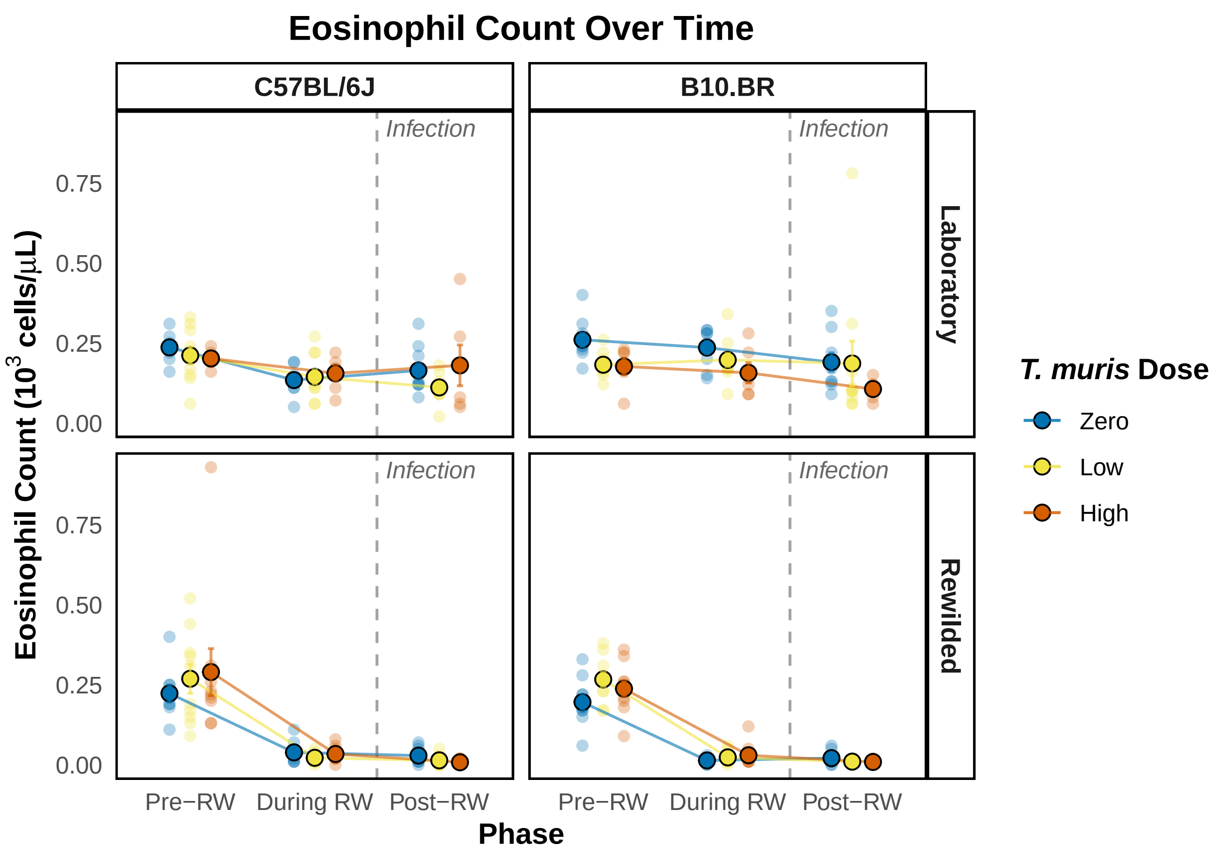
```r
  summarise(mean = mean(number),
            sd = sd(number),
            n = n(),
            se = sd / sqrt(n),
            .groups = "drop")

mon_plot <- cbc_plotter(mon_plot_data,
                        standard_error_mon,
                        expression(bold("Monocyte Count (10"^3~"cells/"*mu*"L)")),
                        "Monocyte Count Over Time")

mon_plot
```




```r
plot_saver("mon_number.png", plot = mon_plot)
```

```r
eos_plot_data <- cleaned_combined_cbc_data_long %>%
  filter(cell_type == "eos")

standard_error_eos <- eos_plot_data %>%
  group_by(phase, t_muris_dose, strain, lab_or_rw) %>%
  summarise(mean = mean(number),
            sd = sd(number),
            n = n(),
            se = sd / sqrt(n),
            .groups = "drop")
```

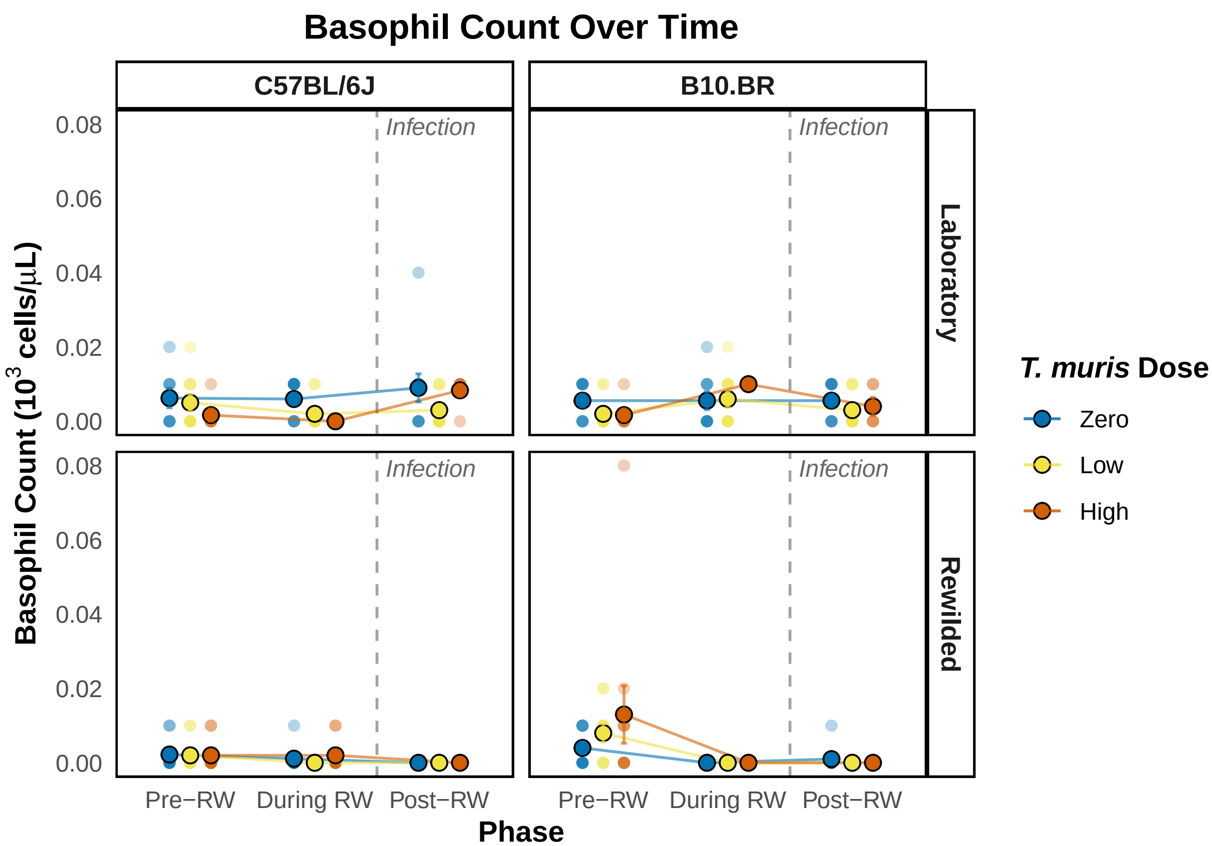
```r
eos_plot <- cbc_plotter(eos_plot_data,
                        standard_error_eos,
                        expression(bold("Eosinophil Count (10"^3~"cells/"*mu*"L)")),
                        "Eosinophil Count Over Time")

eos_plot
```




```r
plot_saver("eos_number.png", plot = eos_plot)
```

```r
bas_plot_data <- cleaned_combined_cbc_data_long %>%
  filter(cell_type == "bas")

standard_error_bas <- bas_plot_data %>%
  group_by(phase, t_muris_dose, strain, lab_or_rw) %>%
  summarise(mean = mean(number),
            sd = sd(number),
            n = n(),
            se = sd / sqrt(n),
            .groups = "drop")

bas_plot <- cbc_plotter(bas_plot_data,
                        standard_error_bas,
                        expression(bold("Basophil Count (10"^3~"cells/"*mu*"L)")),
                        "Basophil Count Over Time")
```

```r
bas_plot
```




```r
plot_saver("bas_number.png", plot = bas_plot)
```

Phase-split models and EMM plots for each CBC readout

I'm thinking that since the pre-rw phase is where everyone is still just lab, there is no environment effect yet — so lab_or_rw is excluded.

We still include t_muris_dose * strain so that the emmeans reflect the baseline differences between groups that will later receive different doses.

Similarly, during-rw we add lab_or_rw since mice are now in different environments, giving us the full lab_or_rw * t_muris_dose * strain.

Post-rw is the same full three-way interaction.

We will also remove the mouse random effect since we're not sampling the same mouse over and over.

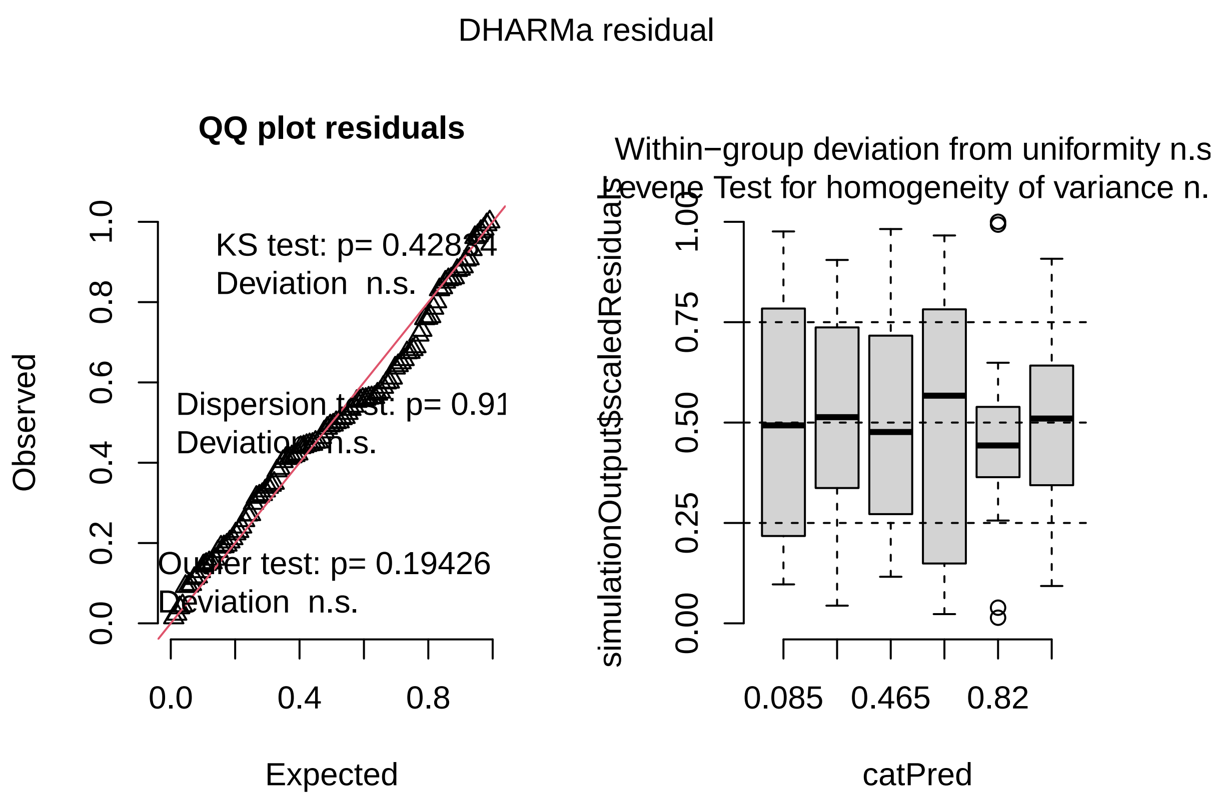
Neutrophil

```
# Pre-RW: dose x strain (no environment - all mice are lab-housed)
neu_pre <- lmer(log(neu_number) ~ t_muris_dose * strain +
                  (1 | cage_wedge) + (1 | cohort),
                data = filter(cleaned_combined_cbc_data,
                              phase == "Pre-RW"))

print(summary(neu_pre), correlation = FALSE)
```

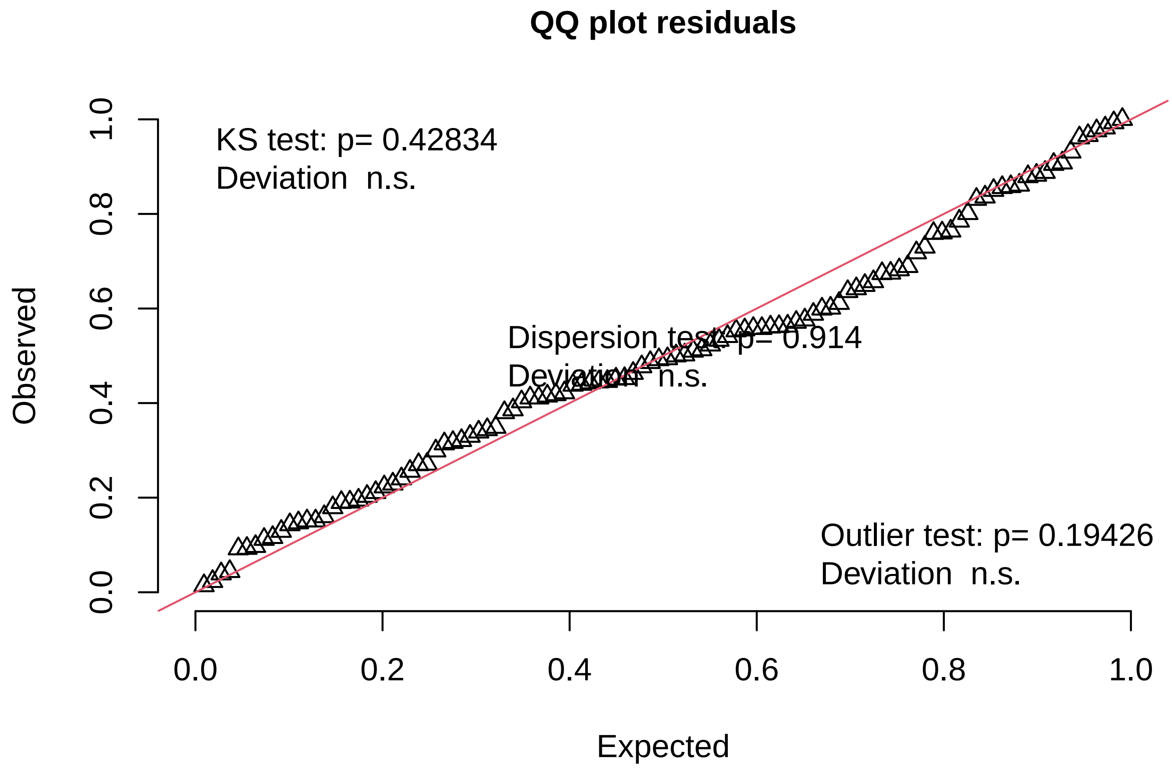
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(neu_number) ~ t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Pre-RW")
##
## REML criterion at convergence: 54.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.5687 -0.6285 -0.0339  0.4809  3.5828
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.002092 0.04574
##  cohort     (Intercept) 0.012449 0.11157
##  Residual               0.080682 0.28405
## Number of obs: 108, groups:  cage_wedge, 21; cohort, 2
##
## Fixed effects:
```

```
##                               Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                   -0.10175    0.10567   2.43624  -0.963   0.4212
## t_muris_doseLow               -0.18443    0.09390 93.51049  -1.964   0.0525 .
## t_muris_doseHigh              -0.21202    0.09926 95.33065  -2.136   0.0352 *
## strainB10.BR                  -0.05203    0.09736 75.63859  -0.534   0.5947
## t_muris_doseLow:strainB10.BR   0.03640    0.13084 92.96658   0.278   0.7815
## t_muris_doseHigh:strainB10.BR  0.20931    0.13897 98.71156   1.506   0.1352
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
neu_pre_sim <- simulateResiduals(fittedModel = neu_pre, n = 1000)
plot(neu_pre_sim)
```

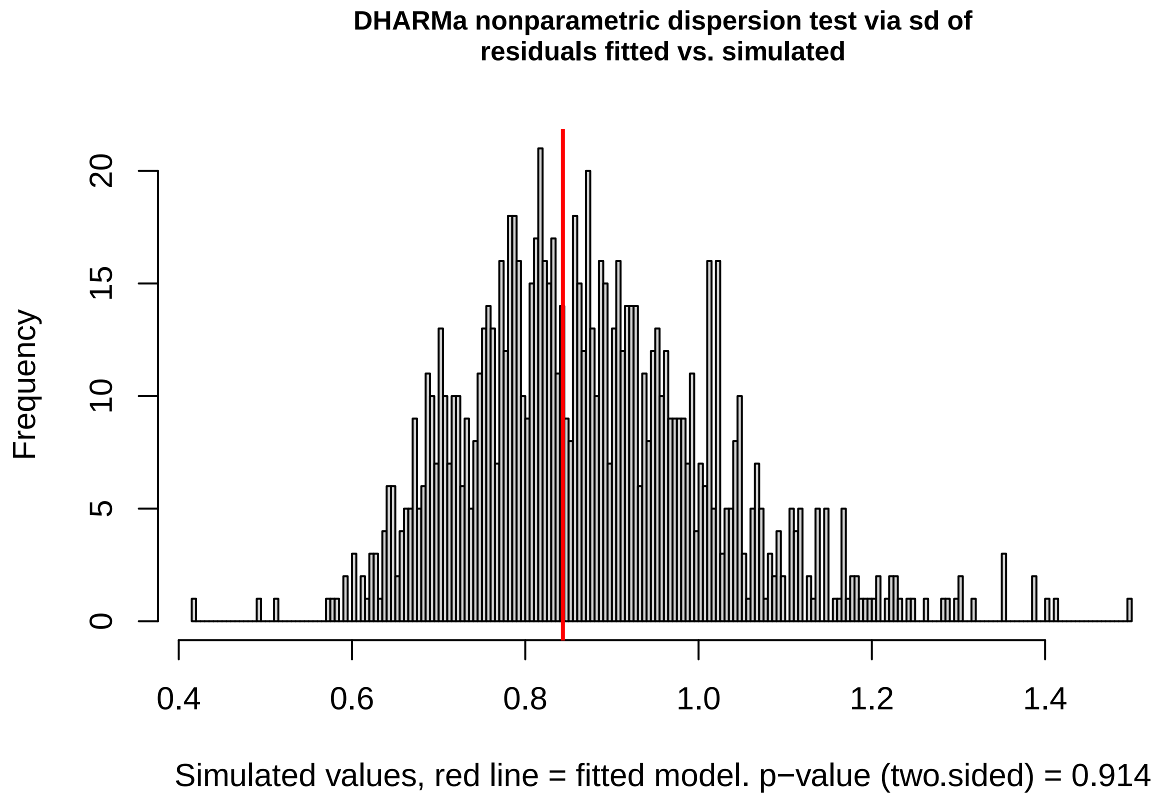



```r
testUniformity(neu_pre_sim)
```



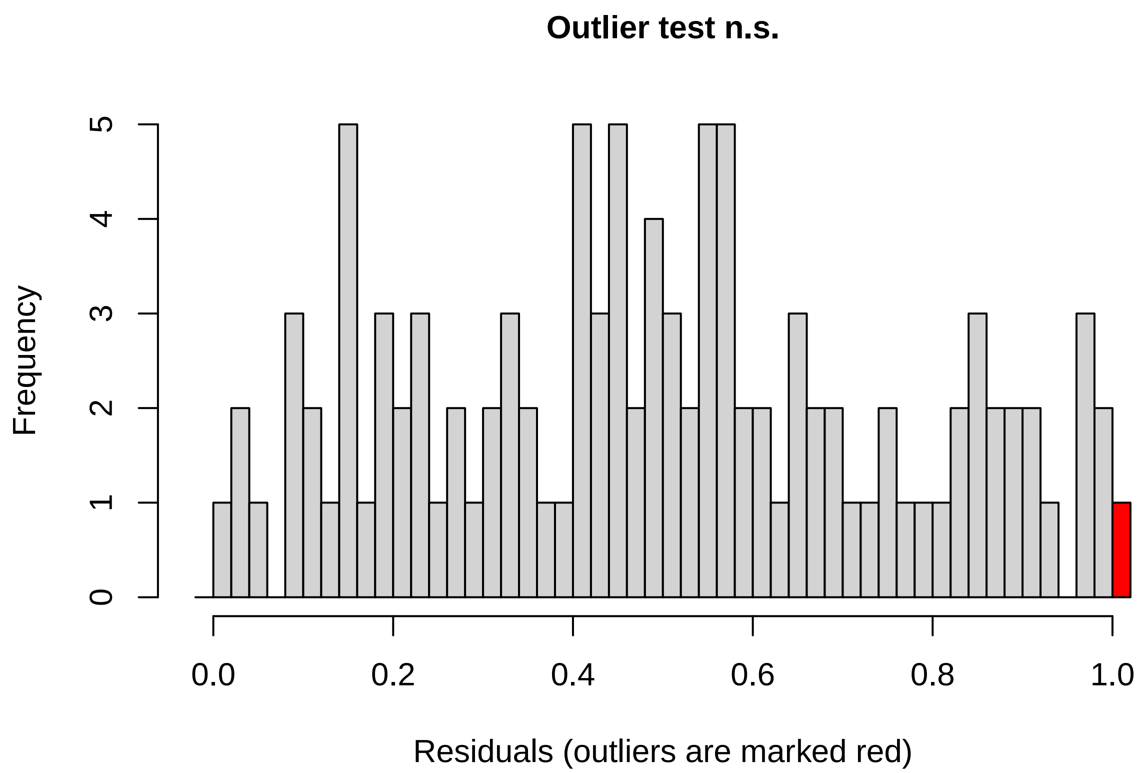

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.084185, p-value = 0.4283
## alternative hypothesis: two-sided
```

```
testDispersion(neu_pre_sim)
```



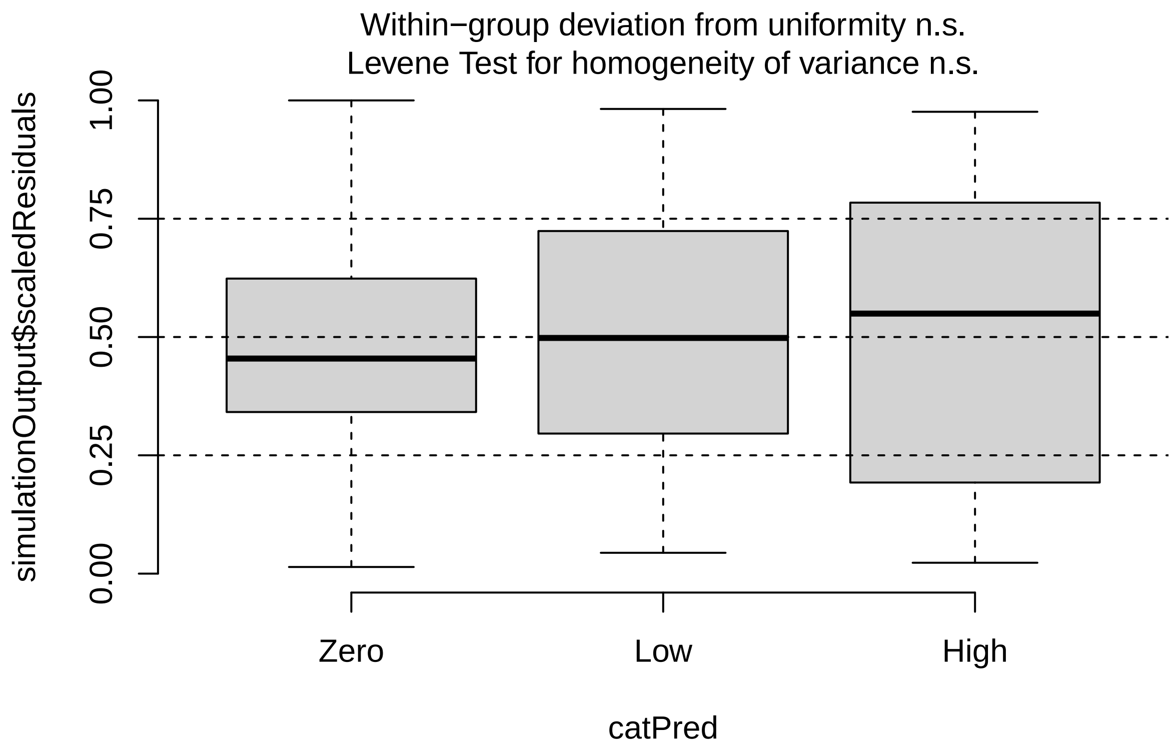

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.96506, p-value = 0.914
## alternative hypothesis: two.sided
```

```
testOutliers(neu_pre_sim)
```

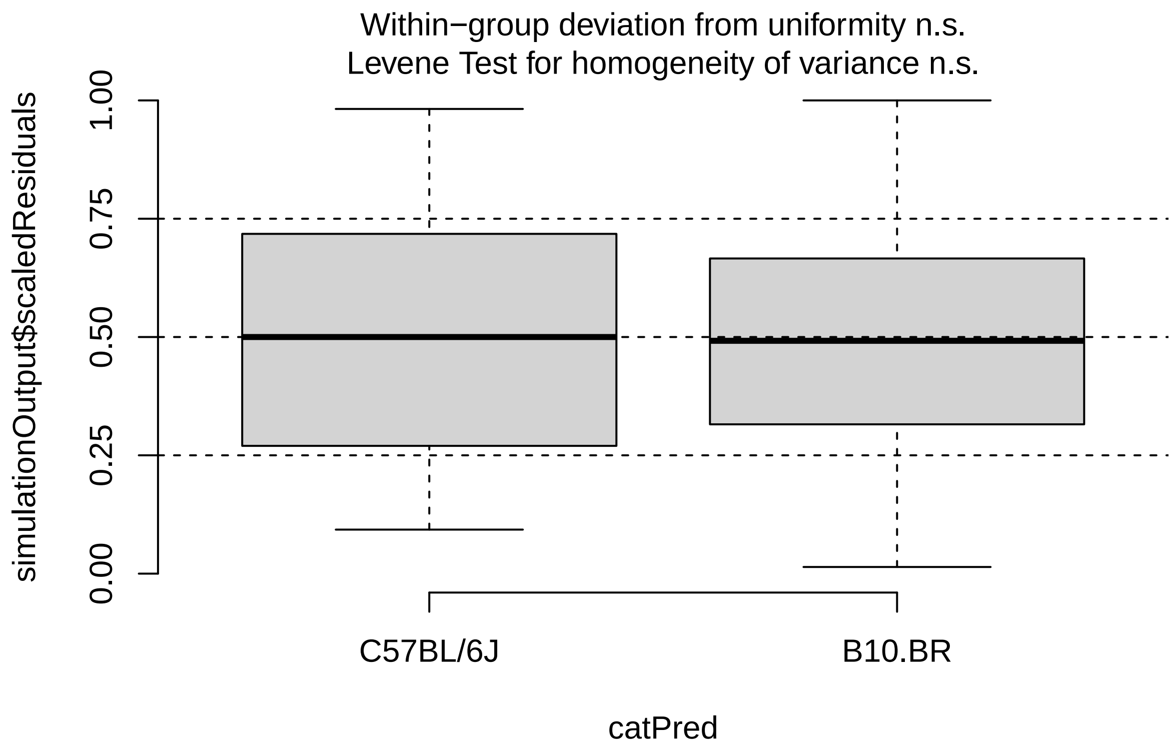



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  neu_pre_sim
## outliers at both margin(s) = 1, observations = 108, p-value = 0.1943
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002343967 0.0505105337
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                         0.009259259
```

```
plotResiduals(neu_pre_sim, form = filter(cleaned_combined_cbc_data,
                                         phase == "Pre-RW")$t_muris_dose)
```




```r
plotResiduals(neu_pre_sim, form = filter(cleaned_combined_cbc_data,
                                         phase == "Pre-RW")$strain)
```




```r
# During RW: environment x dose x strain
neu_during <- lmer(log(neu_number) ~ lab_or_rw * t_muris_dose * strain +
                     (1 | cage_wedge) + (1 | cohort),
                   data = filter(cleaned_combined_cbc_data, phase == "During RW"))
```

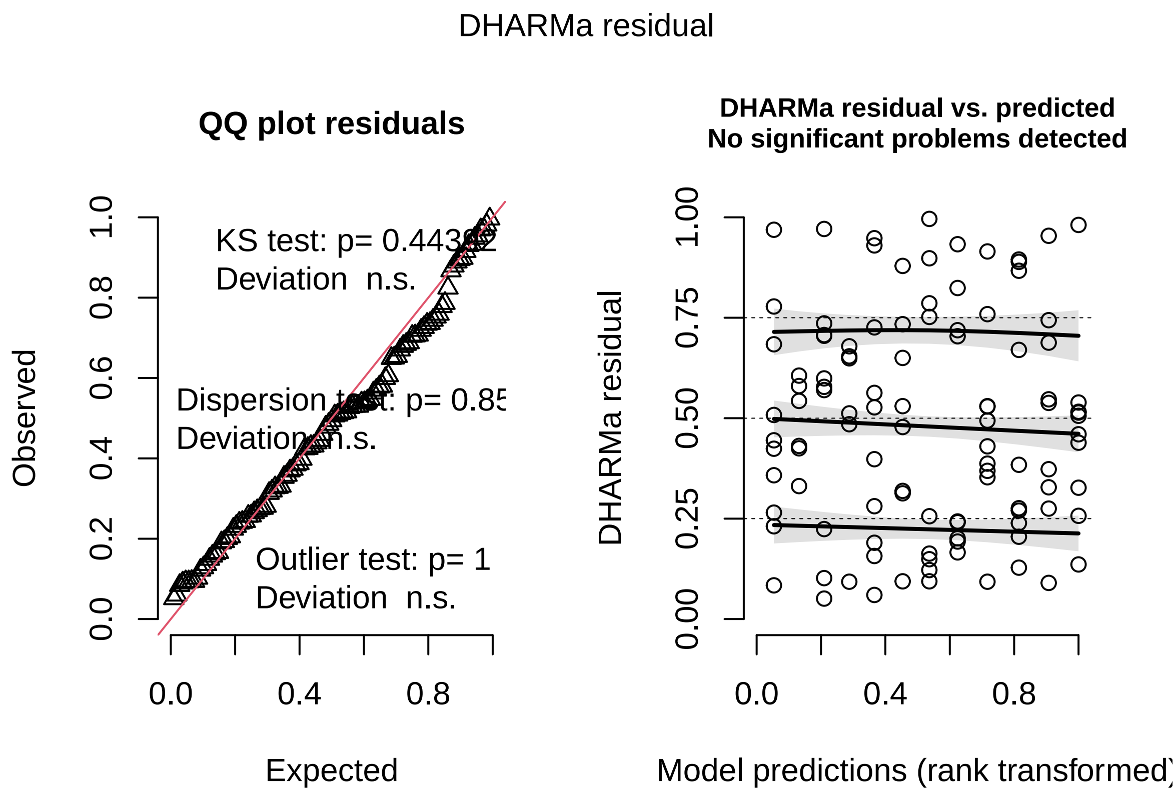
```
## boundary (singular) fit: see help('isSingular')
```

```r
print(summary(neu_during), correlation = FALSE)
```

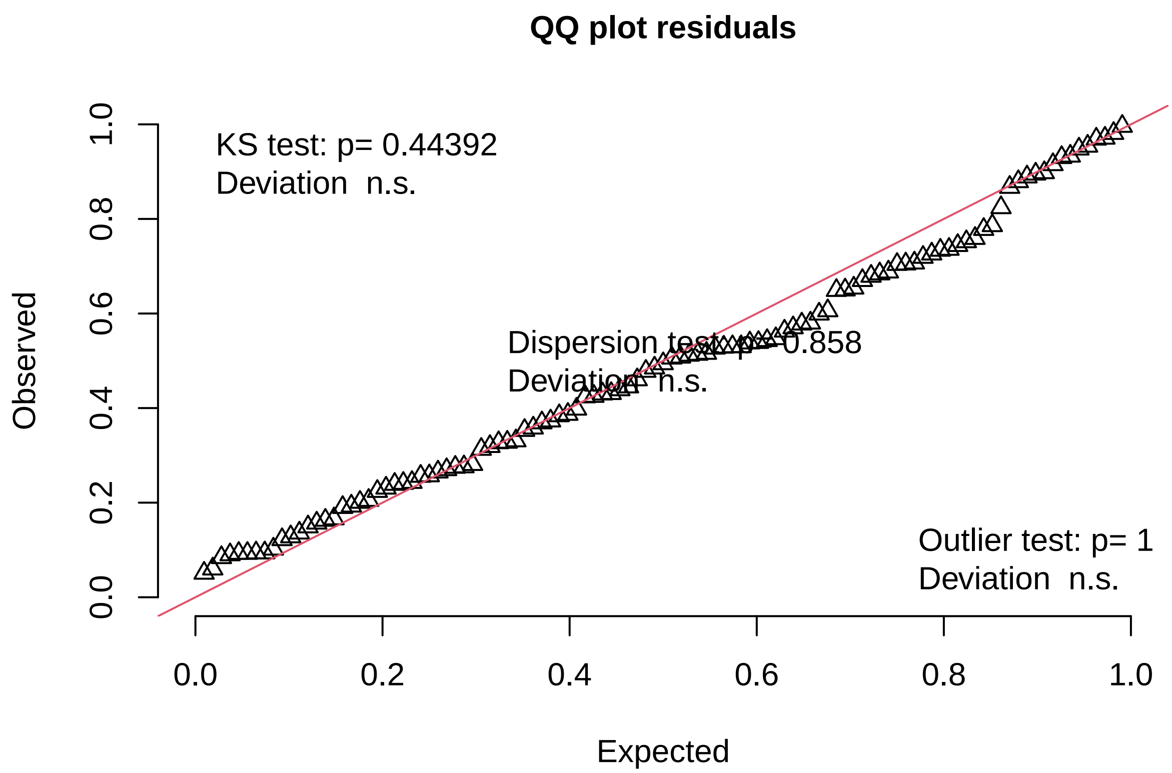
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(neu_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "During RW")
##
## REML criterion at convergence: 141.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2518 -0.5555 -0.0492  0.4258  2.8069
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.0000   0.0000
```

```
##  cohort     (Intercept) 0.1286   0.3586
##  Residual               0.1905   0.4365
## Number of obs: 107, groups:  cage_wedge, 29; cohort, 2
##
## Fixed effects:
##                                           Estimate Std. Error        df
## (Intercept)                              -0.698556   0.288812  1.592009
## lab_or_rwRW                               1.165703   0.200587 94.001786
## t_muris_doseLow                          -0.156864   0.195382 94.009551
## t_muris_doseHigh                          0.002688   0.225556 94.007168
## strainB10.BR                              0.711701   0.200587 94.001786
## lab_or_rwRW:t_muris_doseLow               0.174646   0.279891 94.000913
## lab_or_rwRW:t_muris_doseHigh             -0.043271   0.301729 94.000785
## lab_or_rwRW:strainB10.BR                 -1.181717   0.291901 94.000044
## t_muris_doseLow:strainB10.BR             -0.338131   0.279891 94.000913
## t_muris_doseHigh:strainB10.BR            -0.801285   0.322090 94.000688
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  0.380085   0.404407 94.000433
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR 0.997723   0.432224 94.000015
##                                          t value Pr(>|t|)
## (Intercept)                               -2.419 0.168044
## lab_or_rwRW                                5.811 8.43e-08 ***
## t_muris_doseLow                           -0.803 0.424082
## t_muris_doseHigh                           0.012 0.990516
## strainB10.BR                               3.548 0.000608 ***
## lab_or_rwRW:t_muris_doseLow                0.624 0.534154
## lab_or_rwRW:t_muris_doseHigh              -0.143 0.886273
## lab_or_rwRW:strainB10.BR                  -4.048 0.000106 ***
## t_muris_doseLow:strainB10.BR              -1.208 0.230047
## t_muris_doseHigh:strainB10.BR             -2.488 0.014616 *
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   0.940 0.349699
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  2.308 0.023172 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
neu_during_sim <- simulateResiduals(fittedModel = neu_during, n = 1000)
plot(neu_during_sim)
```

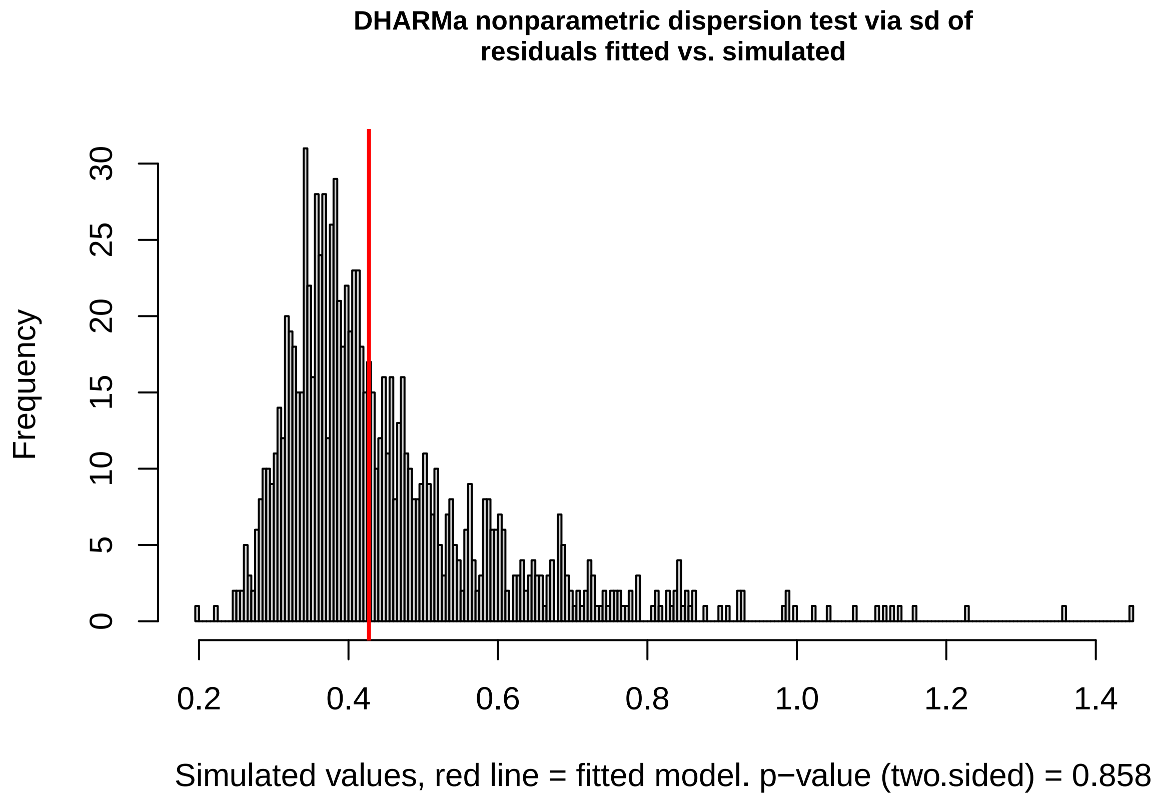



```
testUniformity(neu_during_sim)
```



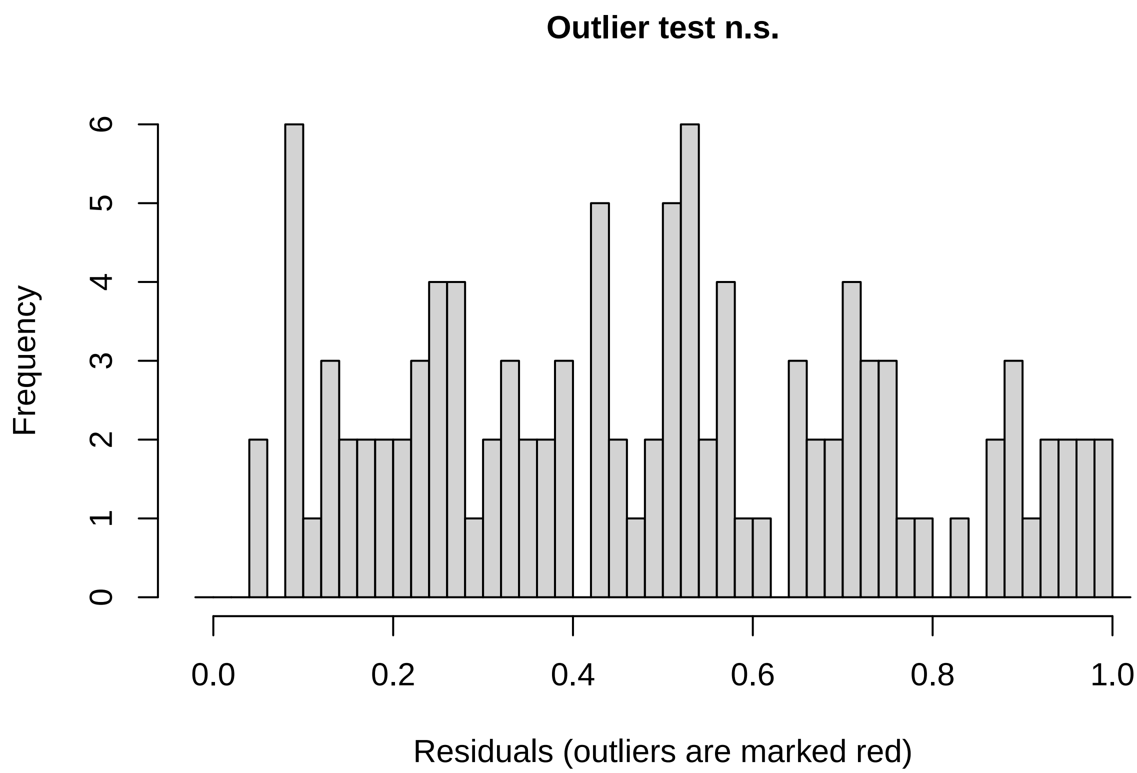

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.083551, p-value = 0.4439
## alternative hypothesis: two-sided
```

```
testDispersion(neu_during_sim)
```



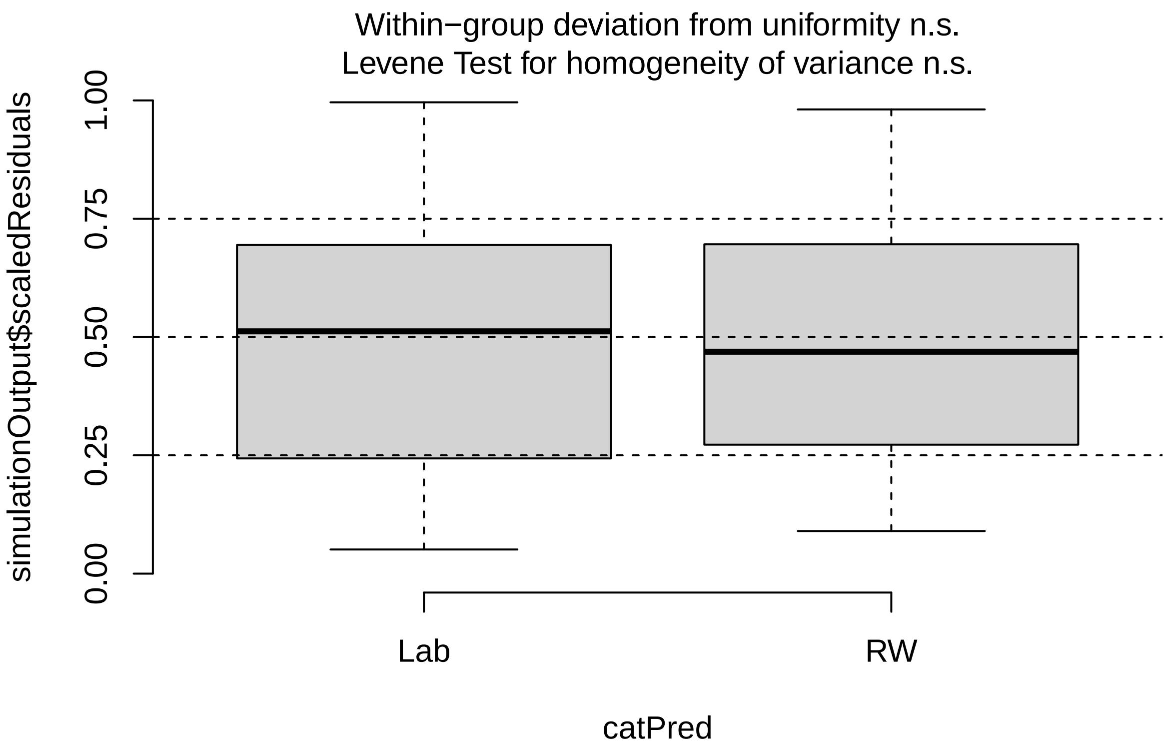

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.9426, p-value = 0.858
## alternative hypothesis: two.sided
```

```
testOutliers(neu_during_sim)
```

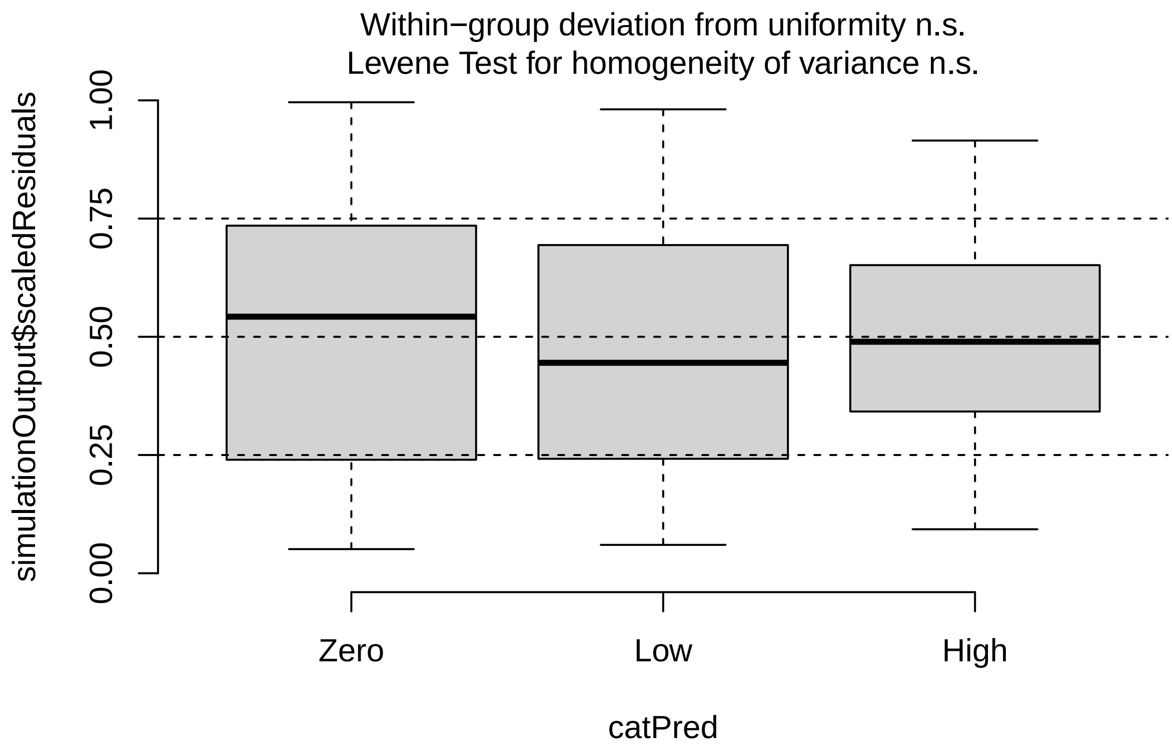



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  neu_during_sim
## outliers at both margin(s) = 0, observations = 107, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.000000 0.033888
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                    0
```

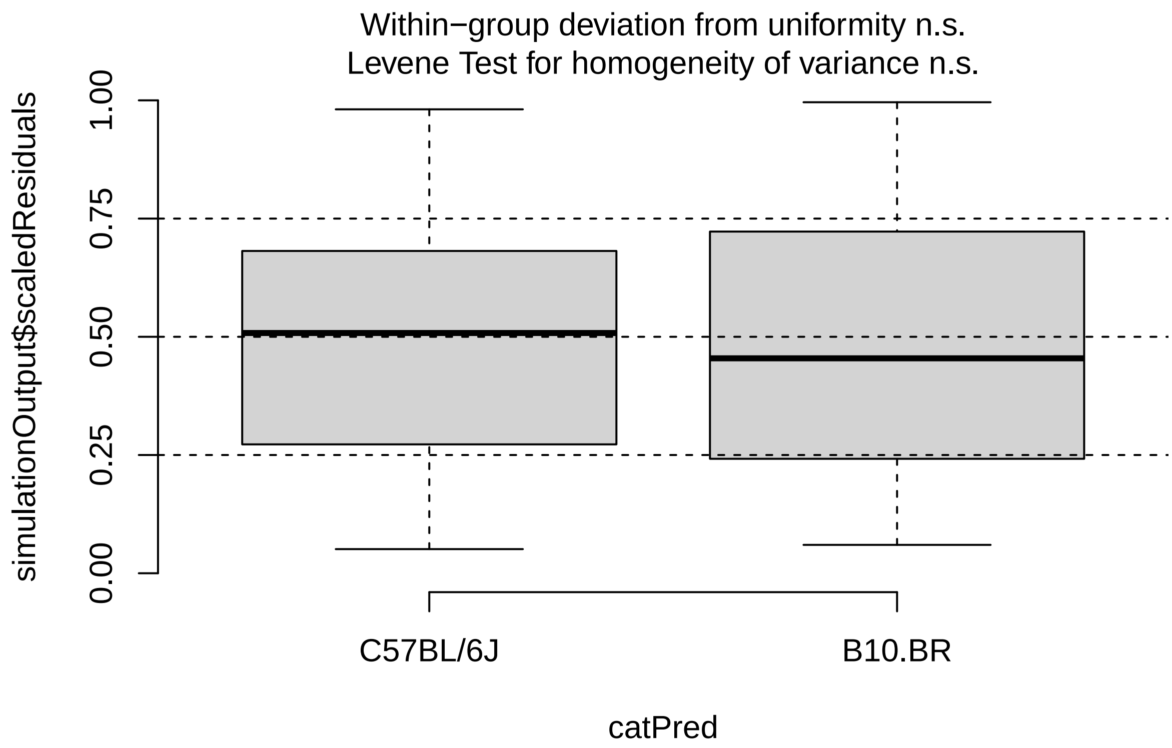
```
plotResiduals(neu_during_sim, form = filter(cleaned_combined_cbc_data,
                                             phase == "During RW")$lab_or_rw)
```




```
plotResiduals(neu_during_sim, form = filter(cleaned_combined_cbc_data,
                                            phase == "During RW")$t_muris_dose)
```




```r
plotResiduals(neu_during_sim, form = filter(cleaned_combined_cbc_data,
                                            phase == "During RW")$strain)
```

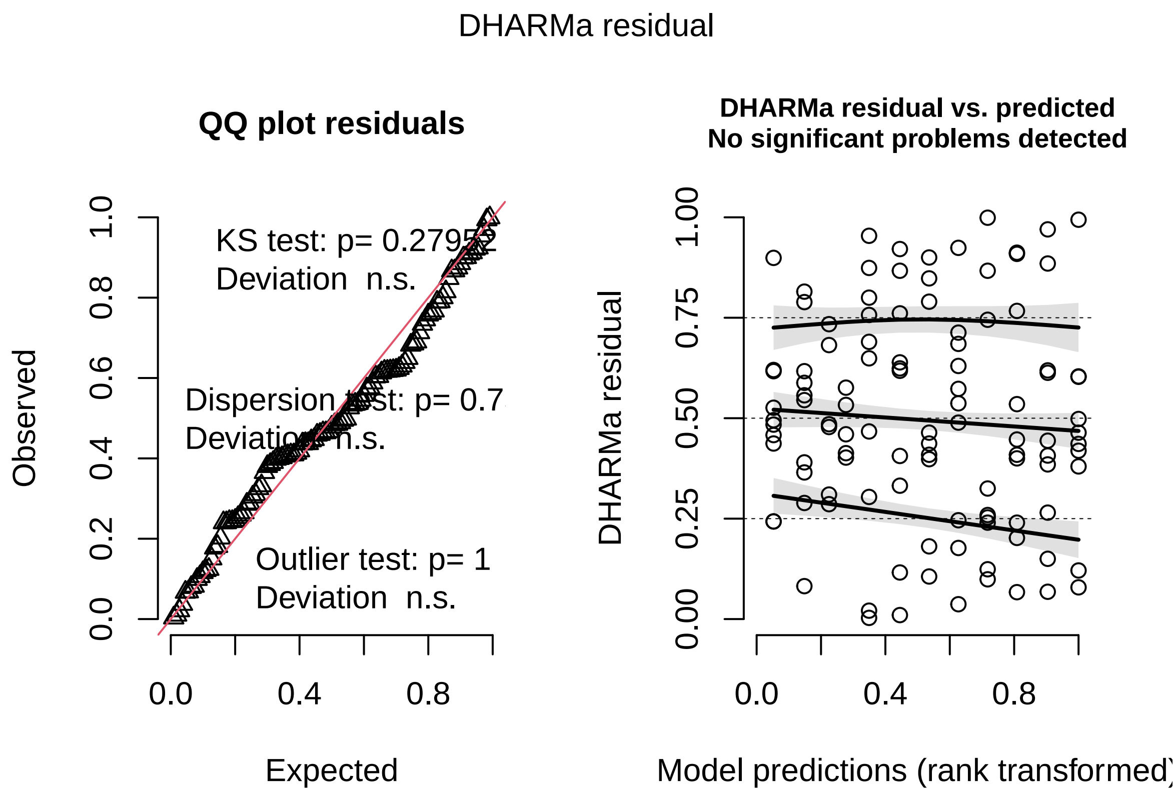



```r
# Post-RW: environment x infection x strain
neu_post <- lmer(log(neu_number) ~ lab_or_rw * t_muris_dose * strain +
                   (1 | cage_wedge) + (1 | cohort),
                 data = filter(cleaned_combined_cbc_data, phase == "Post-RW"))

print(summary(neu_post), correlation = FALSE)
```

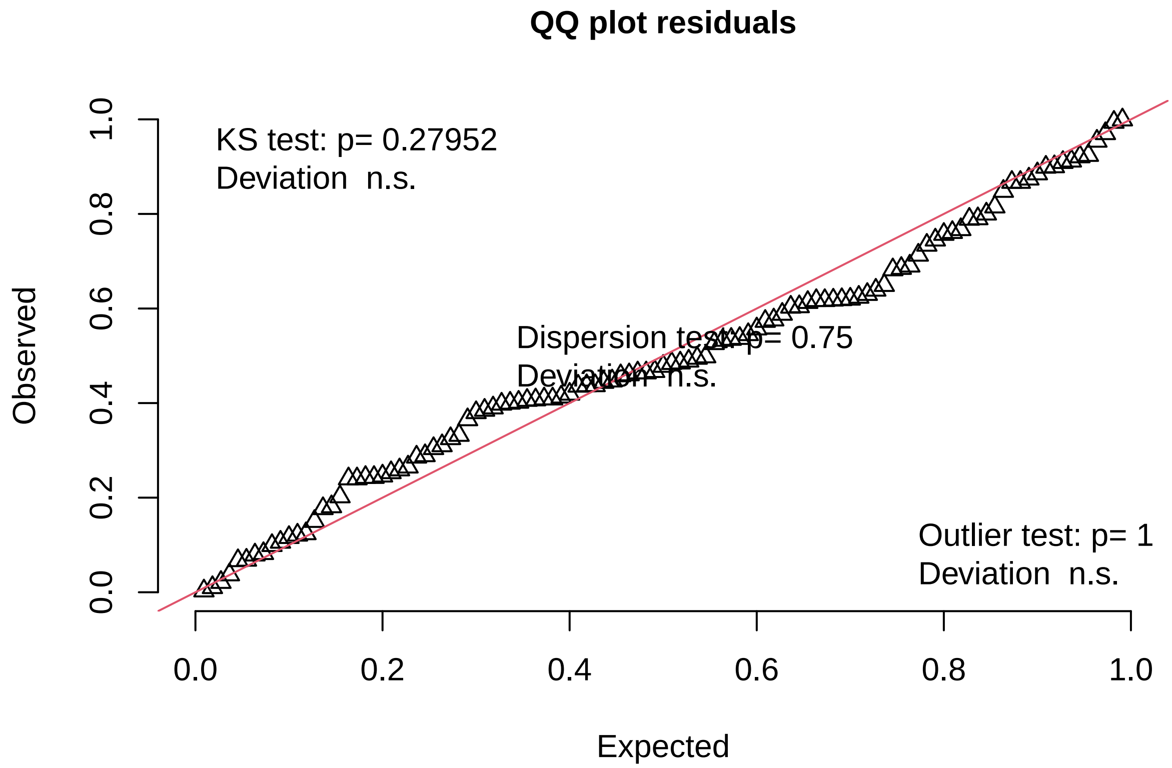
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(neu_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Post-RW")
##
## REML criterion at convergence: 159.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.70241 -0.54506  0.02463  0.53796  2.76363
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.01425  0.1194
##  cohort     (Intercept) 0.03475  0.1864
##  Residual               0.21820  0.4671
## Number of obs: 109, groups:  cage_wedge, 29; cohort, 2
```

```
##
## Fixed effects:
##                                         Estimate Std. Error       df t value
## (Intercept)                             -0.47338    0.20237  3.61437  -2.339
## lab_or_rwRW                              0.45071    0.23440 31.51021   1.923
## t_muris_doseLow                          0.08241    0.21286 96.03636   0.387
## t_muris_doseHigh                         0.11744    0.24736 93.87016   0.475
## strainB10.BR                             0.62328    0.22203 84.61849   2.807
## lab_or_rwRW:t_muris_doseLow             -0.18649    0.30227 93.70850  -0.617
## lab_or_rwRW:t_muris_doseHigh            -0.24494    0.32749 95.72959  -0.748
## lab_or_rwRW:strainB10.BR                -0.39087    0.33112 31.88454  -1.180
## t_muris_doseLow:strainB10.BR            -0.21876    0.30333 95.30002  -0.721
## t_muris_doseHigh:strainB10.BR           -0.60453    0.36323 94.74247  -1.664
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  0.28339    0.42631 91.98623   0.665
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.83273    0.47087 95.56978   1.768
##                                         Pr(>|t|)
## (Intercept)                               0.0864 .
## lab_or_rwRW                               0.0636 .
## t_muris_doseLow                           0.6995
## t_muris_doseHigh                          0.6361
## strainB10.BR                              0.0062 **
## lab_or_rwRW:t_muris_doseLow               0.5388
## lab_or_rwRW:t_muris_doseHigh              0.4563
## lab_or_rwRW:strainB10.BR                  0.2466
## t_muris_doseLow:strainB10.BR              0.4726
## t_muris_doseHigh:strainB10.BR             0.0993 .
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   0.5079
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.0802 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
neu_post_sim <- simulateResiduals(fittedModel = neu_post, n = 1000)
plot(neu_post_sim)
```

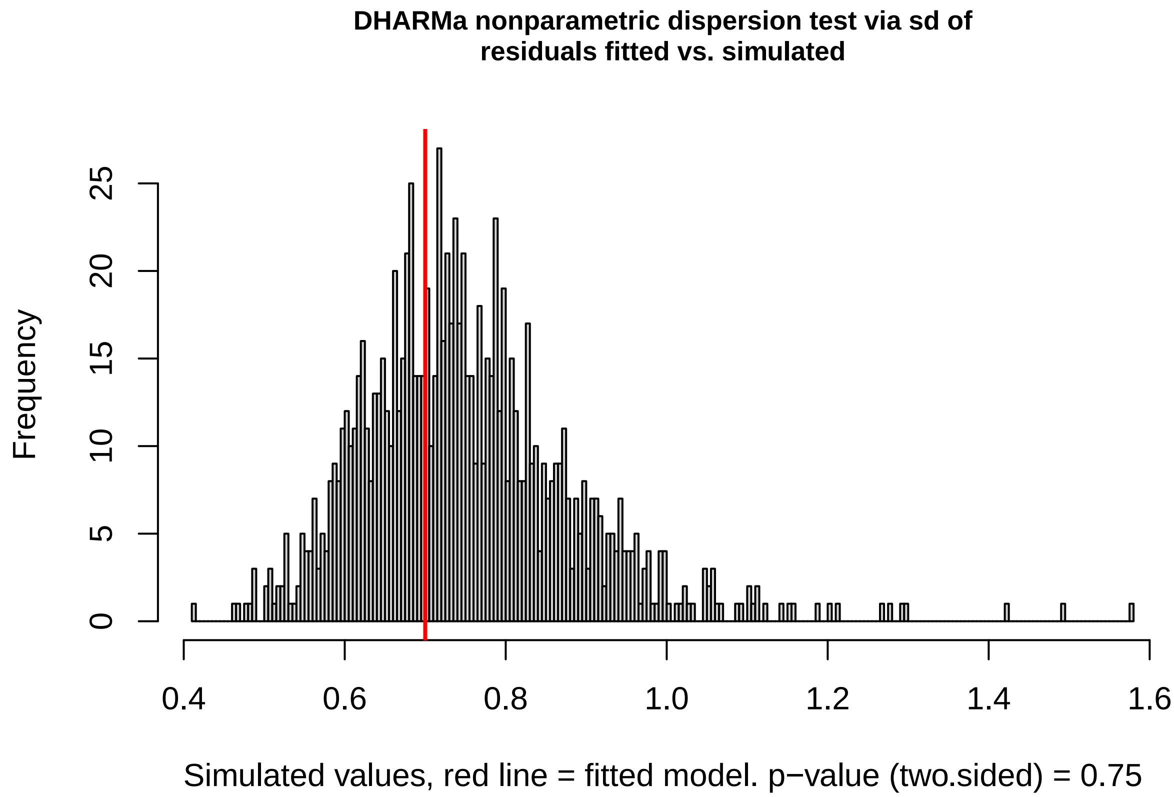



```
testUniformity(neu_post_sim)
```



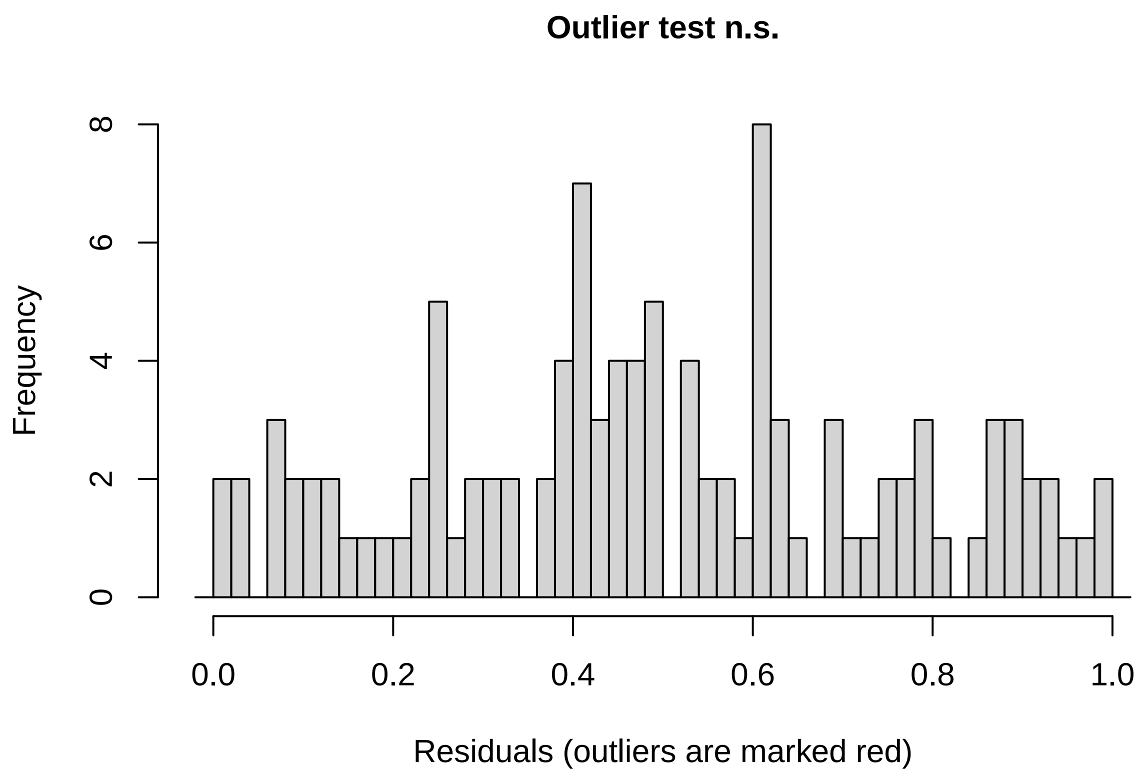

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.094945, p-value = 0.2795
## alternative hypothesis: two-sided
```

```r
testDispersion(neu_post_sim)
```



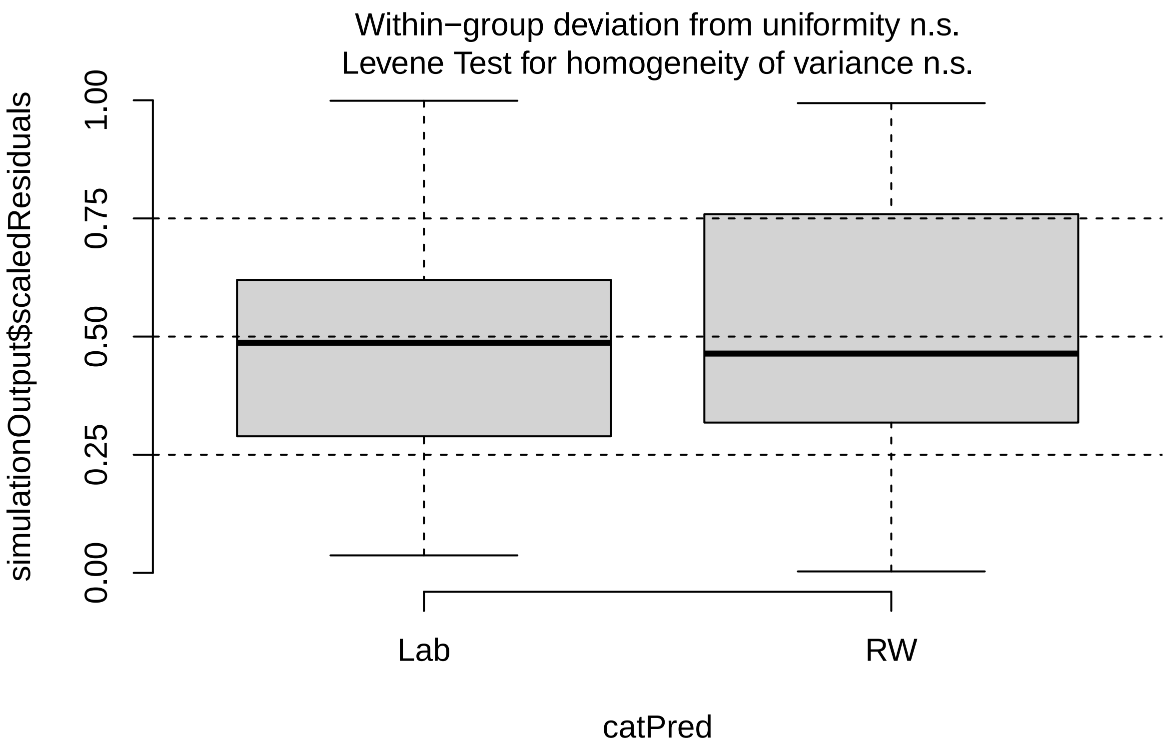

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.93189, p-value = 0.75
## alternative hypothesis: two.sided
```

```
testOutliers(neu_post_sim)
```

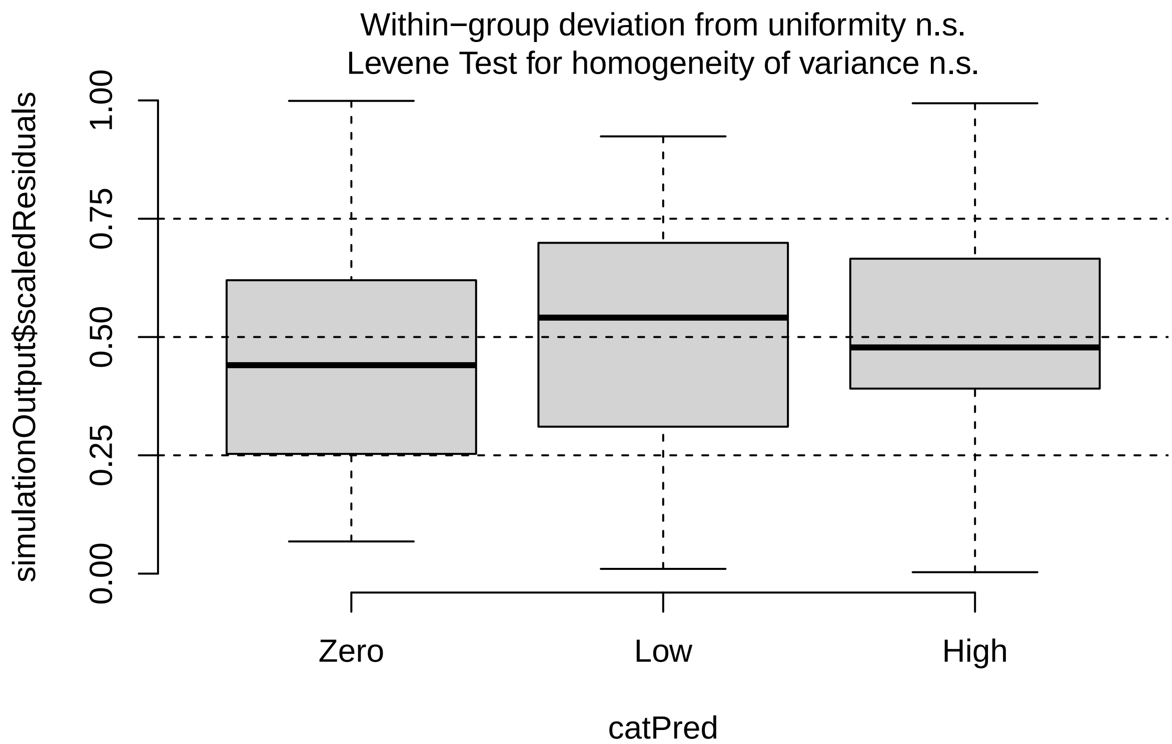



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  neu_post_sim
## outliers at both margin(s) = 0, observations = 109, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.00000000 0.03327666
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                     0
```

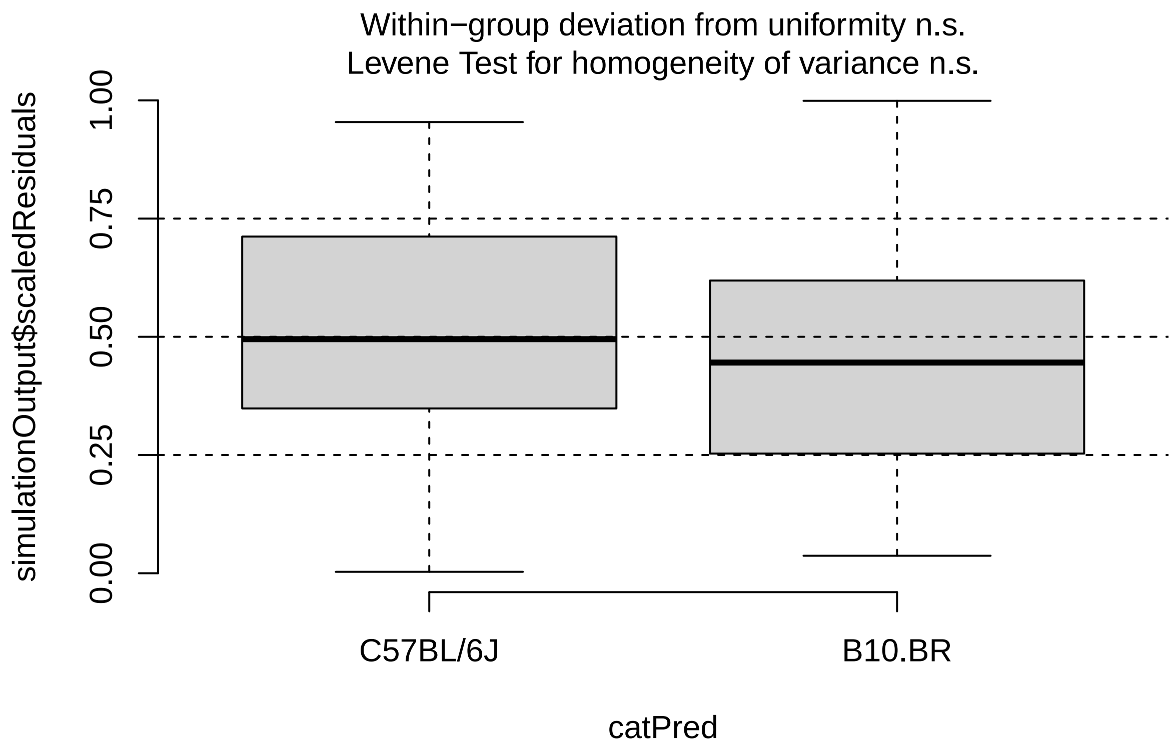
```
plotResiduals(neu_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$lab_or_rw)
```




```
plotResiduals(neu_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$t_muris_dose)
```



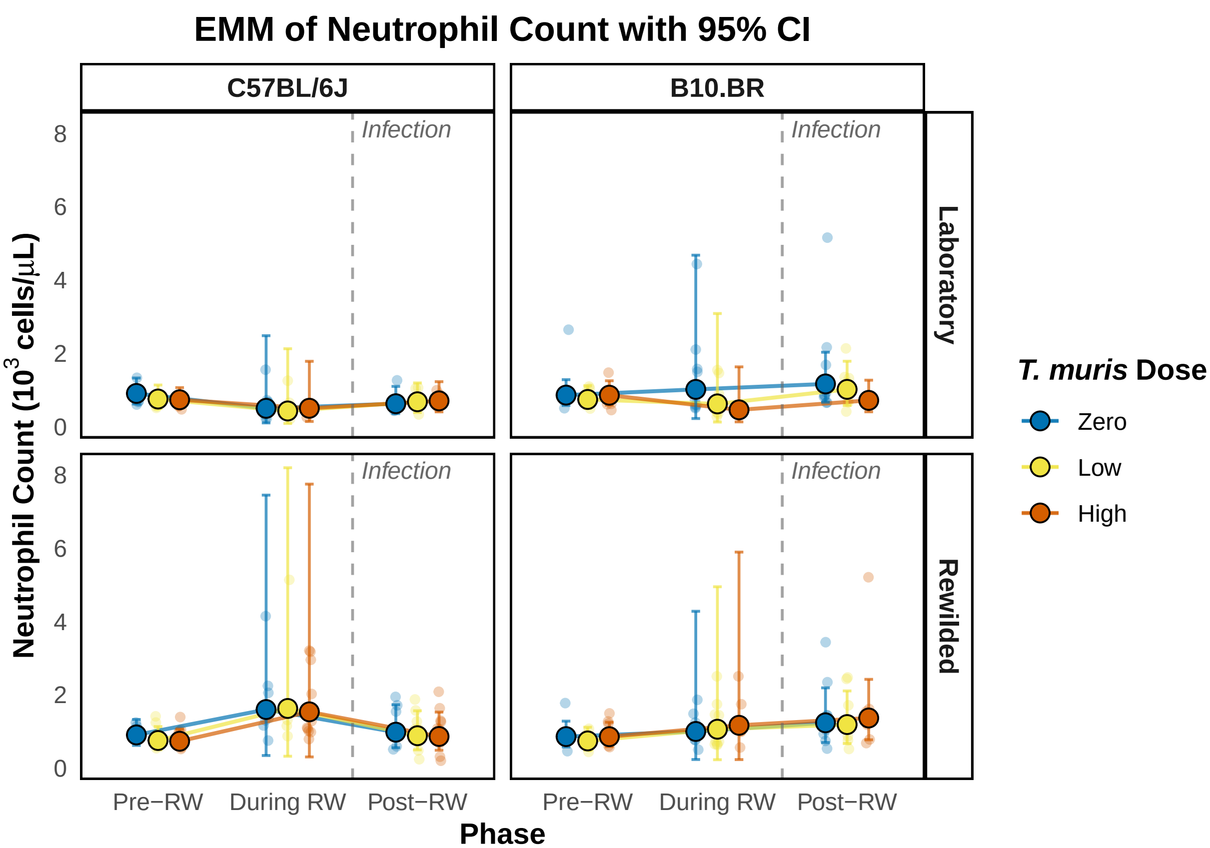

```r
plotResiduals(neu_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$strain)
```




```r
# Pre-RW emmeans: extract ~ t_muris_dose * strain, duplicate across environment
emm_neu_pre <- as.data.frame(emmeans(neu_pre,
                                     specs = "t_muris_dose",
                                     by = "strain",
                                     type = "response")) %>%
  crossing(lab_or_rw = factor(c("Lab", "RW"),
                              levels = c("Lab", "RW"))) %>%
  mutate(phase = factor("Pre-RW",
                        levels = c("Pre-RW", "During RW", "Post-RW")))

# During RW emmeans: full extraction
emm_neu_during <- as.data.frame(emmeans(neu_during,
                                        specs = c("t_muris_dose", "lab_or_rw"),
                                        by = "strain",
                                        type = "response")) %>%
  mutate(phase = factor("During RW", levels = c("Pre-RW", "During RW", "Post-RW")))

# Post-RW emmeans: full extraction
emm_neu_post <- as.data.frame(emmeans(neu_post,
                                      specs = c("t_muris_dose", "lab_or_rw"),
                                      by = "strain",
                                      type = "response")) %>%
  mutate(phase = factor("Post-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

# Combine
emm_neu_combined <- bind_rows(emm_neu_pre, emm_neu_during, emm_neu_post) %>%
```

```r
  mutate(phase = factor(phase, levels = c("Pre-RW", "During RW", "Post-RW")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "Low", "High")),
         strain = factor(strain, levels = c("C57BL/6J", "B10.BR")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

```r
emm_neu_plot <- emm_plotter(
  emm_df = emm_neu_combined,
  cbc_input = cleaned_combined_cbc_data,
  y_col = "neu_number",
  y_label = expression(bold("Neutrophil Count (10"^3~"cells/"*mu*"L)")),
  title = "EMM of Neutrophil Count with 95% CI"
)

emm_neu_plot
```




```r
plot_saver("emm_neu_number.png", emm_neu_plot)
```

Lymphocyte

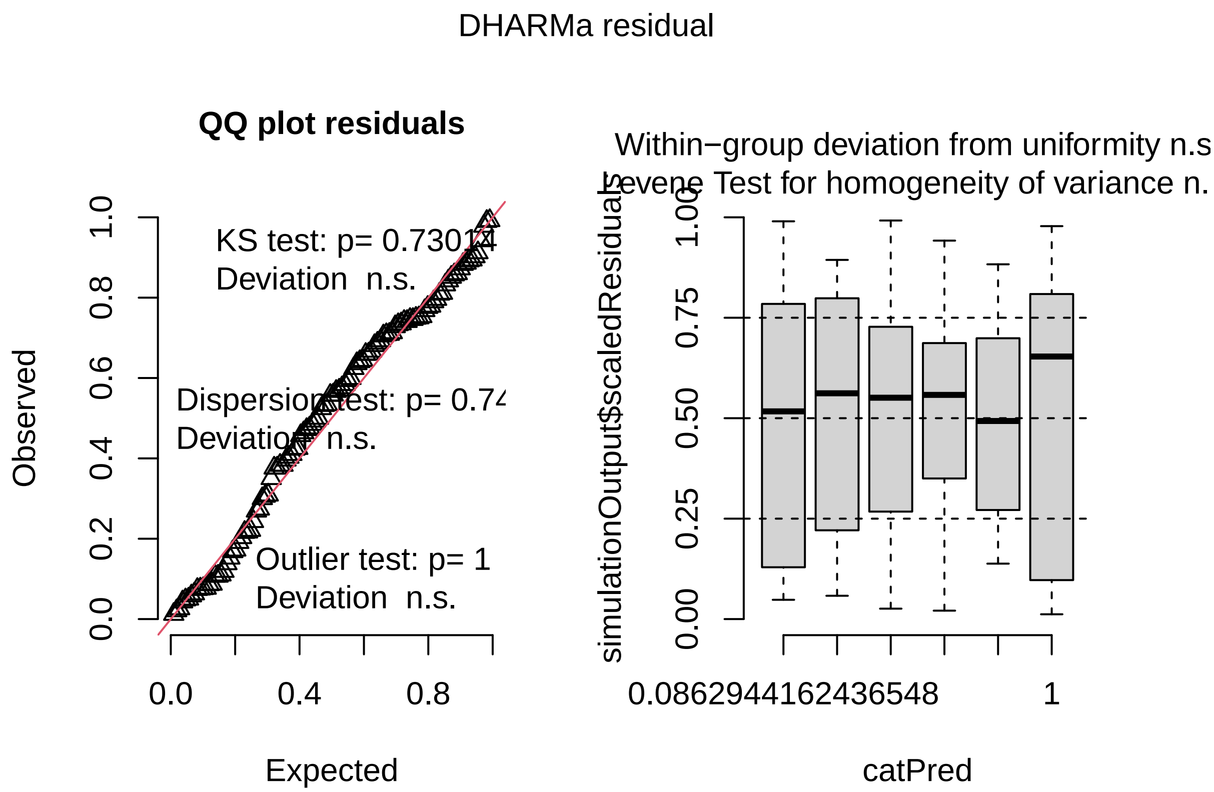
```r
# Pre-RW: dose x strain (no environment - all mice are lab-housed)
lym_pre <- lmer(log(lym_number) ~ t_muris_dose * strain +
                  (1 | cage_wedge) + (1 | cohort),
                data = filter(cleaned_combined_cbc_data, phase == "Pre-RW"))
```

```
## boundary (singular) fit: see help('isSingular')
```

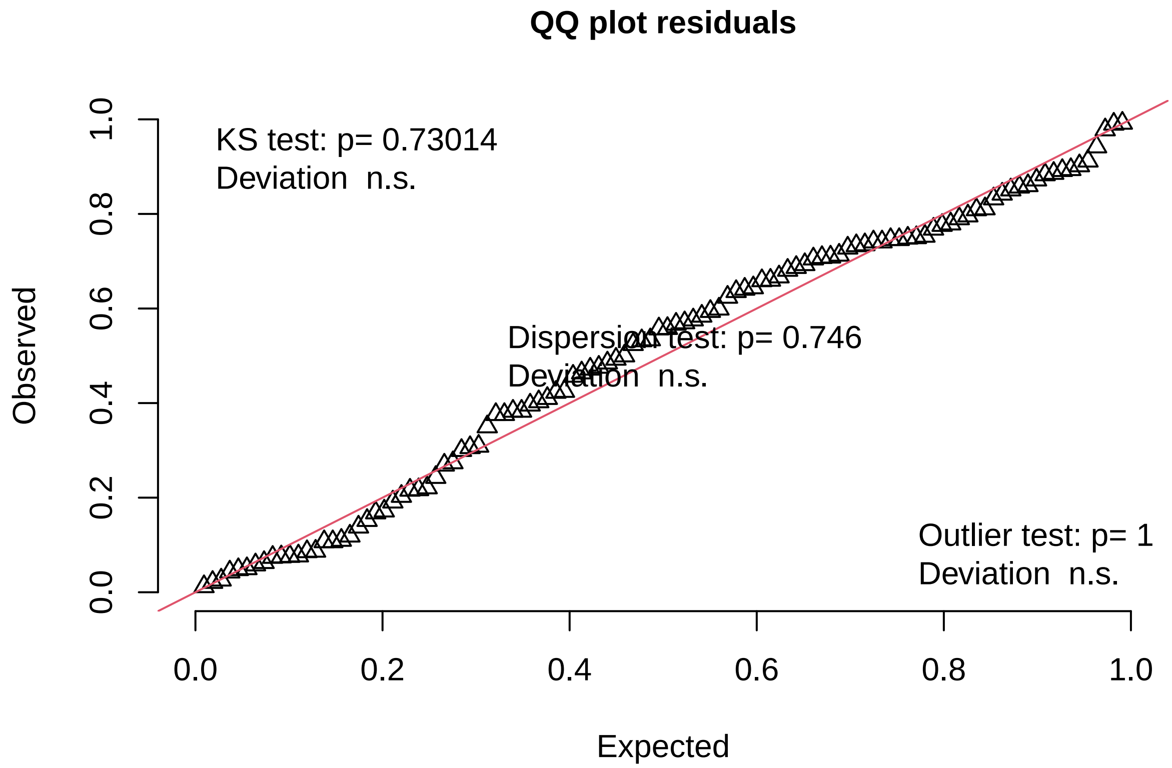
```r
print(summary(lym_pre), correlation = FALSE)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(lym_number) ~ t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Pre-RW")
##
## REML criterion at convergence: 12.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.1539 -0.6782  0.1748  0.6616  2.4973
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 0.0000000 0.00000
##  cohort     (Intercept) 0.0001733 0.01317
##  Residual               0.0558610 0.23635
## Number of obs: 108, groups:  cage_wedge, 21; cohort, 2
##
## Fixed effects:
##                                Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                     2.18018    0.05808  25.68012  37.540   <2e-16
## t_muris_doseLow                -0.03800    0.07797 101.05036  -0.487    0.627
## t_muris_doseHigh               -0.09429    0.08233 101.04523  -1.145    0.255
## strainB10.BR                    0.01372    0.07891 101.00055   0.174    0.862
## t_muris_doseLow:strainB10.BR    0.05232    0.10868 101.00029   0.481    0.631
## t_muris_doseHigh:strainB10.BR   0.07964    0.11493 101.00026   0.693    0.490
##
## (Intercept)                   ***
## t_muris_doseLow
## t_muris_doseHigh
## strainB10.BR
## t_muris_doseLow:strainB10.BR
## t_muris_doseHigh:strainB10.BR
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
lym_pre_sim <- simulateResiduals(fittedModel = lym_pre, n = 1000)
plot(lym_pre_sim)
```

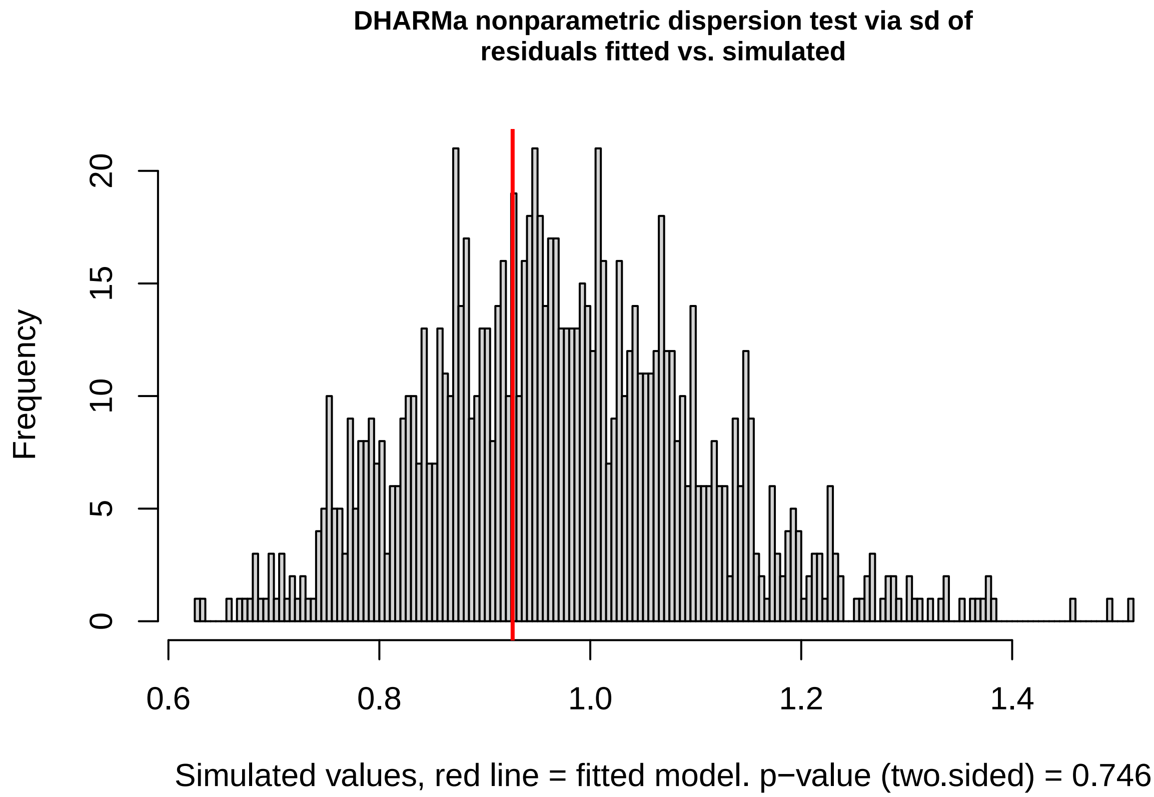



```r
testUniformity(lym_pre_sim)
```



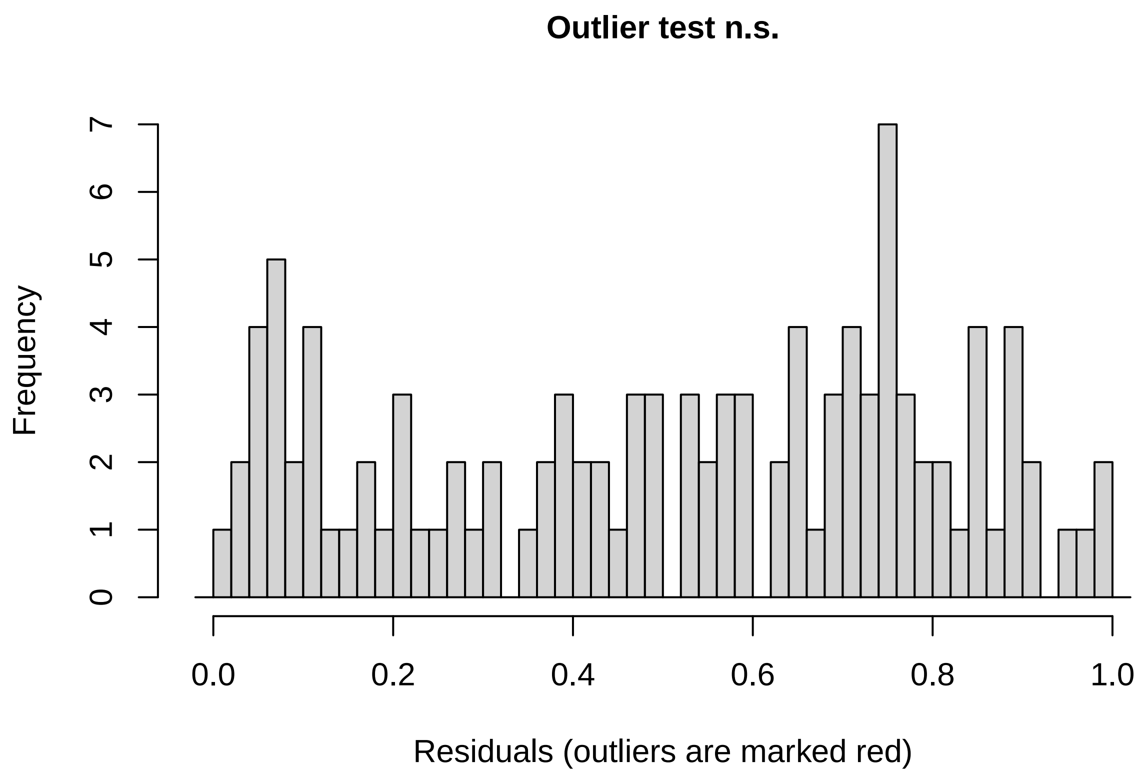

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.066259, p-value = 0.7301
## alternative hypothesis: two-sided
```

```
testDispersion(lym_pre_sim)
```



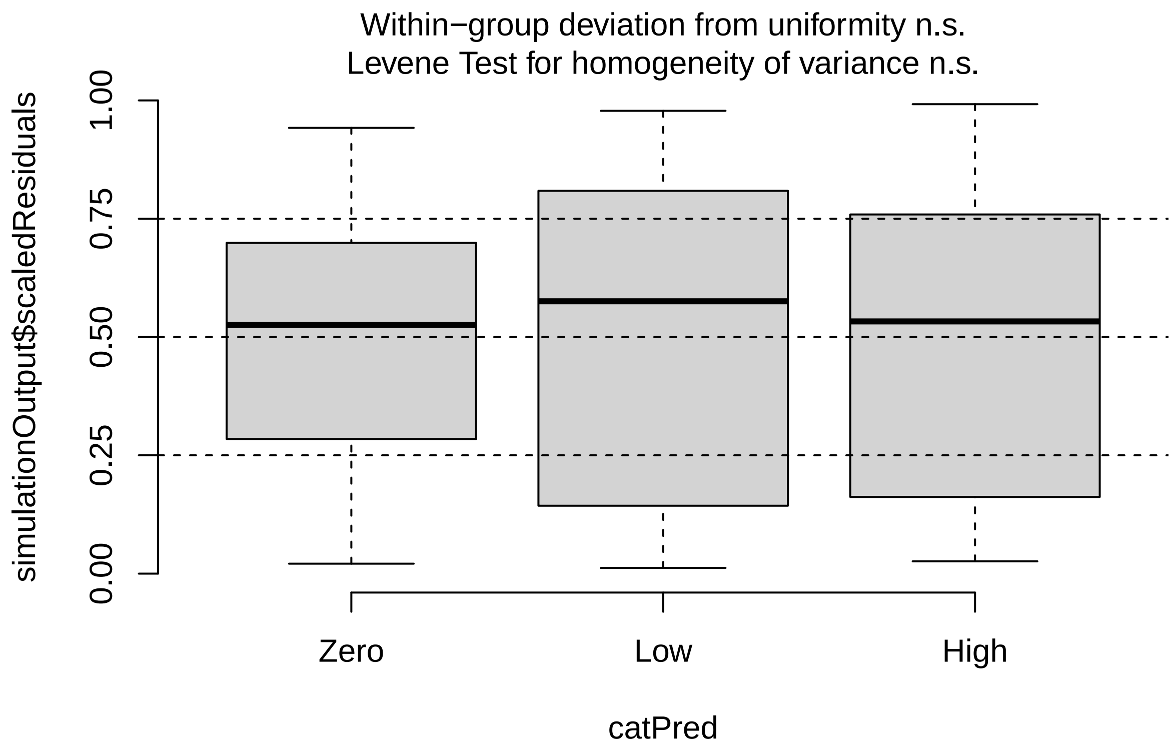

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.95278, p-value = 0.746
## alternative hypothesis: two.sided
```

```r
testOutliers(lym_pre_sim)
```

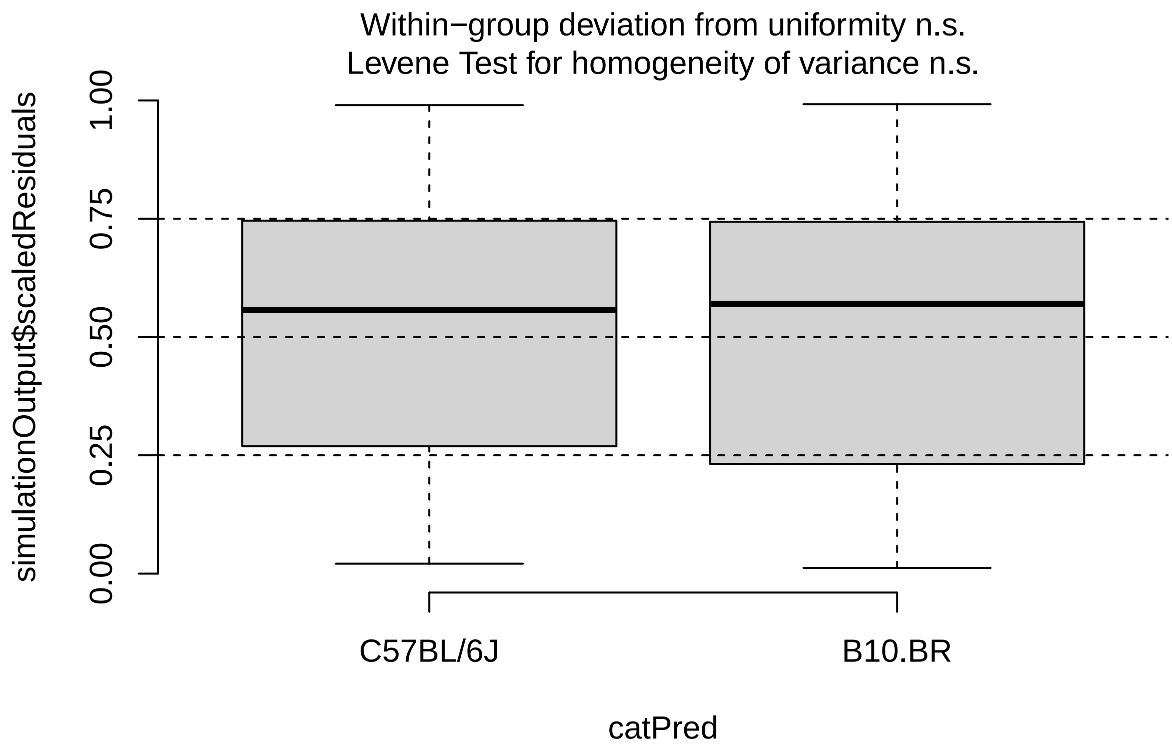



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  lym_pre_sim
## outliers at both margin(s) = 0, observations = 108, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.00000000 0.03357955
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                    0
```

```
plotResiduals(lym_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$t_muris_dose)
```




```r
plotResiduals(lym_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$strain)
```

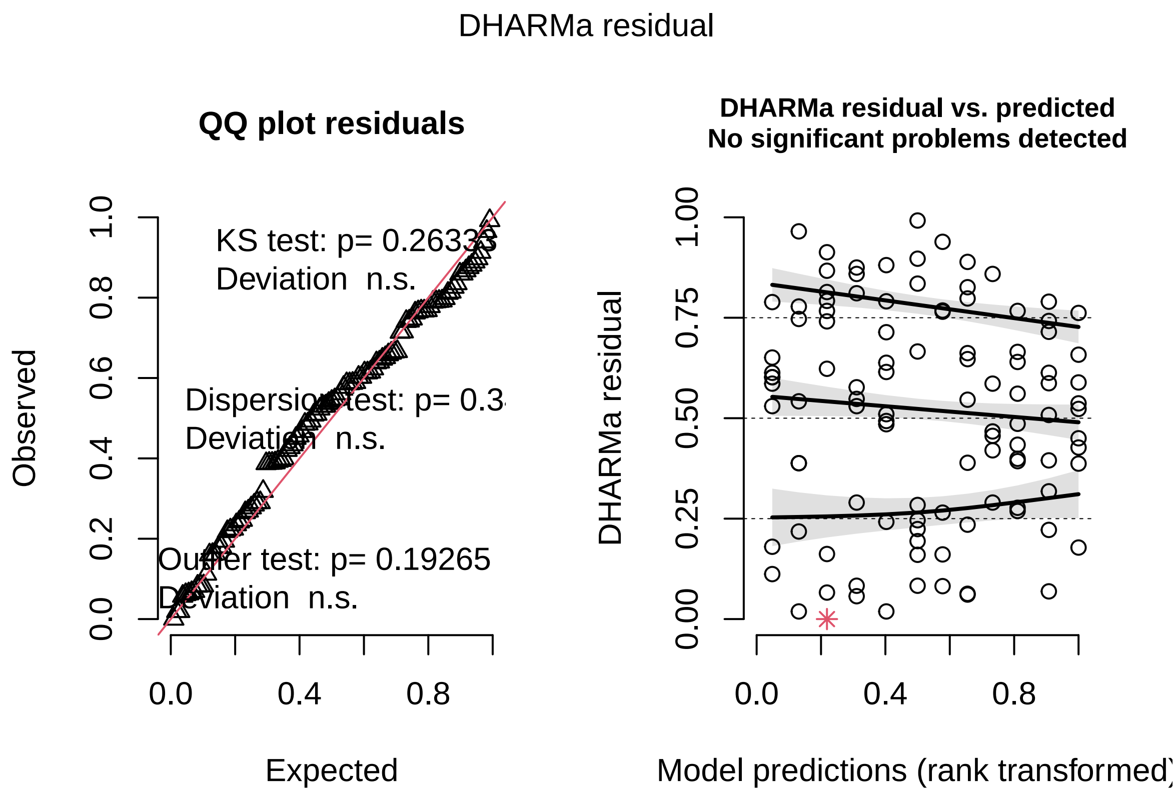



```r
# During RW: environment x dose x strain
lym_during <- lmer(log(lym_number) ~ lab_or_rw * t_muris_dose * strain +
                     (1 | cage_wedge) + (1 | cohort),
                   data = filter(cleaned_combined_cbc_data, phase == "During RW"))

print(summary(lym_during), correlation = FALSE)
```

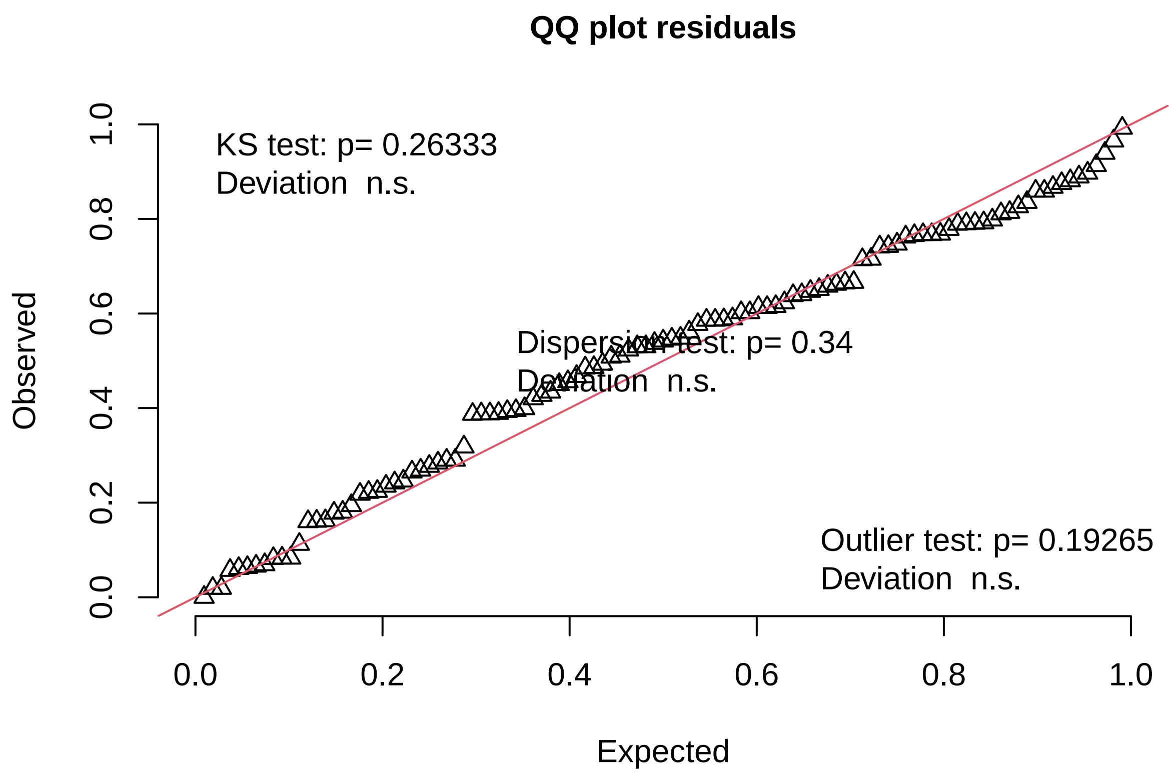
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(lym_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "During RW")
##
## REML criterion at convergence: 94.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5508 -0.4082  0.0517  0.5497  2.8406
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.01865  0.1366
##  cohort     (Intercept) 0.01190  0.1091
##  Residual               0.10773  0.3282
## Number of obs: 107, groups:  cage_wedge, 29; cohort, 2
```

```
##
## Fixed effects:
##                                          Estimate Std. Error       df t value
## (Intercept)                               1.88516    0.13736  5.04939  13.724
## lab_or_rwRW                              -1.09542    0.18336 41.38778  -5.974
## t_muris_doseLow                          -0.12543    0.15194 92.40962  -0.826
## t_muris_doseHigh                         -0.13912    0.17939 93.88714  -0.776
## strainB10.BR                              0.10834    0.16375 84.98760   0.662
## lab_or_rwRW:t_muris_doseLow               0.13628    0.21447 88.46036   0.635
## lab_or_rwRW:t_muris_doseHigh              0.29005    0.23440 92.28280   1.237
## lab_or_rwRW:strainB10.BR                 -0.33349    0.26166 50.87108  -1.275
## t_muris_doseLow:strainB10.BR              0.05956    0.21586 90.31386   0.276
## t_muris_doseHigh:strainB10.BR             0.03007    0.25412 93.96701   0.118
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  -0.07612    0.30812 86.46508  -0.247
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -0.06660    0.33404 91.05119  -0.199
##                                          Pr(>|t|)
## (Intercept)                              3.43e-05 ***
## lab_or_rwRW                              4.56e-07 ***
## t_muris_doseLow                             0.411
## t_muris_doseHigh                            0.440
## strainB10.BR                                0.510
## lab_or_rwRW:t_muris_doseLow                 0.527
## lab_or_rwRW:t_muris_doseHigh                0.219
## lab_or_rwRW:strainB10.BR                    0.208
## t_muris_doseLow:strainB10.BR                0.783
## t_muris_doseHigh:strainB10.BR               0.906
## lab_or_rwRW:t_muris_doseLow:strainB10.BR    0.805
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR   0.842
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
lym_during_sim <- simulateResiduals(fittedModel = lym_during, n = 1000)
plot(lym_during_sim)
```

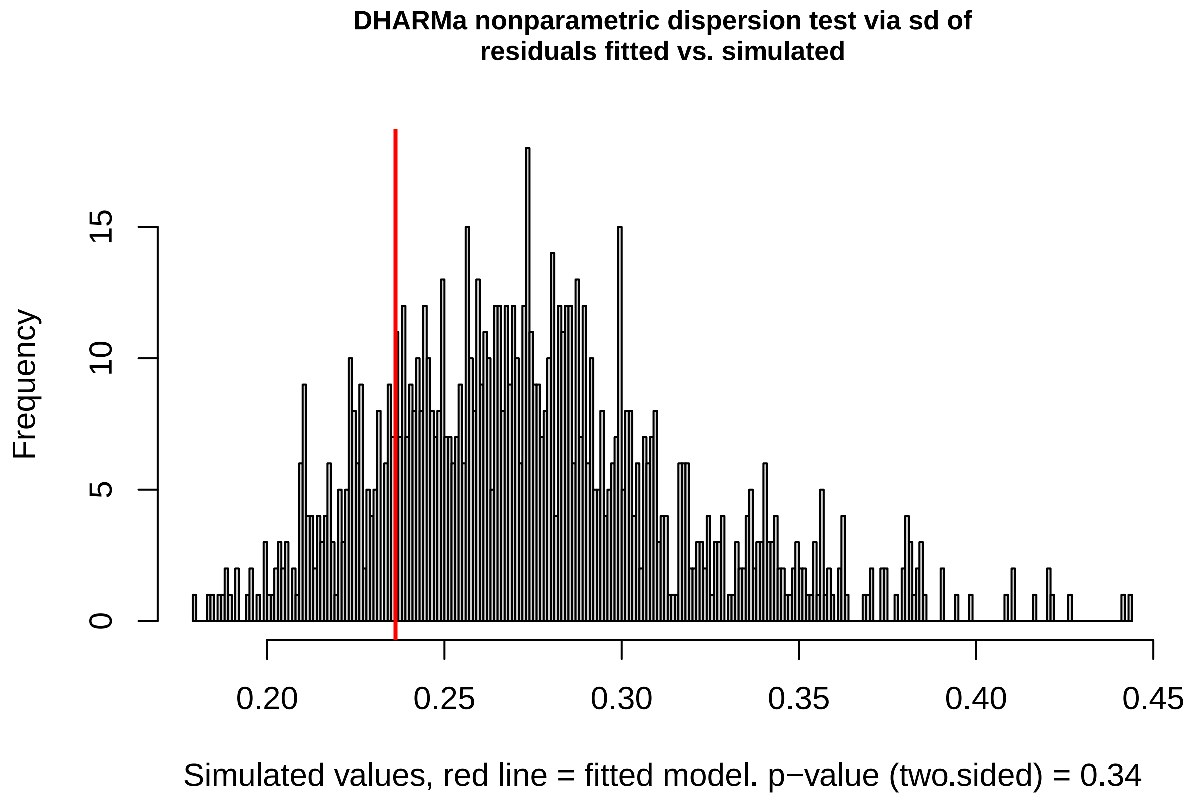



```
testUniformity(lym_during_sim)
```



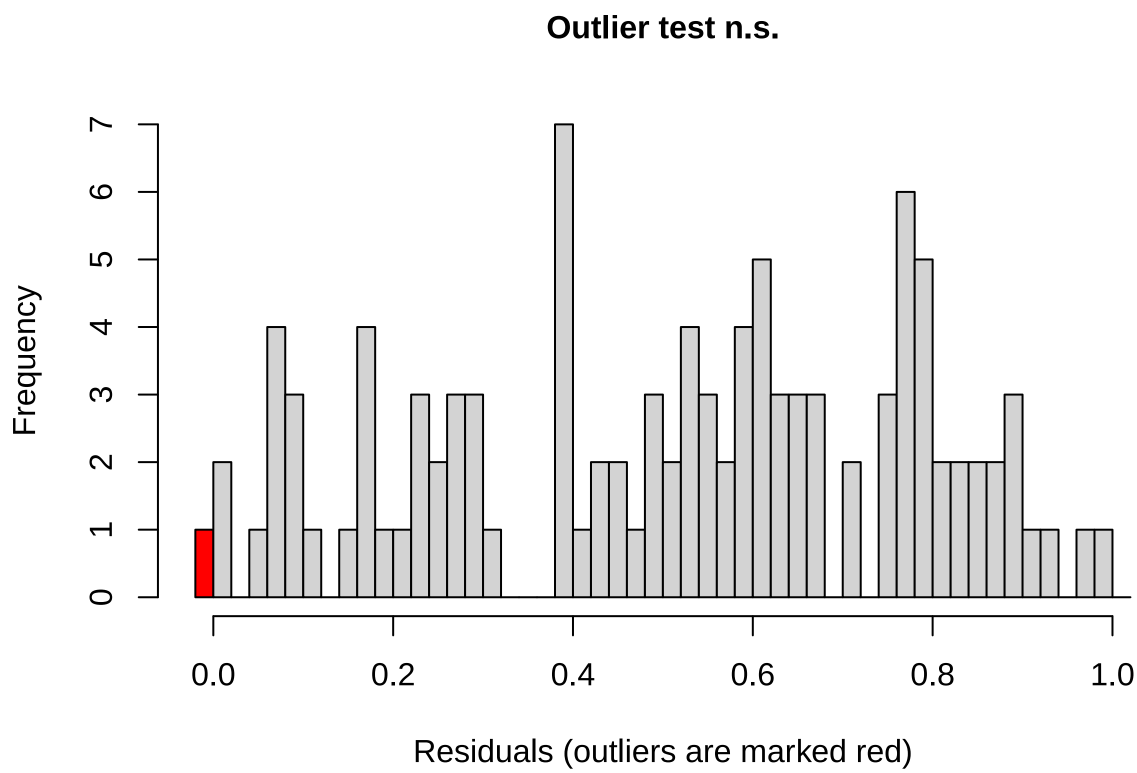

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.09728, p-value = 0.2633
## alternative hypothesis: two-sided
```

```
testDispersion(lym_during_sim)
```



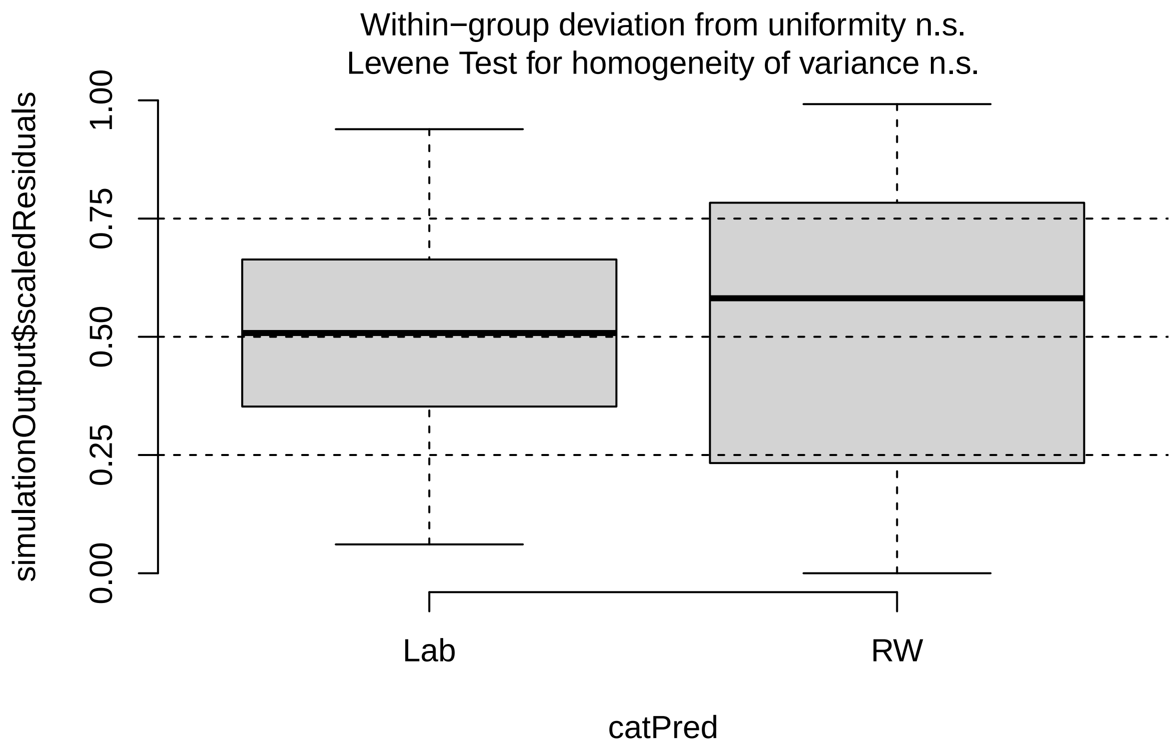

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.85722, p-value = 0.34
## alternative hypothesis: two.sided
```

```r
testOutliers(lym_during_sim)
```

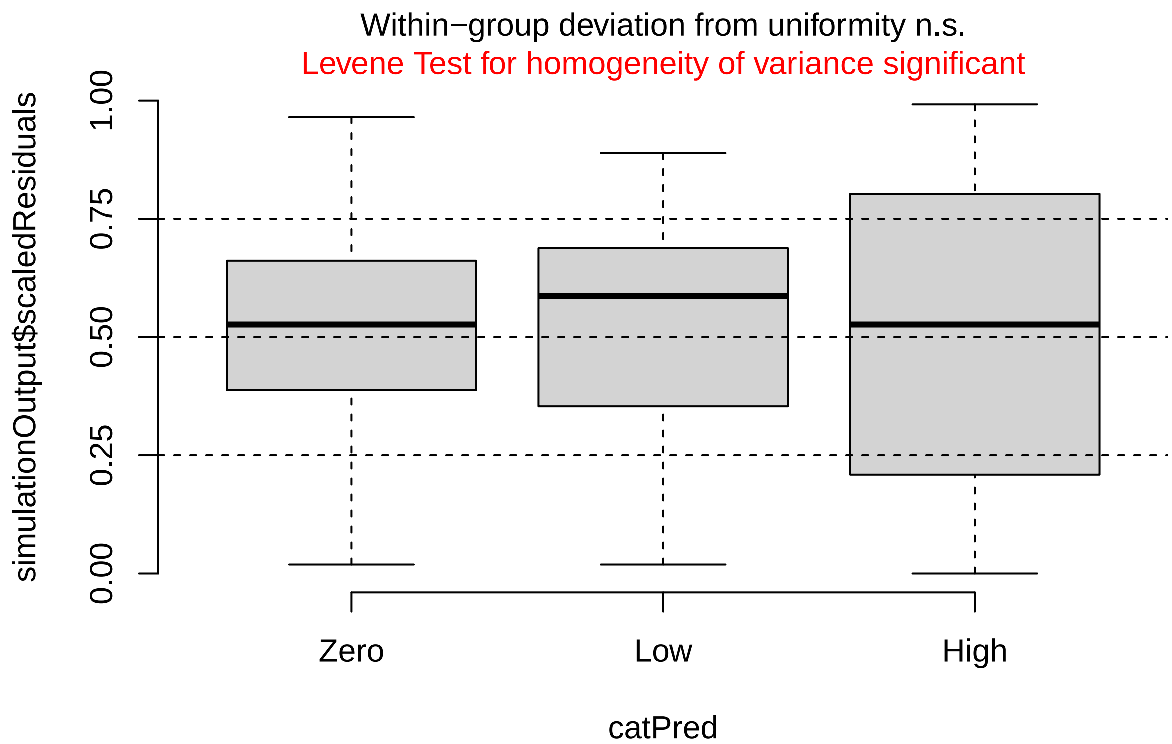



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  lym_during_sim
## outliers at both margin(s) = 1, observations = 107, p-value = 0.1927
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.000236587 0.050972543
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                         0.009345794
```

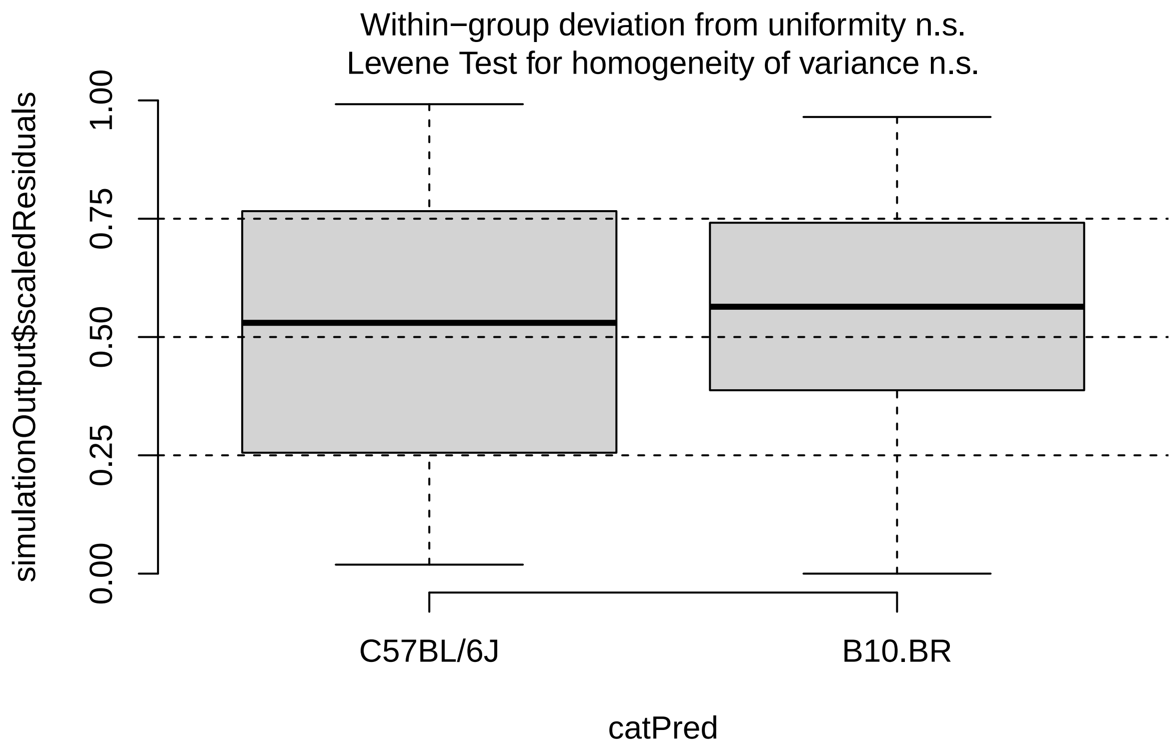
```
plotResiduals(lym_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$lab_or_rw)
```




```
plotResiduals(lym_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$t_muris_dos
```




```
plotResiduals(lym_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$strain)
```




```r
# Post-RW: environment x infection x strain
lym_post <- lmer(log(lym_number) ~ lab_or_rw * t_muris_dose * strain +
                   (1 | cage_wedge) + (1 | cohort),
                 data = filter(cleaned_combined_cbc_data, phase == "Post-RW"))
```

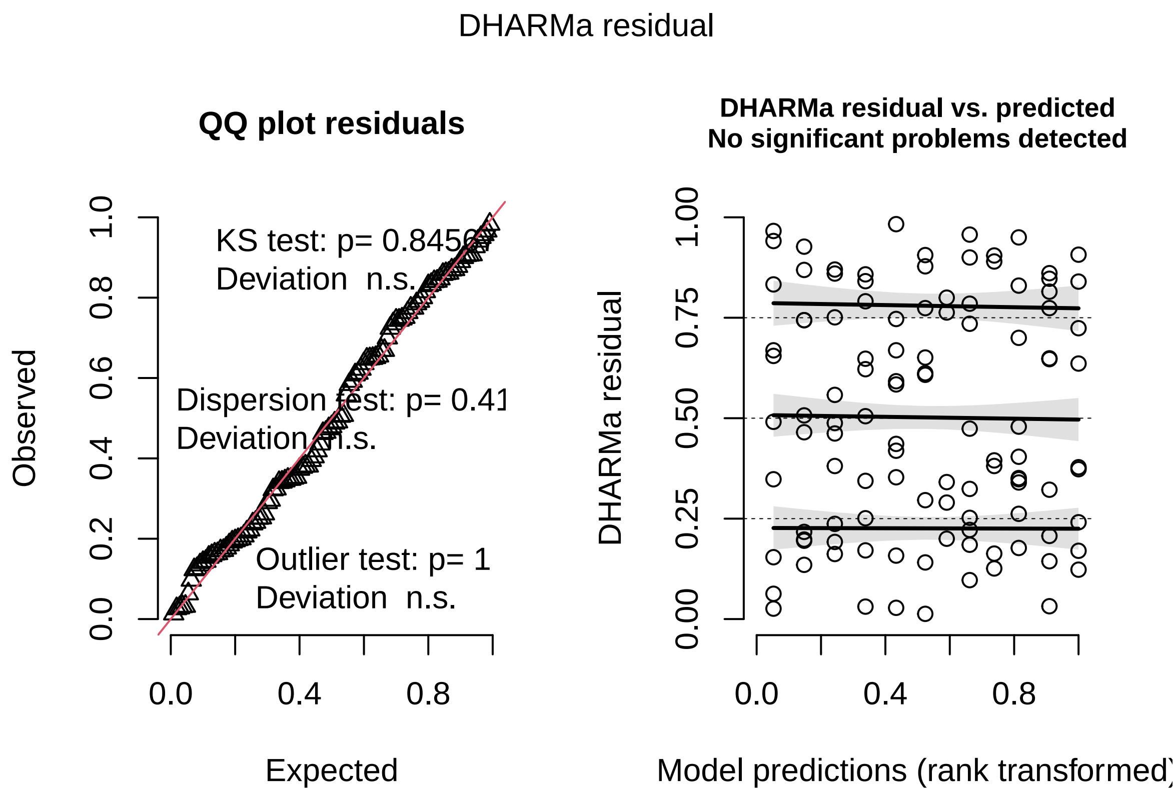
```
## boundary (singular) fit: see help('isSingular')
```

```r
print(summary(lym_post), correlation = FALSE)
```

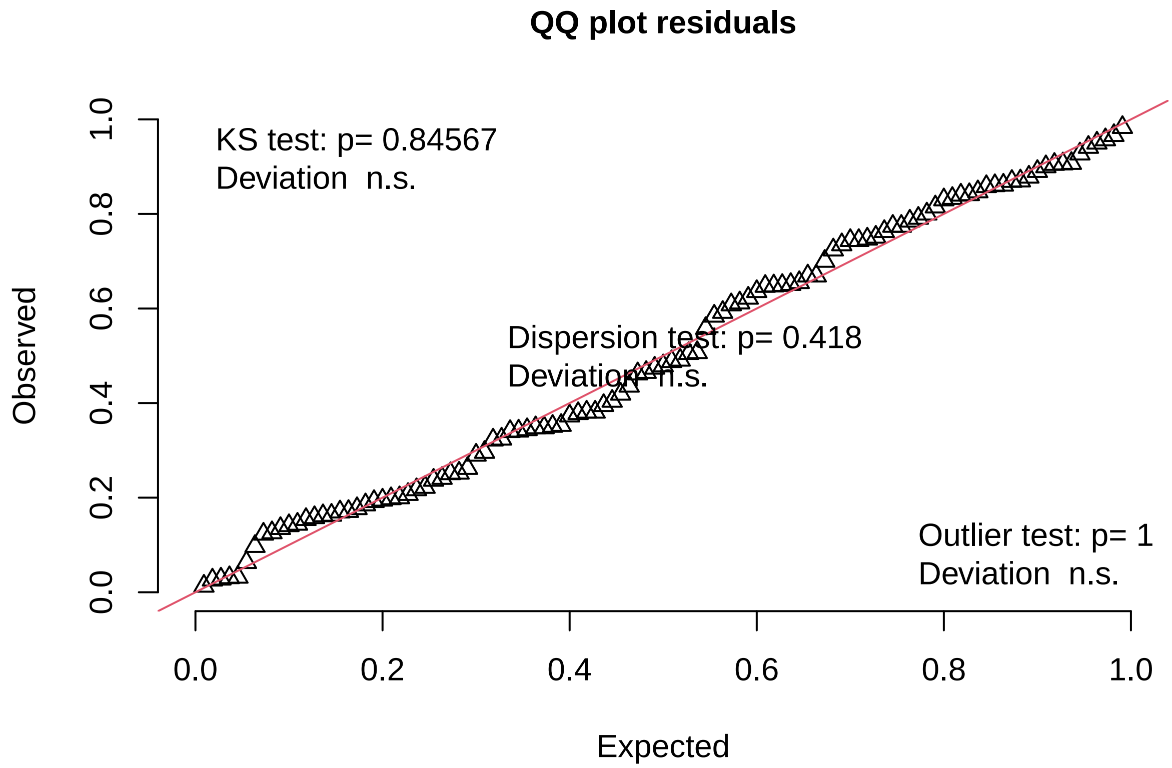
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(lym_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Post-RW")
##
## REML criterion at convergence: 75.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.10991 -0.66180 -0.09366  0.78983  2.16476
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.01628  0.1276
```

```
##  cohort     (Intercept) 0.00000  0.0000
##  Residual               0.08859  0.2976
## Number of obs: 109, groups:  cage_wedge, 29; cohort, 2
##
## Fixed effects:
##                                            Estimate Std. Error       df t value
## (Intercept)                                 2.11679    0.10337 75.50528  20.478
## lab_or_rwRW                                -1.64206    0.16940 25.40144  -9.693
## t_muris_doseLow                            -0.33887    0.13890 96.62874  -2.440
## t_muris_doseHigh                           -0.20524    0.16303 96.04183  -1.259
## strainB10.BR                                0.03091    0.14914 79.40367   0.207
## lab_or_rwRW:t_muris_doseLow                 0.02023    0.19500 90.66304   0.104
## lab_or_rwRW:t_muris_doseHigh               -0.02129    0.21286 95.21608  -0.100
## lab_or_rwRW:strainB10.BR                   -0.07910    0.23928 25.50464  -0.331
## t_muris_doseLow:strainB10.BR                0.12015    0.19668 93.81390   0.611
## t_muris_doseHigh:strainB10.BR              -0.18886    0.23891 96.71507  -0.790
## lab_or_rwRW:t_muris_doseLow:strainB10.BR    0.09615    0.27411 87.74001   0.351
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR   0.24435    0.30583 94.82007   0.799
##                                            Pr(>|t|)
## (Intercept)                                 < 2e-16 ***
## lab_or_rwRW                                5.11e-10 ***
## t_muris_doseLow                              0.0165 *
## t_muris_doseHigh                             0.2111
## strainB10.BR                                 0.8363
## lab_or_rwRW:t_muris_doseLow                  0.9176
## lab_or_rwRW:t_muris_doseHigh                 0.9205
## lab_or_rwRW:strainB10.BR                     0.7437
## t_muris_doseLow:strainB10.BR                 0.5427
## t_muris_doseHigh:strainB10.BR                0.4312
## lab_or_rwRW:t_muris_doseLow:strainB10.BR     0.7266
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR    0.4263
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
lym_post_sim <- simulateResiduals(fittedModel = lym_post, n = 1000)
plot(lym_post_sim)
```

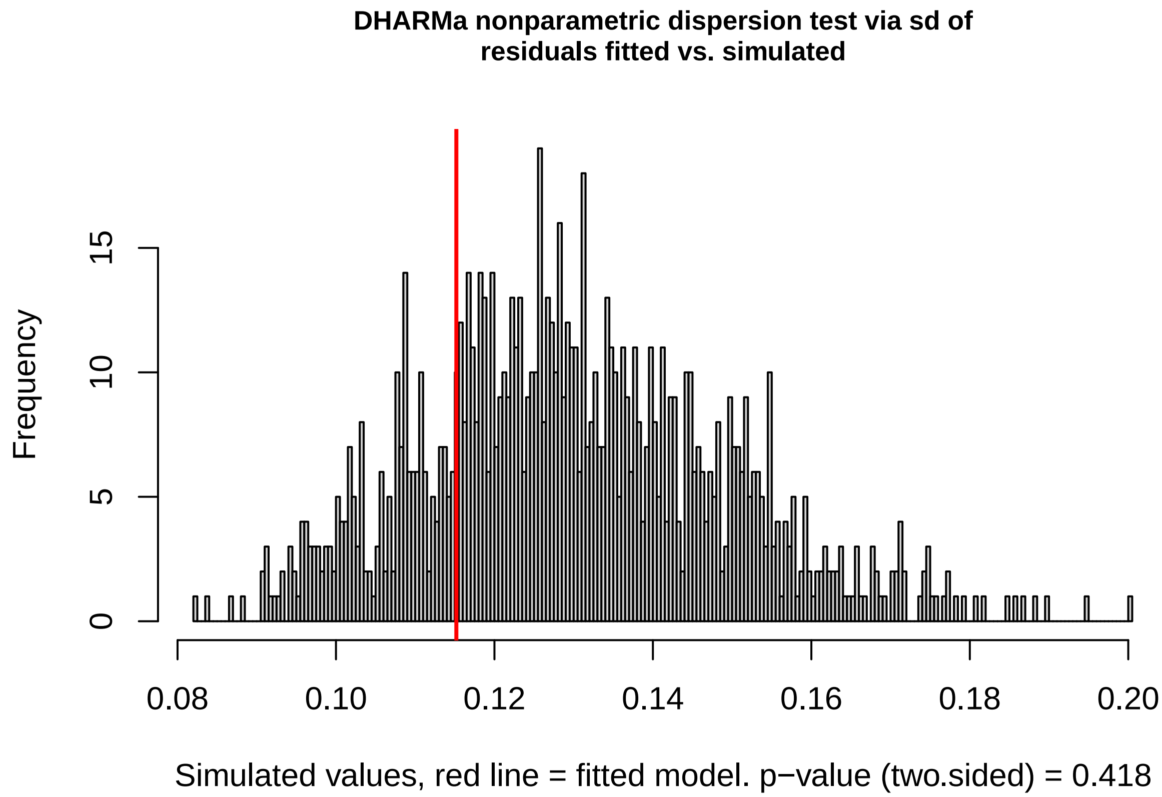



```
testUniformity(lym_post_sim)
```



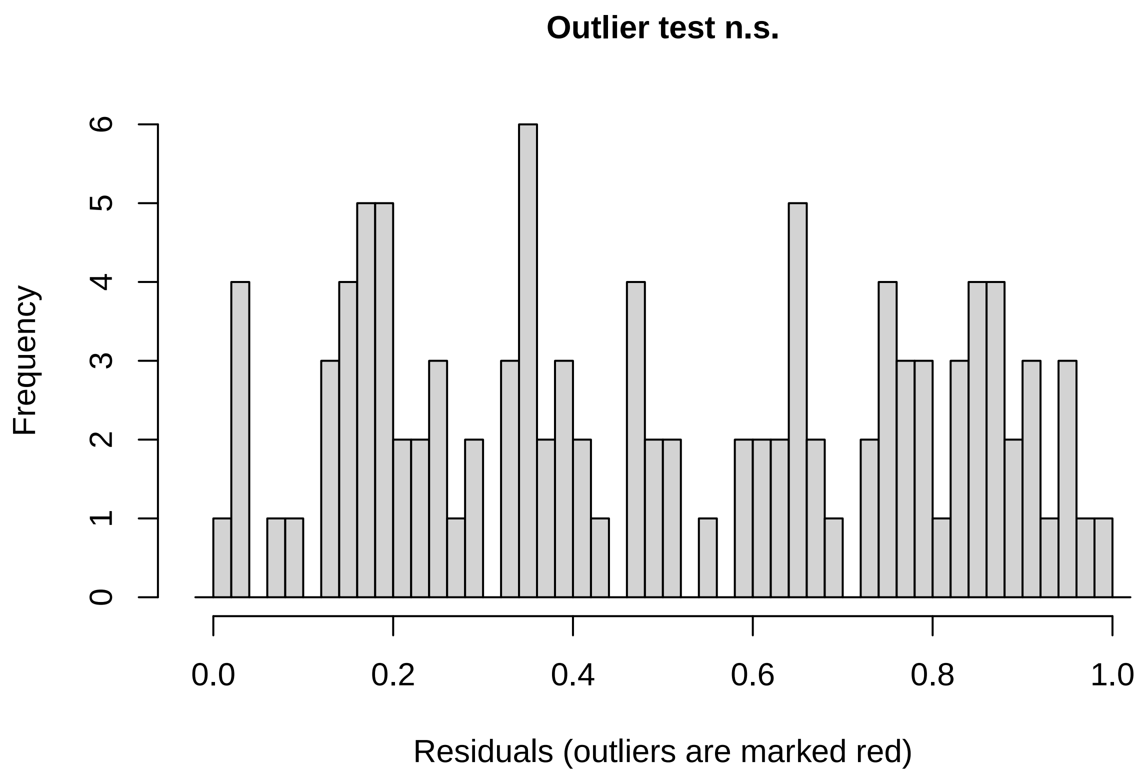

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.05878, p-value = 0.8457
## alternative hypothesis: two-sided
```

```
testDispersion(lym_post_sim)
```



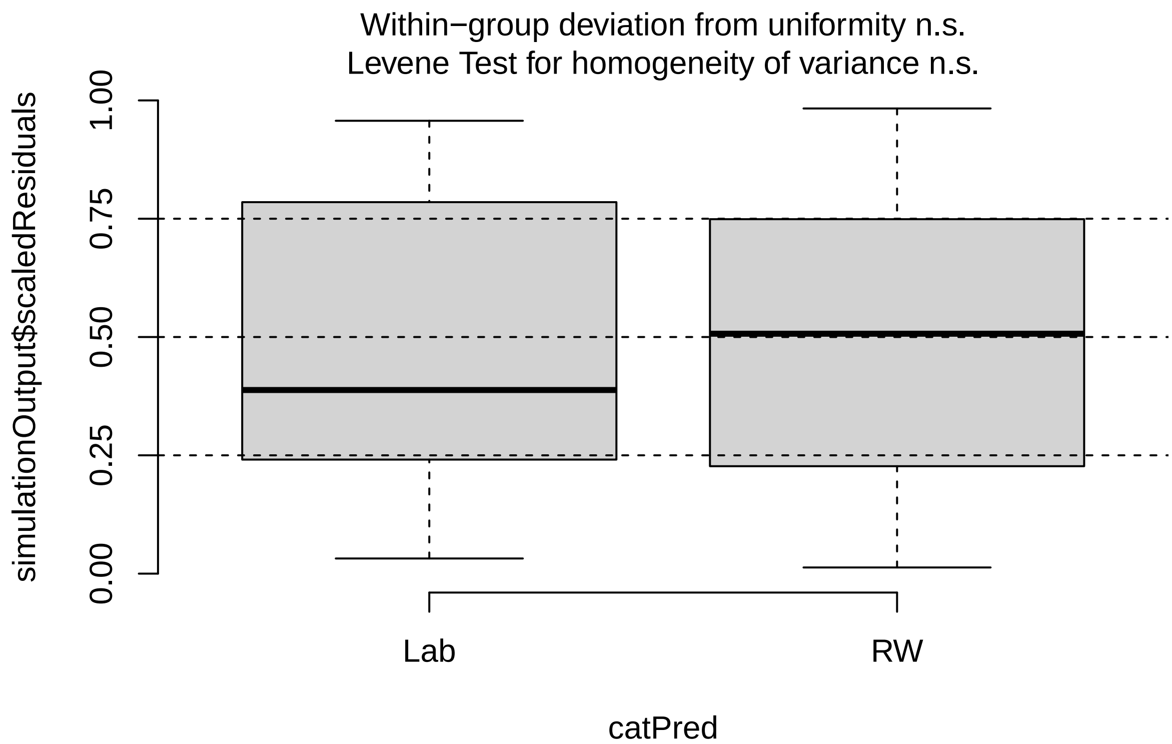

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.88562, p-value = 0.418
## alternative hypothesis: two.sided
```

```
testOutliers(lym_post_sim)
```

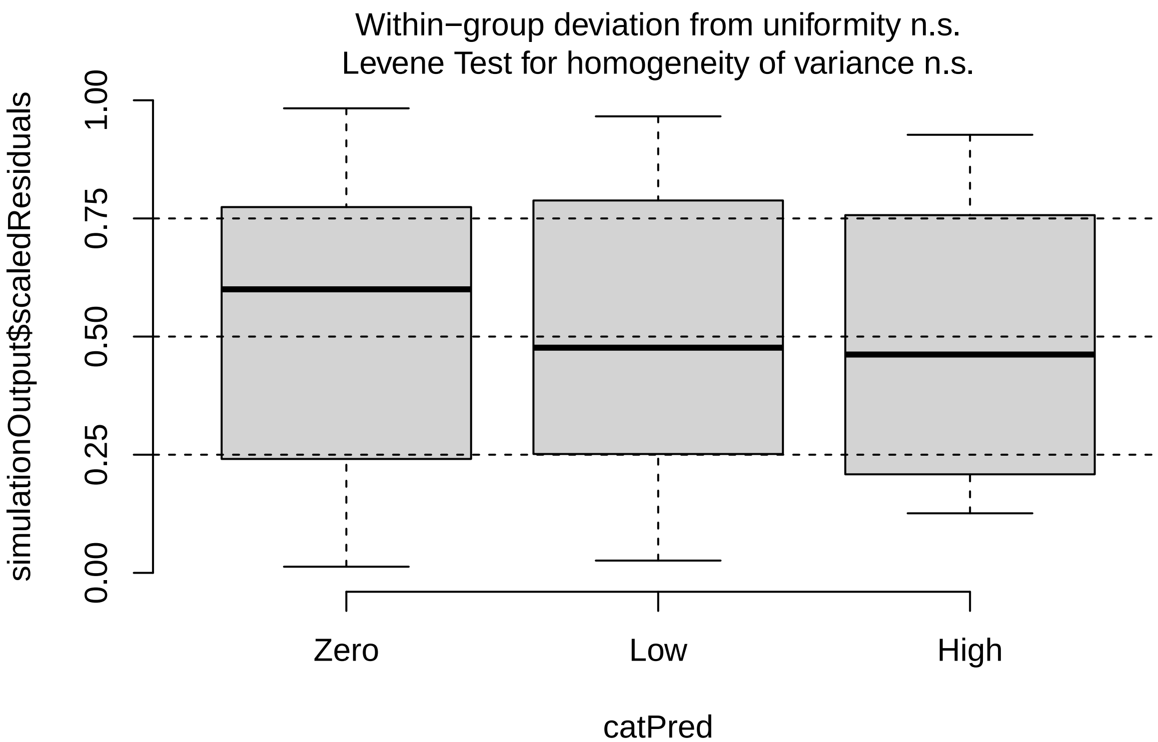



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  lym_post_sim
## outliers at both margin(s) = 0, observations = 109, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.00000000 0.03327666
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                     0
```

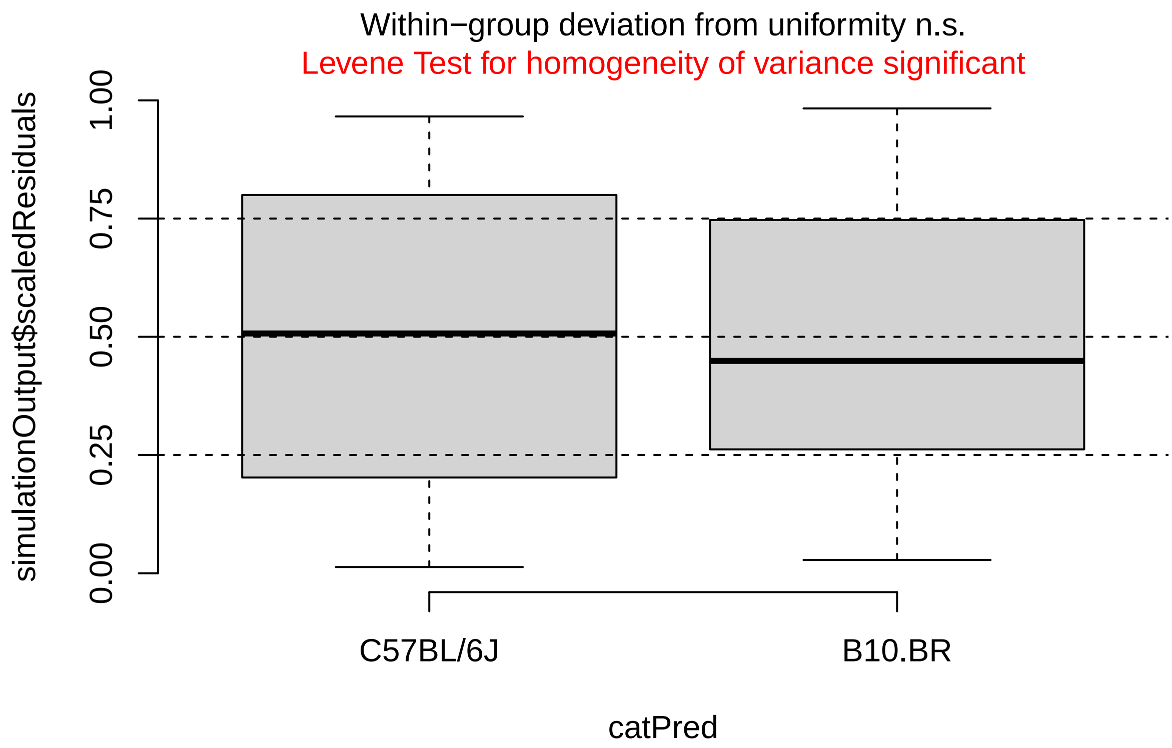
```
plotResiduals(lym_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$lab_or_rw)
```




```r
plotResiduals(lym_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$t_muris_dose)
```

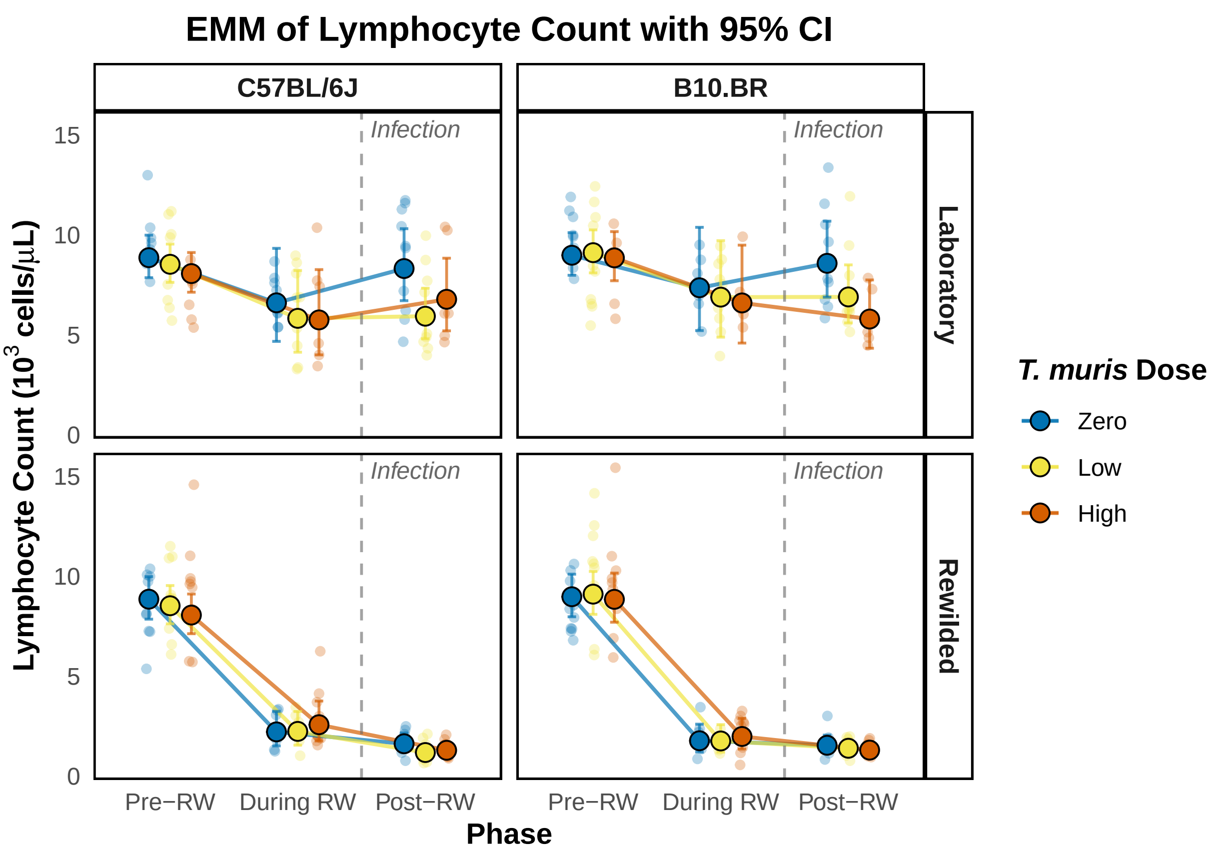



```
plotResiduals(lym_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$strain)
```




```
emm_lym_pre <- as.data.frame(emmeans(lym_pre,
                                     specs = "t_muris_dose",
                                     by = "strain",
                                     type = "response")) %>%
  crossing(lab_or_rw = factor(c("Lab", "RW"),
                              levels = c("Lab", "RW"))) %>%
  mutate(phase = factor("Pre-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_lym_during <- as.data.frame(emmeans(lym_during,
                                        specs = c("t_muris_dose", "lab_or_rw"),
                                        by = "strain",
                                        type = "response")) %>%
  mutate(phase = factor("During RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_lym_post <- as.data.frame(emmeans(lym_post,
                                      specs = c("t_muris_dose", "lab_or_rw"),
                                      by = "strain",
                                      type = "response")) %>%
  mutate(phase = factor("Post-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_lym_combined <- bind_rows(emm_lym_pre, emm_lym_during, emm_lym_post) %>%
  mutate(phase = factor(phase, levels = c("Pre-RW", "During RW", "Post-RW")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "Low", "High")),
         strain = factor(strain, levels = c("C57BL/6J", "B10.BR")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

```r
emm_lym_plot <- emm_plotter(
  emm_df = emm_lym_combined,
  cbc_input = cleaned_combined_cbc_data,
  y_col = "lym_number",
  y_label = expression(bold("Lymphocyte Count (10"^3~"cells/"*mu*"L)")),
  title = "EMM of Lymphocyte Count with 95% CI"
)

emm_lym_plot
```




```r
plot_saver("emm_lym_number.png", emm_lym_plot)
```

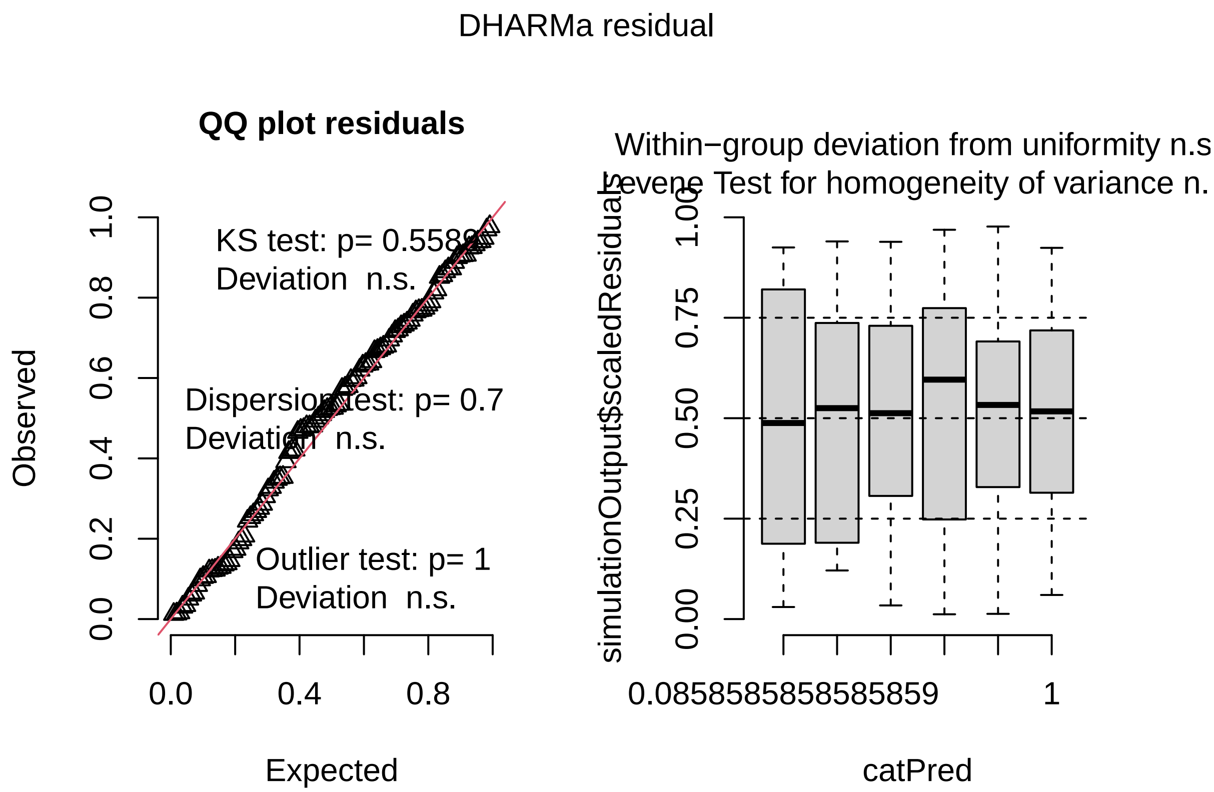
Monocyte

```r
# Pre-RW: dose x strain (no environment - all mice are lab-housed)
mon_pre <- lmer(log(mon_number) ~ t_muris_dose * strain +
                  (1 | cage_wedge) + (1 | cohort),
                data = filter(cleaned_combined_cbc_data, phase == "Pre-RW"))

print(summary(mon_pre), correlation = FALSE)
```

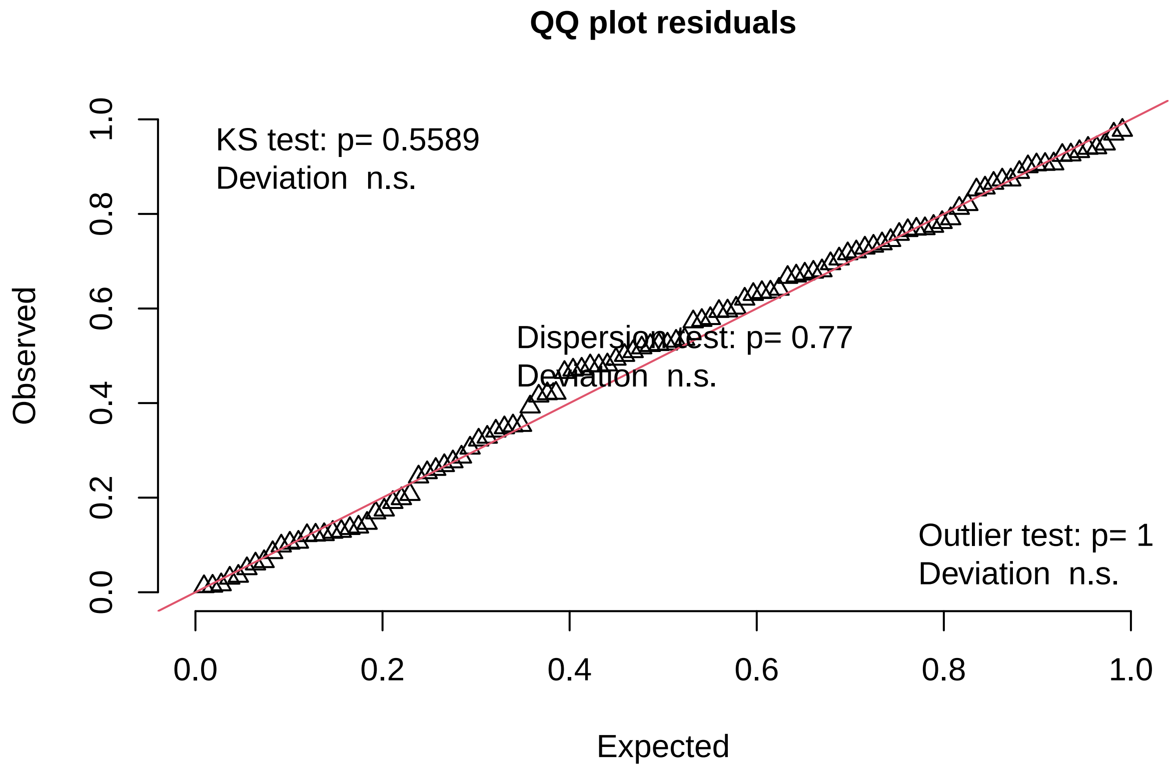
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
## Formula: log(mon_number) ~ t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Pre-RW")
##
## REML criterion at convergence: 137.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.33513 -0.62461  0.00306  0.68369  2.09113
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.004136 0.06431
##  cohort     (Intercept) 0.005562 0.07458
##  Residual               0.184593 0.42964
## Number of obs: 108, groups:  cage_wedge, 21; cohort, 2
##
## Fixed effects:
##                             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                 -1.37235    0.11843  8.37235 -11.588 1.92e-06 ***
## t_muris_doseLow              0.02997    0.14199 88.69599   0.211   0.8333
## t_muris_doseHigh            -0.16776    0.15008 91.55945  -1.118   0.2666
## strainB10.BR                 0.25696    0.14676 65.88316   1.751   0.0846 .
## t_muris_doseLow:strainB10.BR  -0.18139    0.19786 87.87145  -0.917   0.3618
## t_muris_doseHigh:strainB10.BR -0.02456    0.21005 97.09195  -0.117   0.9072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mon_pre_sim <- simulateResiduals(fittedModel = mon_pre, n = 1000)
plot(mon_pre_sim)
```

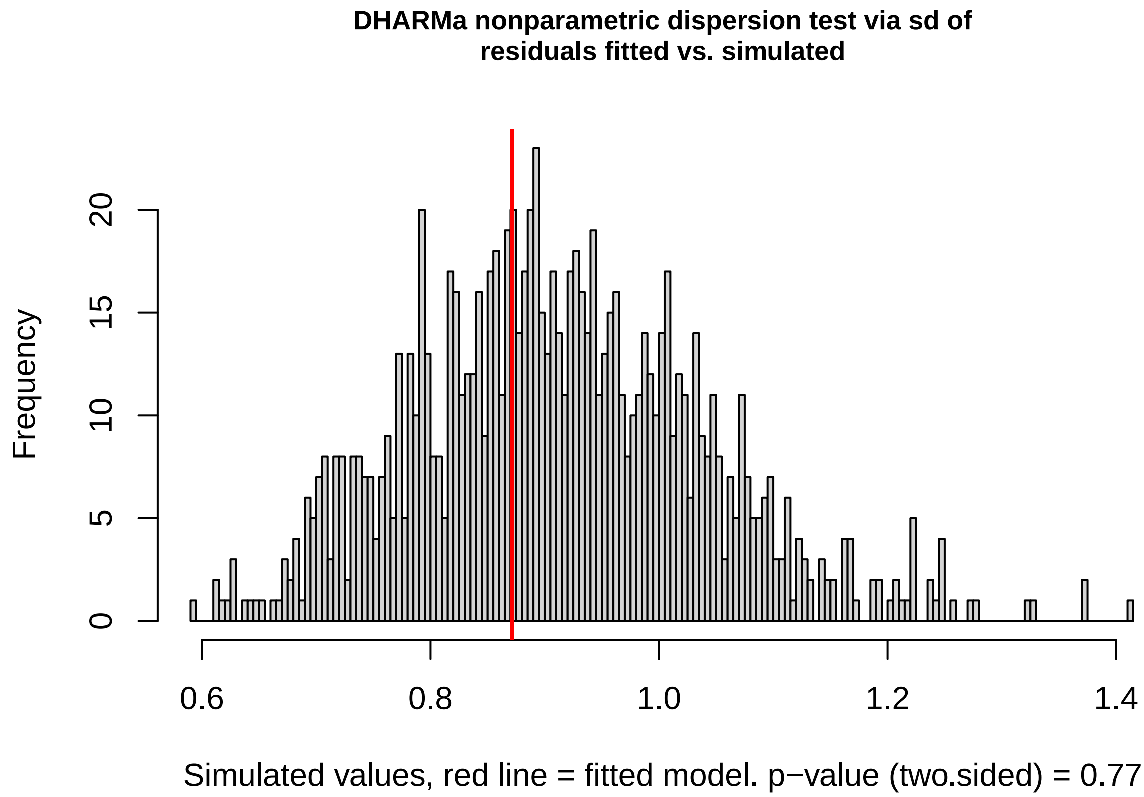



```r
testUniformity(mon_pre_sim)
```



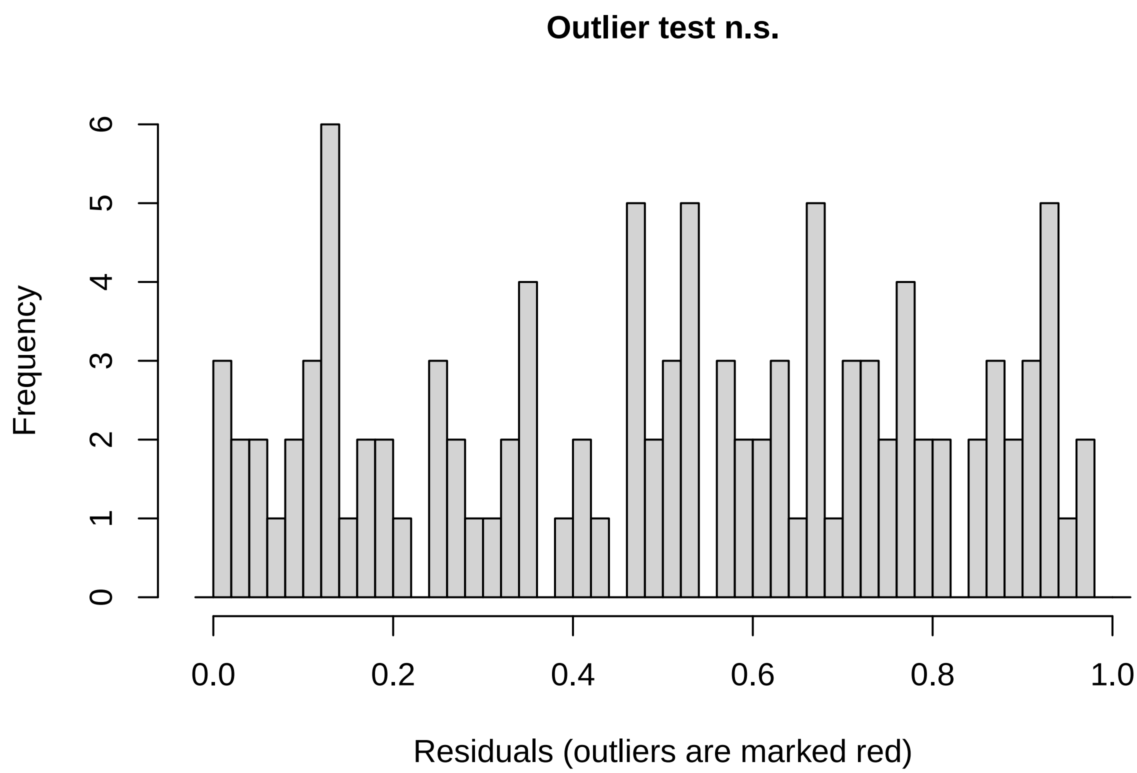

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.076111, p-value = 0.5589
## alternative hypothesis: two-sided
```

```
testDispersion(mon_pre_sim)
```



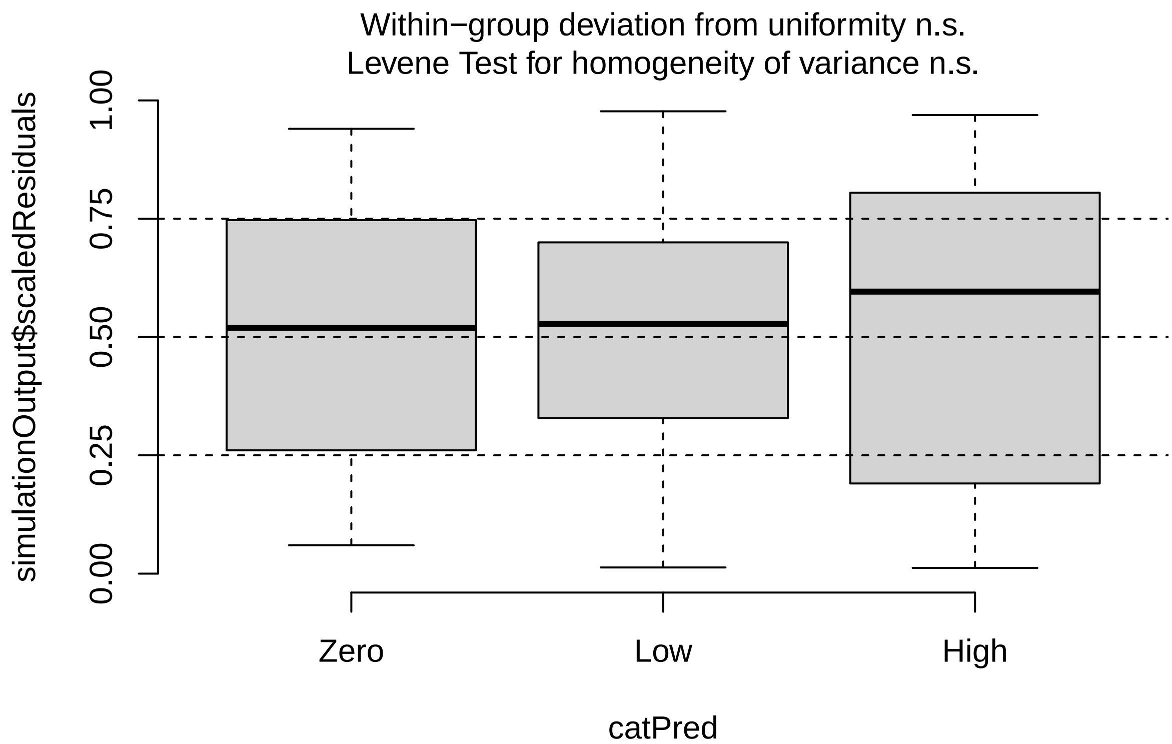

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.9554, p-value = 0.77
## alternative hypothesis: two.sided
```

```
testOutliers(mon_pre_sim)
```

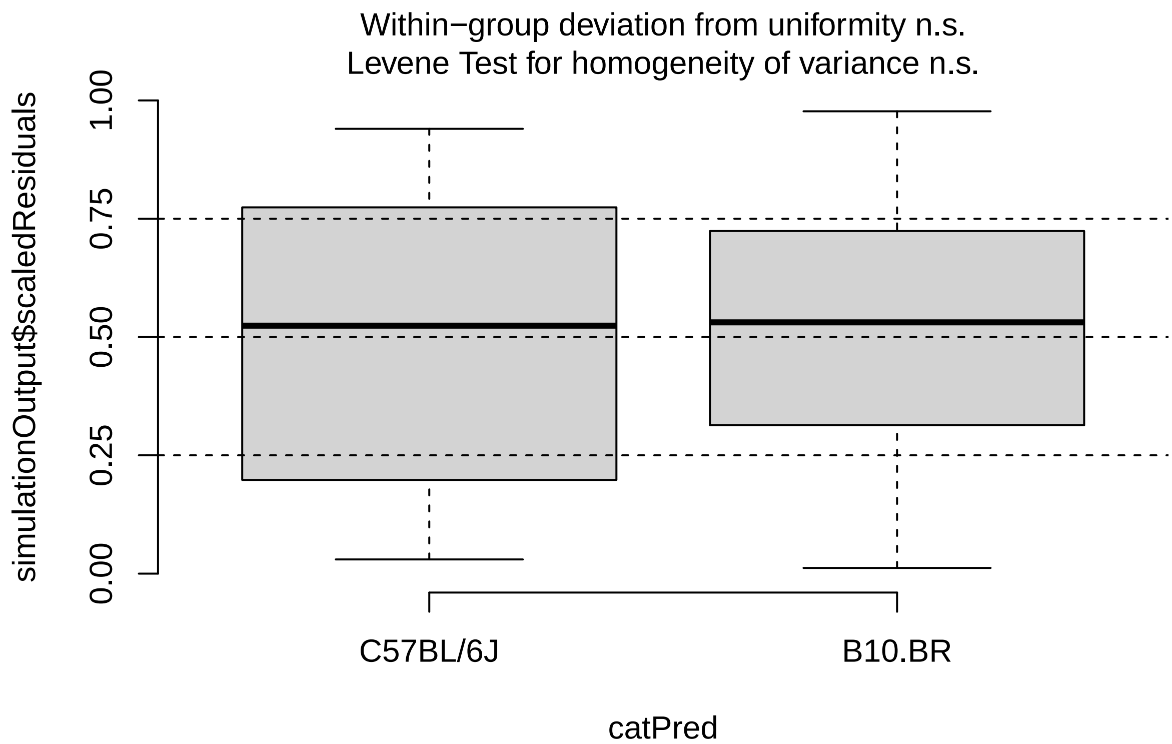



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  mon_pre_sim
## outliers at both margin(s) = 0, observations = 108, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.00000000 0.03357955
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                     0
```

```
plotResiduals(mon_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$t_muris_dose)
```




```r
plotResiduals(mon_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$strain)
```

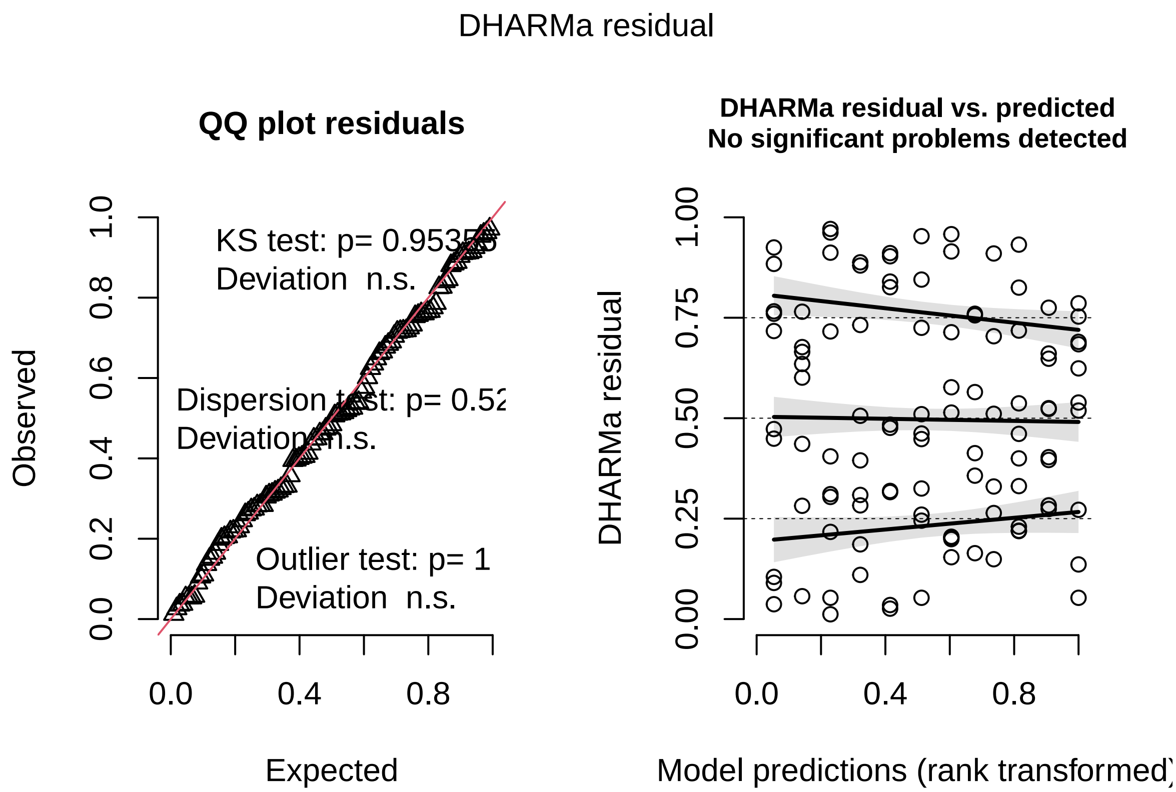



```r
# During RW: environment x dose x strain
mon_during <- lmer(log(mon_number) ~ lab_or_rw * t_muris_dose * strain +
                     (1 | cage_wedge) + (1 | cohort),
                   data = filter(cleaned_combined_cbc_data, phase == "During RW"))

print(summary(mon_during), correlation = FALSE)
```

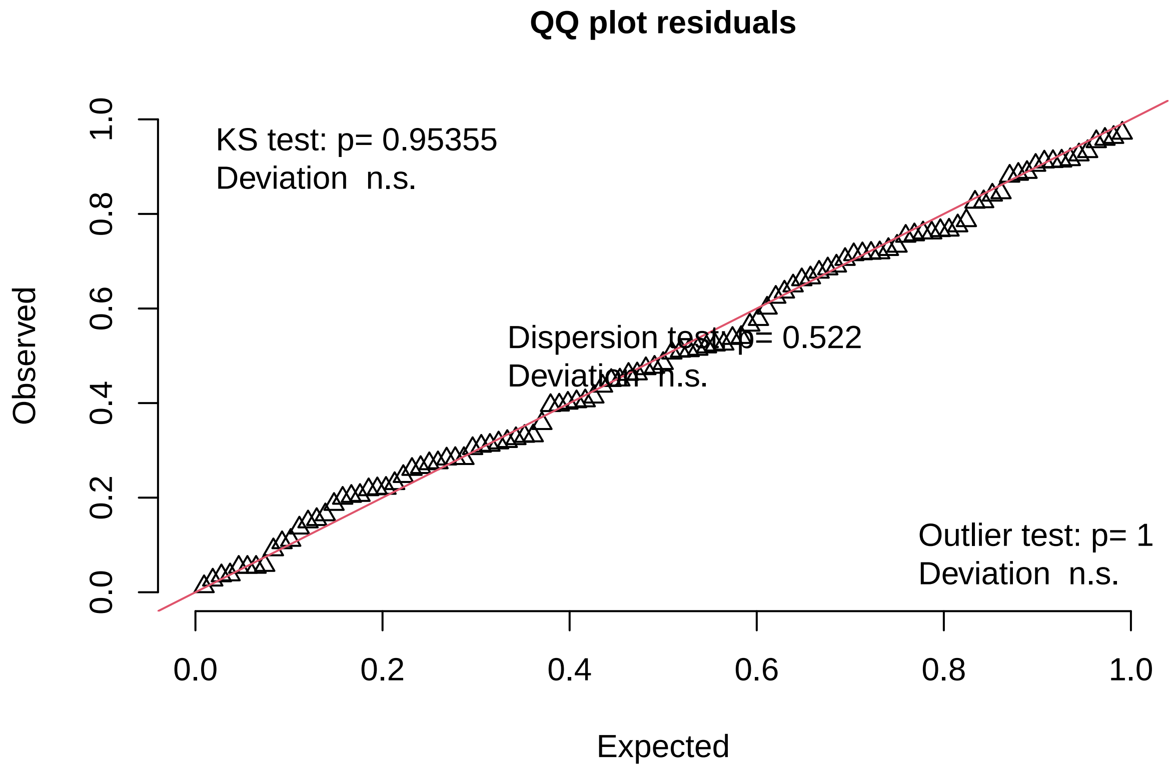
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(mon_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "During RW")
##
## REML criterion at convergence: 185.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.21372 -0.54257 -0.07154  0.71547  2.43821
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.008879 0.09423
##  cohort     (Intercept) 0.050111 0.22385
##  Residual               0.301062 0.54869
## Number of obs: 107, groups:  cage_wedge, 29; cohort, 2
```

```
##
## Fixed effects:
##                                          Estimate Std. Error      df t value
## (Intercept)                               -1.9854     0.2373  3.8499  -8.365
## lab_or_rwRW                               -0.1810     0.2628 52.3163  -0.689
## t_muris_doseLow                           -0.2764     0.2473 93.5599  -1.117
## t_muris_doseHigh                          -0.0709     0.2868 93.7151  -0.247
## strainB10.BR                               0.0970     0.2561 90.9952   0.379
## lab_or_rwRW:t_muris_doseLow               -0.3163     0.3532 91.6388  -0.895
## lab_or_rwRW:t_muris_doseHigh              -0.2929     0.3819 93.3763  -0.767
## lab_or_rwRW:strainB10.BR                  -0.5060     0.3813 57.4842  -1.327
## t_muris_doseLow:strainB10.BR               0.2837     0.3537 92.6203   0.802
## t_muris_doseHigh:strainB10.BR             -0.1143     0.4089 93.9926  -0.279
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   0.3640     0.5098 90.5465   0.714
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.6605     0.5464 92.8233   1.209
##                                          Pr(>|t|)
## (Intercept)                               0.00132 **
## lab_or_rwRW                               0.49410
## t_muris_doseLow                           0.26670
## t_muris_doseHigh                          0.80533
## strainB10.BR                              0.70581
## lab_or_rwRW:t_muris_doseLow               0.37294
## lab_or_rwRW:t_muris_doseHigh              0.44502
## lab_or_rwRW:strainB10.BR                  0.18971
## t_muris_doseLow:strainB10.BR              0.42462
## t_muris_doseHigh:strainB10.BR             0.78058
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  0.47706
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR 0.22981
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mon_during_sim <- simulateResiduals(fittedModel = mon_during, n = 1000)
plot(mon_during_sim)
```

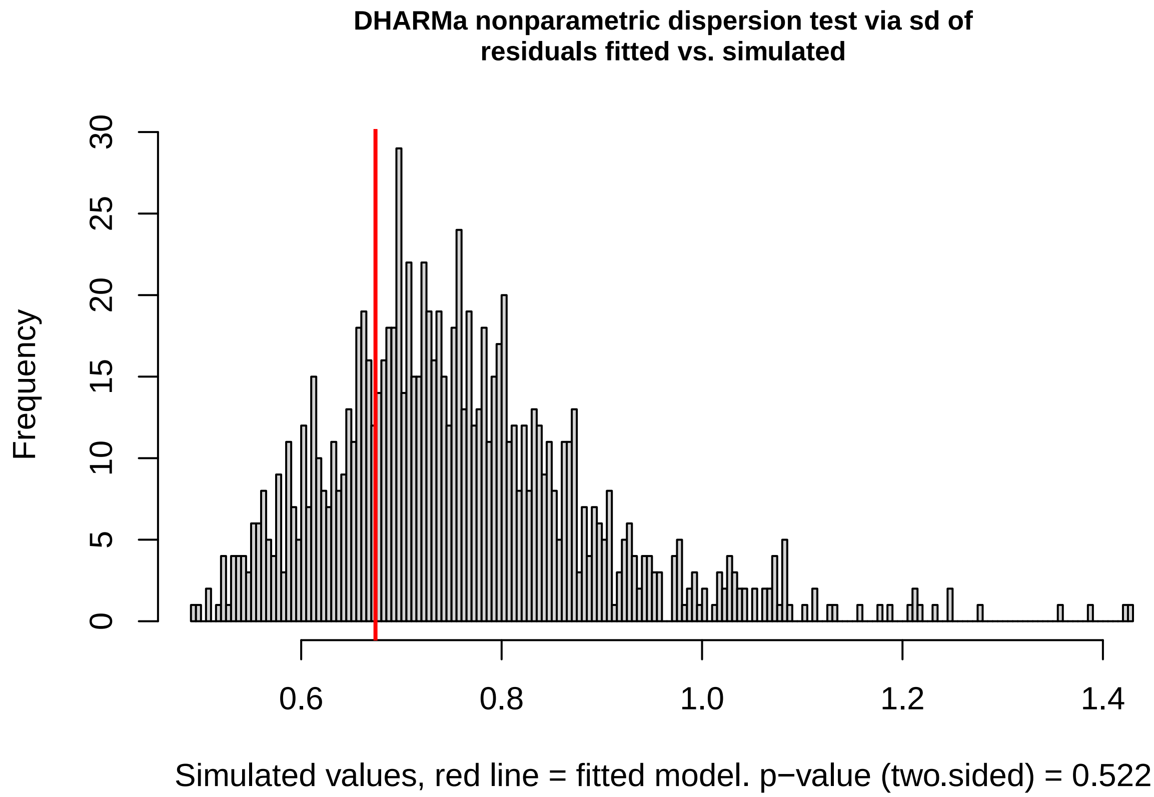



```
testUniformity(mon_during_sim)
```



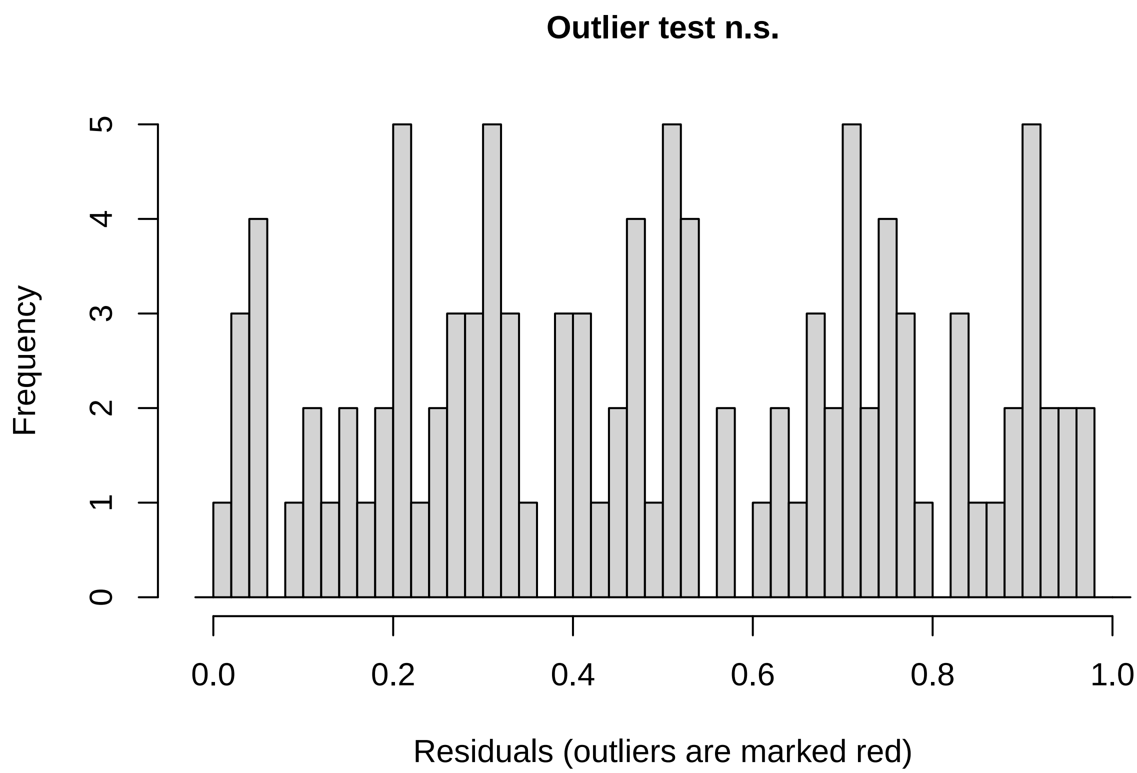

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.049785, p-value = 0.9535
## alternative hypothesis: two-sided
```

```r
testDispersion(mon_during_sim)
```



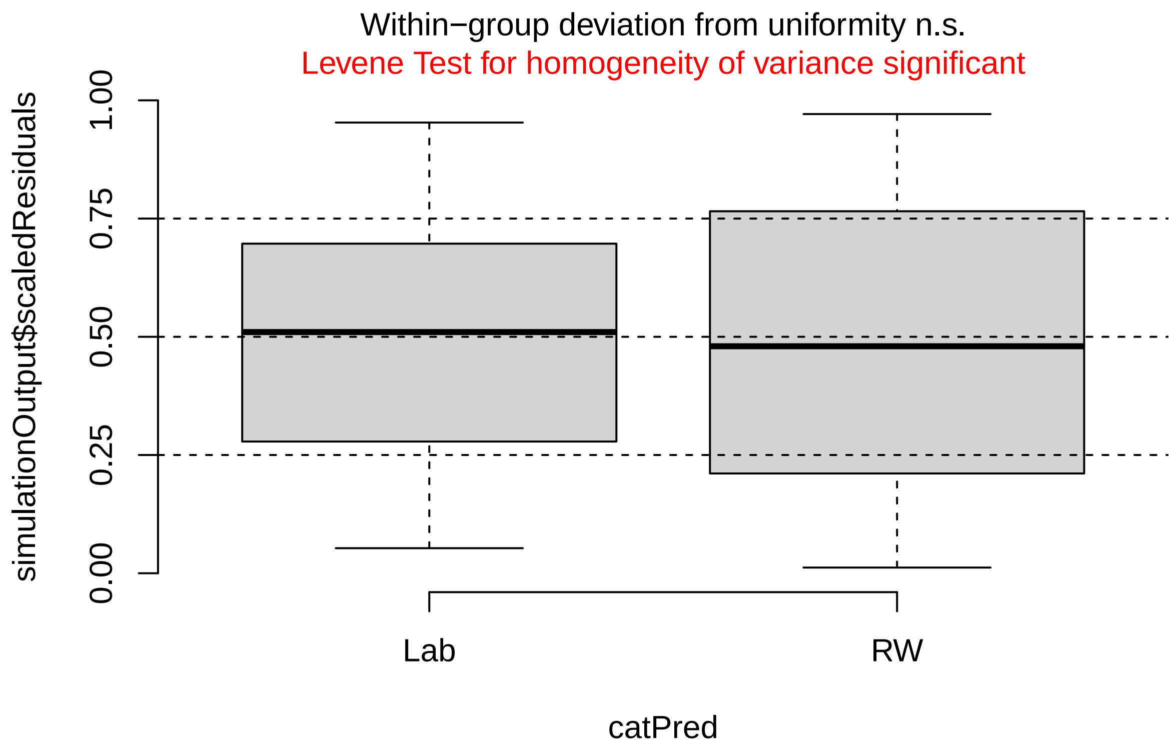

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.88942, p-value = 0.522
## alternative hypothesis: two.sided
```

```
testOutliers(mon_during_sim)
```

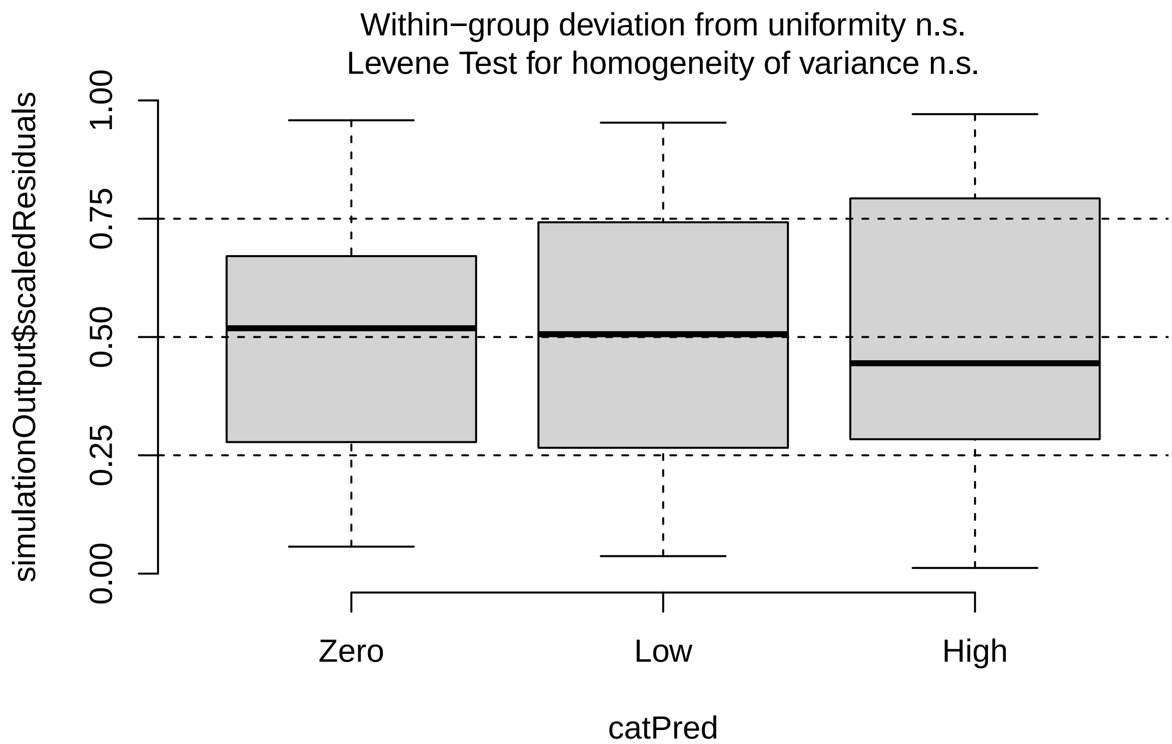



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  mon_during_sim
## outliers at both margin(s) = 0, observations = 107, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.000000 0.033888
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                    0
```

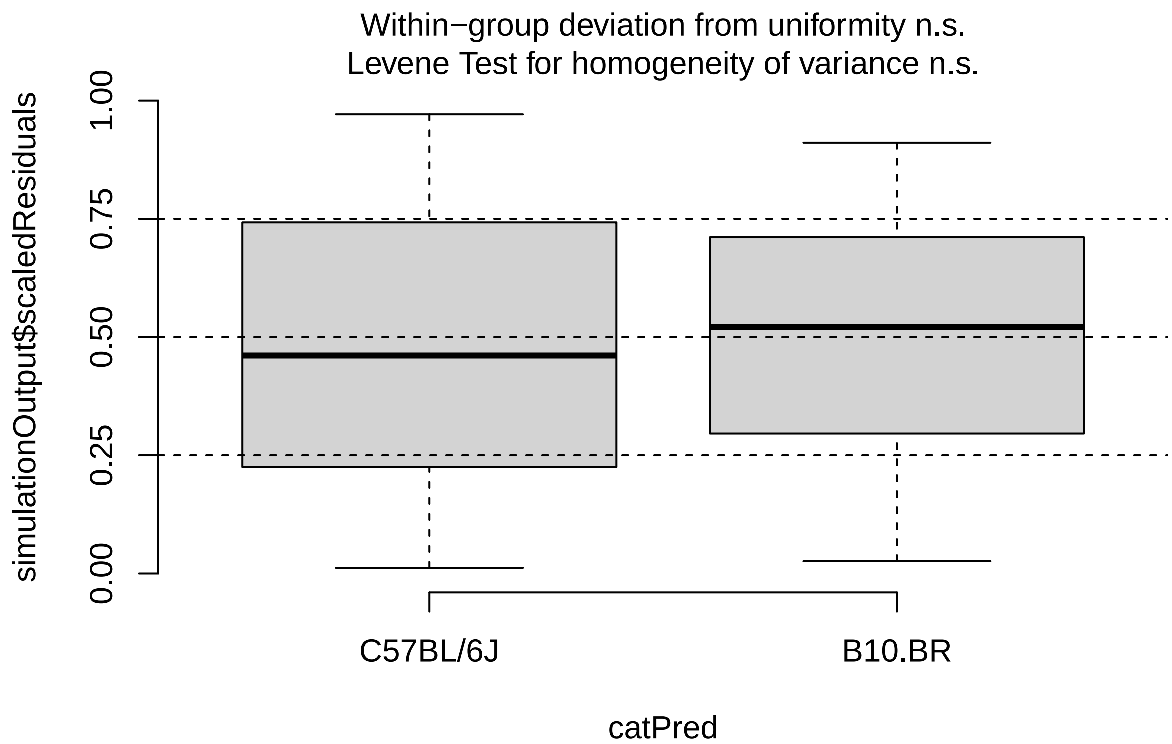
```
plotResiduals(mon_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$lab_or_rw)
```




```
plotResiduals(mon_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$t_muris_dos
```




```r
plotResiduals(mon_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$strain)
```




```r
# Post-RW: environment x infection x strain
mon_post <- lmer(log(mon_number) ~ lab_or_rw * t_muris_dose * strain +
                   (1 | cage_wedge) + (1 | cohort),
                 data = filter(cleaned_combined_cbc_data, phase == "Post-RW"))
```

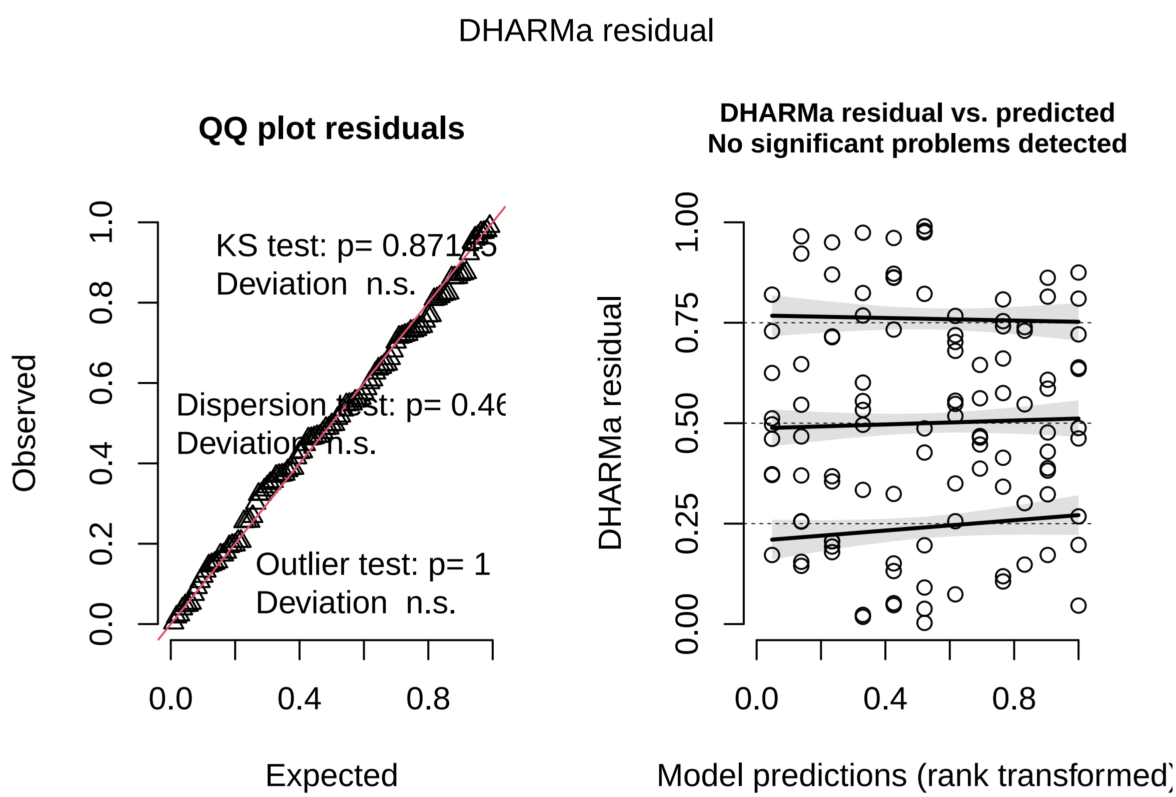
```
## boundary (singular) fit: see help('isSingular')
```

```r
print(summary(mon_post), correlation = FALSE)
```

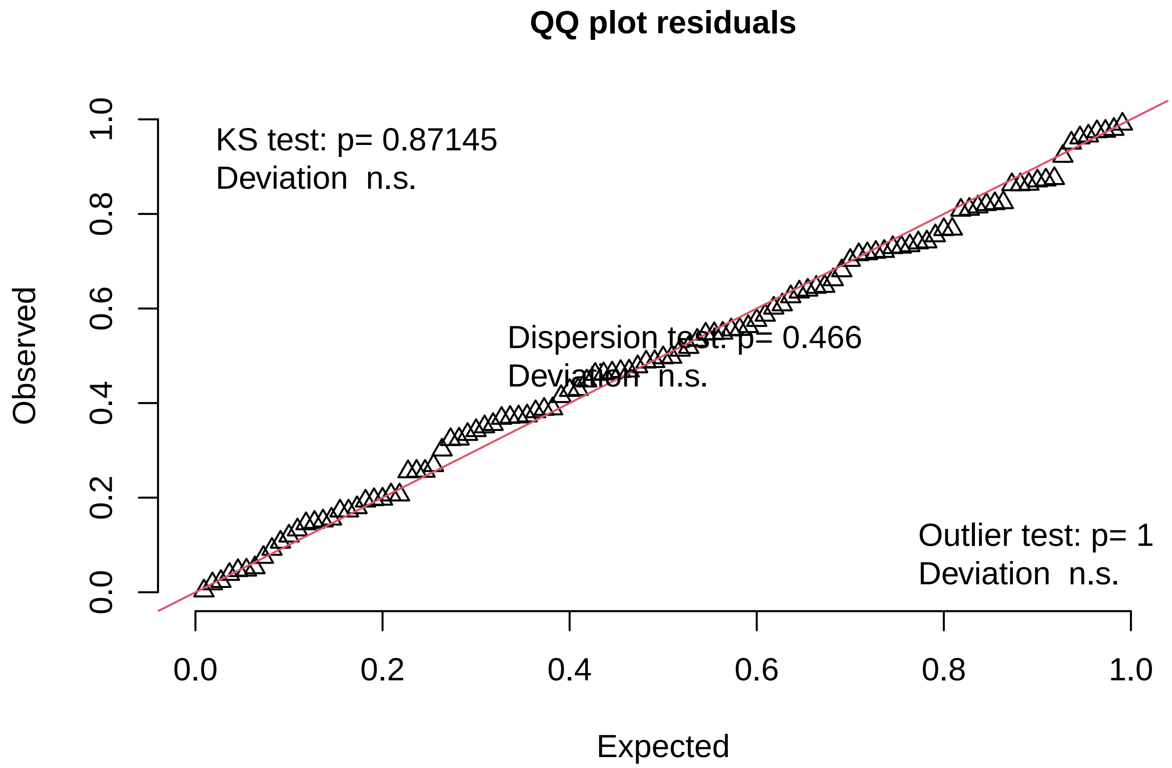
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log(mon_number) ~ lab_or_rw * t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Post-RW")
##
## REML criterion at convergence: 197.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.69577 -0.61911  0.04265  0.57675  2.18509
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cage_wedge (Intercept) 0.02071  0.1439
```

```
##  cohort     (Intercept) 0.00000  0.0000
##  Residual               0.32887  0.5735
## Number of obs: 109, groups:  cage_wedge, 29; cohort, 2
##
## Fixed effects:
##                                            Estimate Std. Error       df t value
## (Intercept)                                -1.22653    0.18791 91.33244  -6.527
## lab_or_rwRW                                -1.76444    0.28683 51.49614  -6.151
## t_muris_doseLow                             0.35590    0.26090 96.90380   1.364
## t_muris_doseHigh                            0.04522    0.30322 96.62883   0.149
## strainB10.BR                                0.13208    0.27222 92.96974   0.485
## lab_or_rwRW:t_muris_doseLow                 0.02291    0.37089 94.78938   0.062
## lab_or_rwRW:t_muris_doseHigh                1.09176    0.40180 96.35925   2.717
## lab_or_rwRW:strainB10.BR                    0.54179    0.40535 51.87007   1.337
## t_muris_doseLow:strainB10.BR               -0.09837    0.37220 95.99216  -0.264
## t_muris_doseHigh:strainB10.BR               0.08484    0.44556 96.86820   0.190
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   -0.02108    0.52326 93.72635  -0.040
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -0.94359    0.57775 96.24047  -1.633
##                                            Pr(>|t|)
## (Intercept)                                3.68e-09 ***
## lab_or_rwRW                                1.15e-07 ***
## t_muris_doseLow                             0.17569
## t_muris_doseHigh                            0.88176
## strainB10.BR                                0.62867
## lab_or_rwRW:t_muris_doseLow                 0.95088
## lab_or_rwRW:t_muris_doseHigh                0.00781 **
## lab_or_rwRW:strainB10.BR                    0.18719
## t_muris_doseLow:strainB10.BR                0.79213
## t_muris_doseHigh:strainB10.BR               0.84939
## lab_or_rwRW:t_muris_doseLow:strainB10.BR    0.96795
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR   0.10569
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
mon_post_sim <- simulateResiduals(fittedModel = mon_post, n = 1000)
plot(mon_post_sim)
```

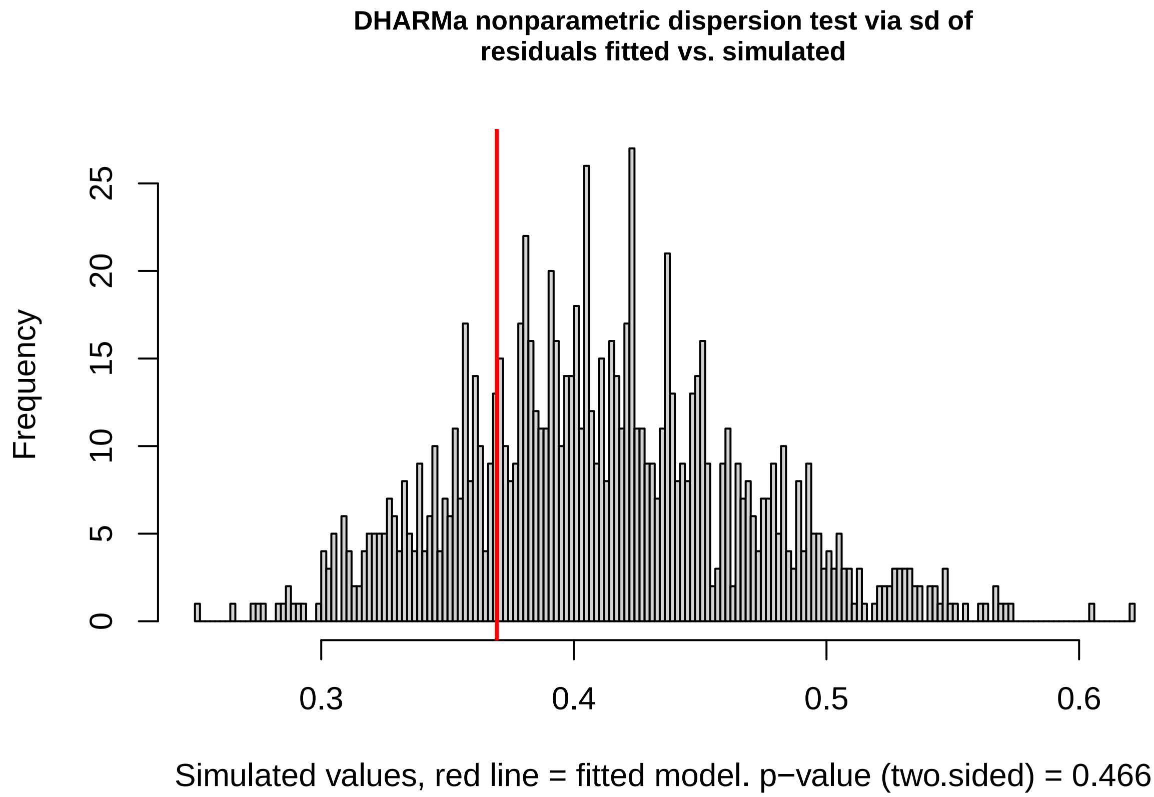



```
testUniformity(mon_post_sim)
```



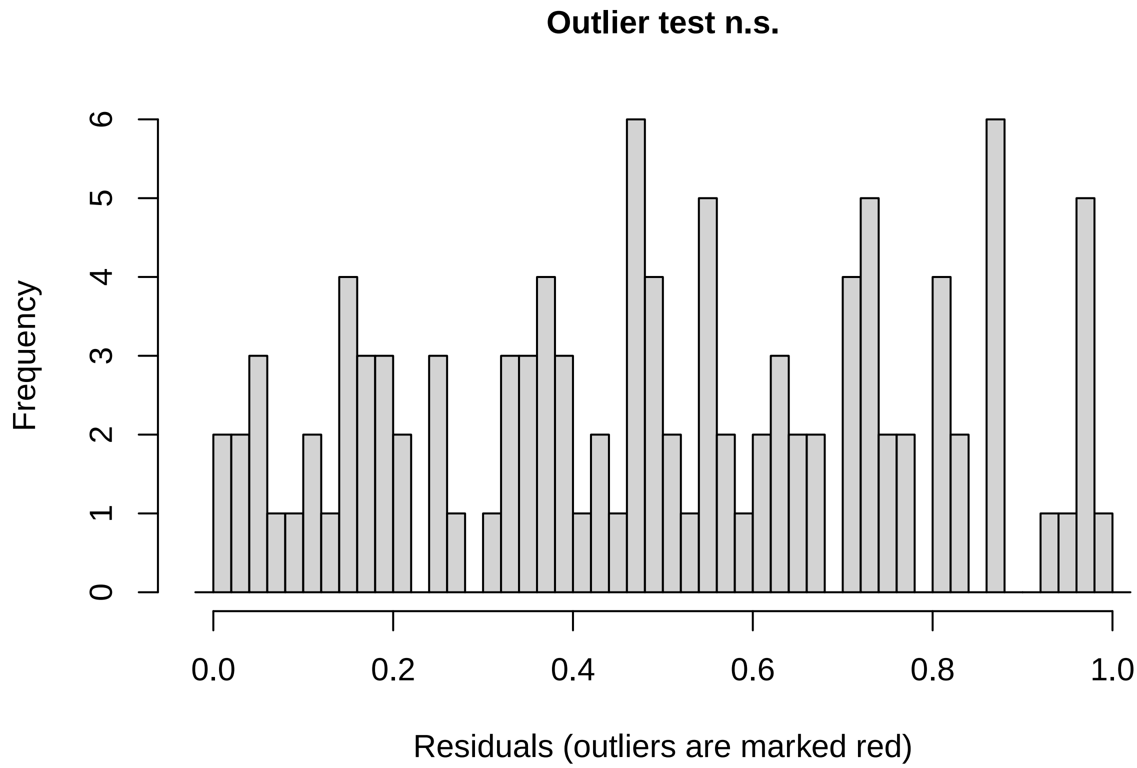

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.056945, p-value = 0.8715
## alternative hypothesis: two-sided
```

```
testDispersion(mon_post_sim)
```



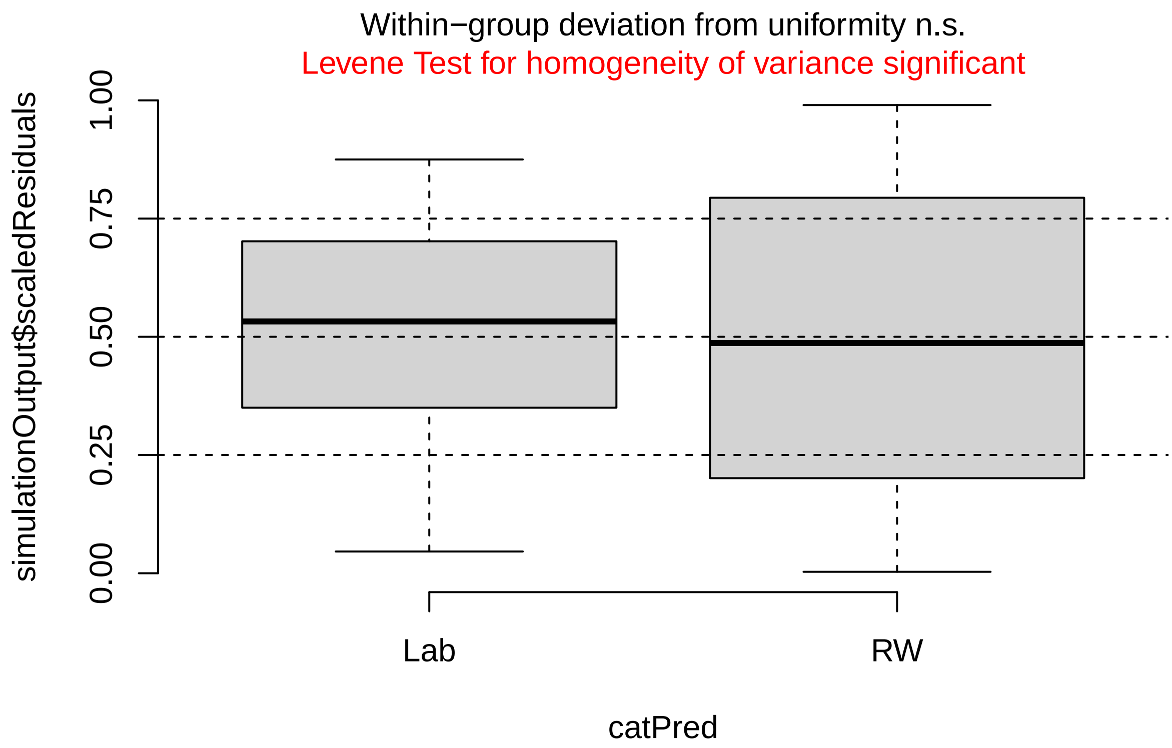

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.89894, p-value = 0.466
## alternative hypothesis: two.sided
```

```
testOutliers(mon_post_sim)
```

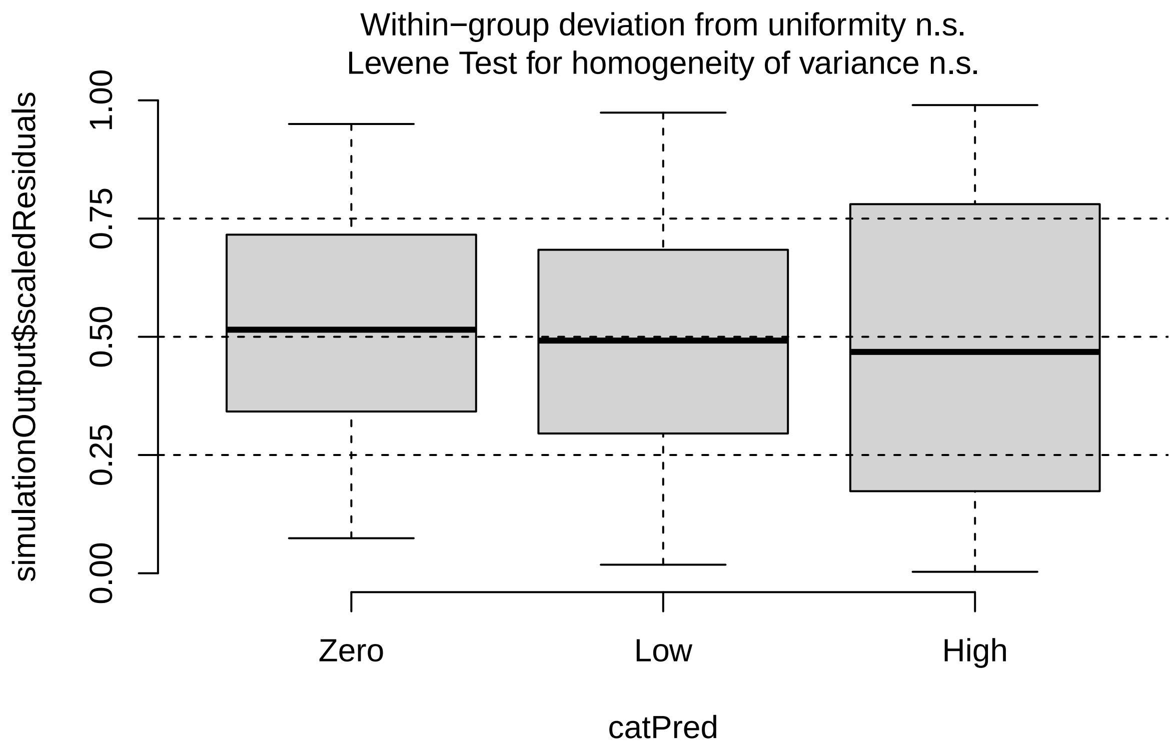



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  mon_post_sim
## outliers at both margin(s) = 0, observations = 109, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.00000000 0.03327666
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                   0
```

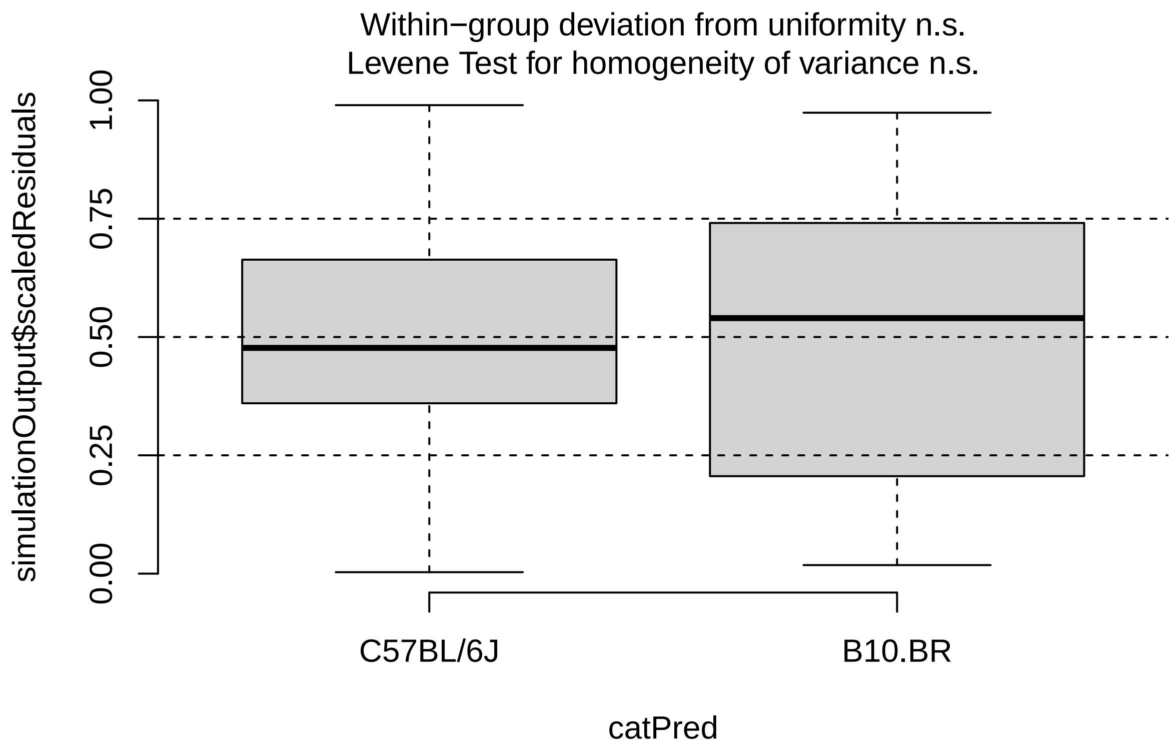
```
plotResiduals(mon_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$lab_or_rw)
```




```
plotResiduals(mon_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$t_muris_dose)
```

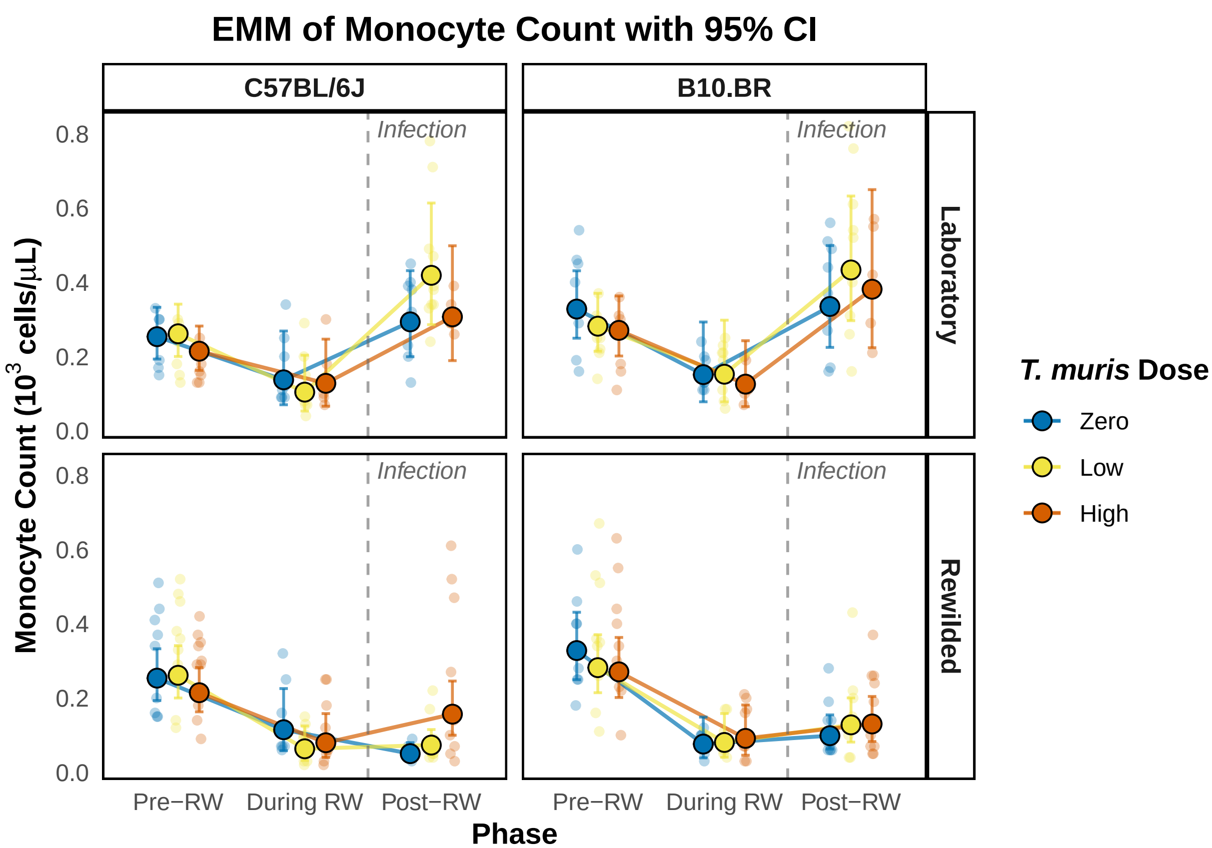



```
plotResiduals(mon_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$strain)
```




```
emm_mon_pre <- as.data.frame(emmeans(mon_pre,
                                     specs = "t_muris_dose",
                                     by = "strain",
                                     type = "response")) %>%
  crossing(lab_or_rw = factor(c("Lab", "RW"),
                              levels = c("Lab", "RW"))) %>%
  mutate(phase = factor("Pre-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_mon_during <- as.data.frame(emmeans(mon_during,
                                        specs = c("t_muris_dose", "lab_or_rw"),
                                        by = "strain",
                                        type = "response")) %>%
  mutate(phase = factor("During RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_mon_post <- as.data.frame(emmeans(mon_post,
                                      specs = c("t_muris_dose", "lab_or_rw"),
                                      by = "strain",
                                      type = "response")) %>%
  mutate(phase = factor("Post-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_mon_combined <- bind_rows(emm_mon_pre, emm_mon_during, emm_mon_post) %>%
  mutate(phase = factor(phase, levels = c("Pre-RW", "During RW", "Post-RW")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "Low", "High")),
         strain = factor(strain, levels = c("C57BL/6J", "B10.BR")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

```r
emm_mon_plot <- emm_plotter(
  emm_df = emm_mon_combined,
  cbc_input = cleaned_combined_cbc_data,
  y_col = "mon_number",
  y_label = expression(bold("Monocyte Count (10"^3~"cells/"*mu*"L)")),
  title = "EMM of Monocyte Count with 95% CI"
)

emm_mon_plot
```




```r
plot_saver("emm_mon_number.png", emm_mon_plot)
```

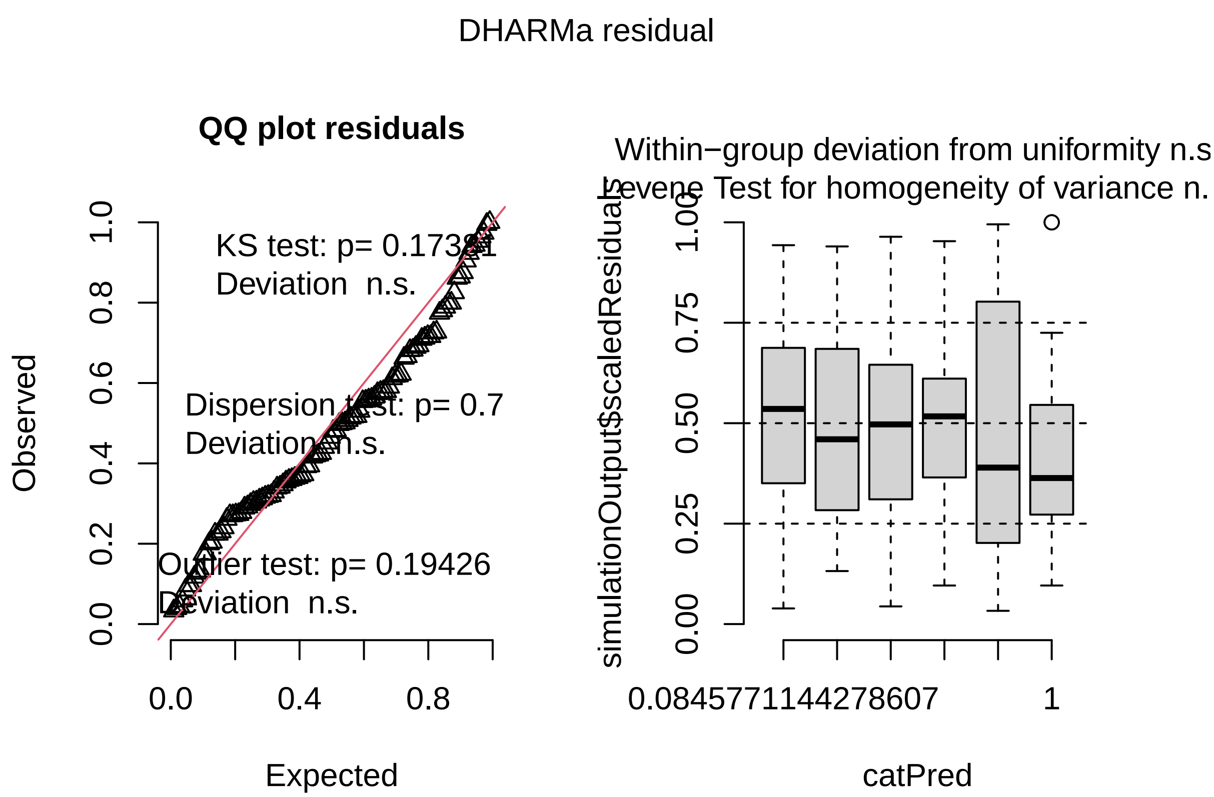
Eosinophil

```r
# Pre-RW: dose x strain (log1p for zeros, no environment - all mice are lab-housed)
eos_pre <- lmer(log1p(eos_number) ~ t_muris_dose * strain +
                  (1 | cage_wedge) + (1 | cohort),
                data = filter(cleaned_combined_cbc_data, phase == "Pre-RW"))

print(summary(eos_pre), correlation = FALSE)
```

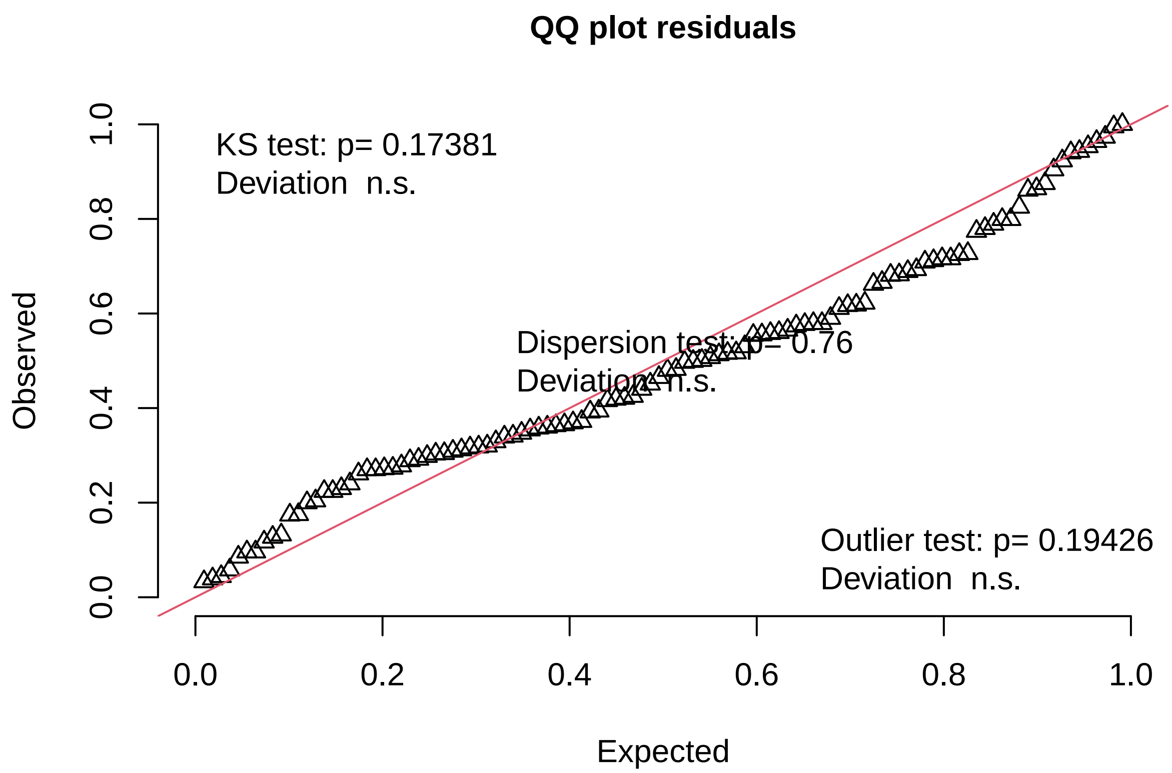
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
## Formula: log1p(eos_number) ~ t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Pre-RW")
##
## REML criterion at convergence: -214.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8922 -0.5025 -0.0841  0.3859  5.5192
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 0.0004496 0.02120
##  cohort     (Intercept) 0.0001596 0.01263
##  Residual               0.0056648 0.07526
## Number of obs: 108, groups:  cage_wedge, 21; cohort, 2
##
## Fixed effects:
##                             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                 0.203689   0.021306  7.921273   9.560 1.27e-05
## t_muris_doseLow             0.006717   0.024958 86.339435   0.269    0.788
## t_muris_doseHigh            0.017955   0.026431 89.029440   0.679    0.499
## strainB10.BR               -0.002248   0.027050 60.090686  -0.083    0.934
## t_muris_doseLow:strainB10.BR  -0.007813   0.034757 85.472658  -0.225    0.823
## t_muris_doseHigh:strainB10.BR -0.028411   0.037147 93.747556  -0.765    0.446
##
## (Intercept)                ***
## t_muris_doseLow
## t_muris_doseHigh
## strainB10.BR
## t_muris_doseLow:strainB10.BR
## t_muris_doseHigh:strainB10.BR
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
eos_pre_sim <- simulateResiduals(fittedModel = eos_pre, n = 1000)
plot(eos_pre_sim)
```

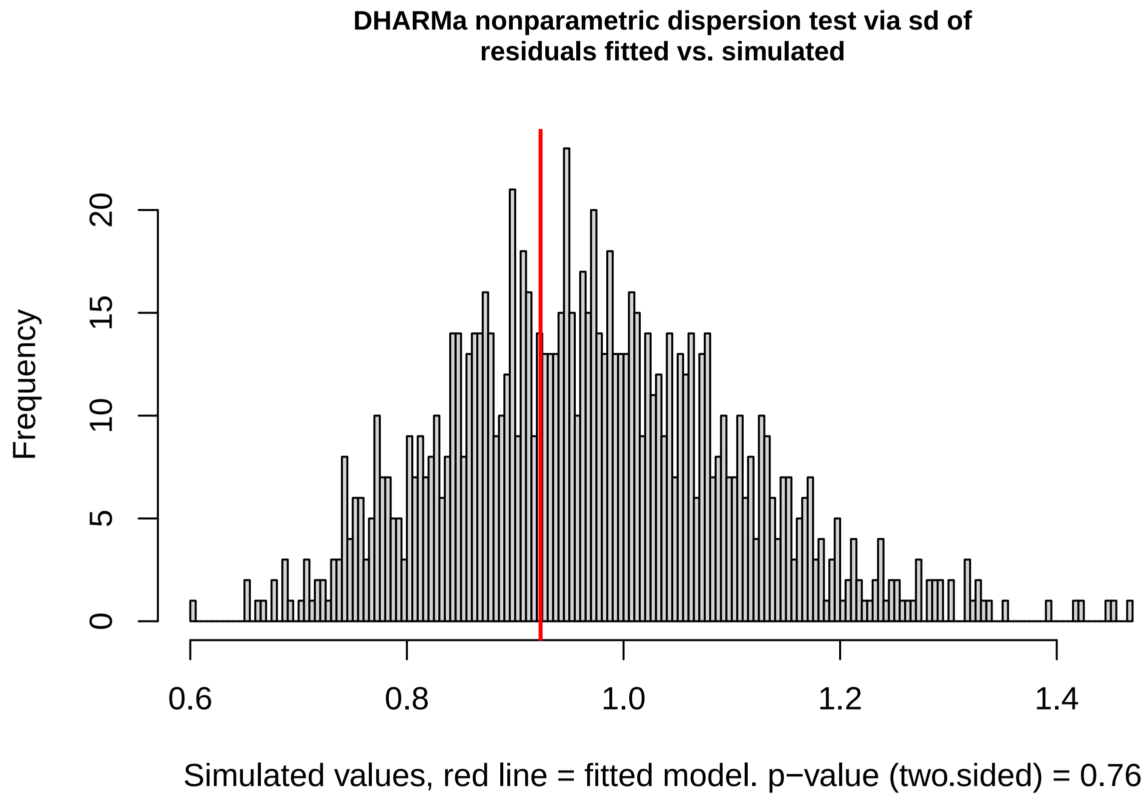



```r
testUniformity(eos_pre_sim)
```



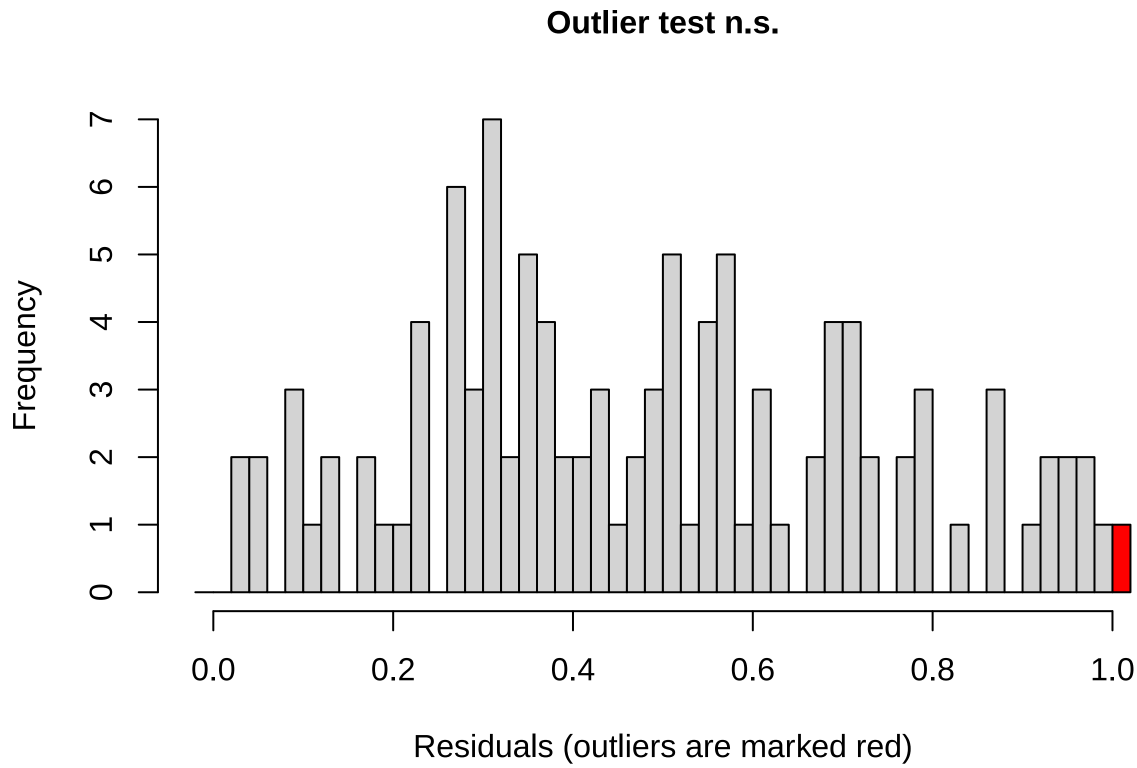

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.10633, p-value = 0.1738
## alternative hypothesis: two-sided
```

```
testDispersion(eos_pre_sim)
```



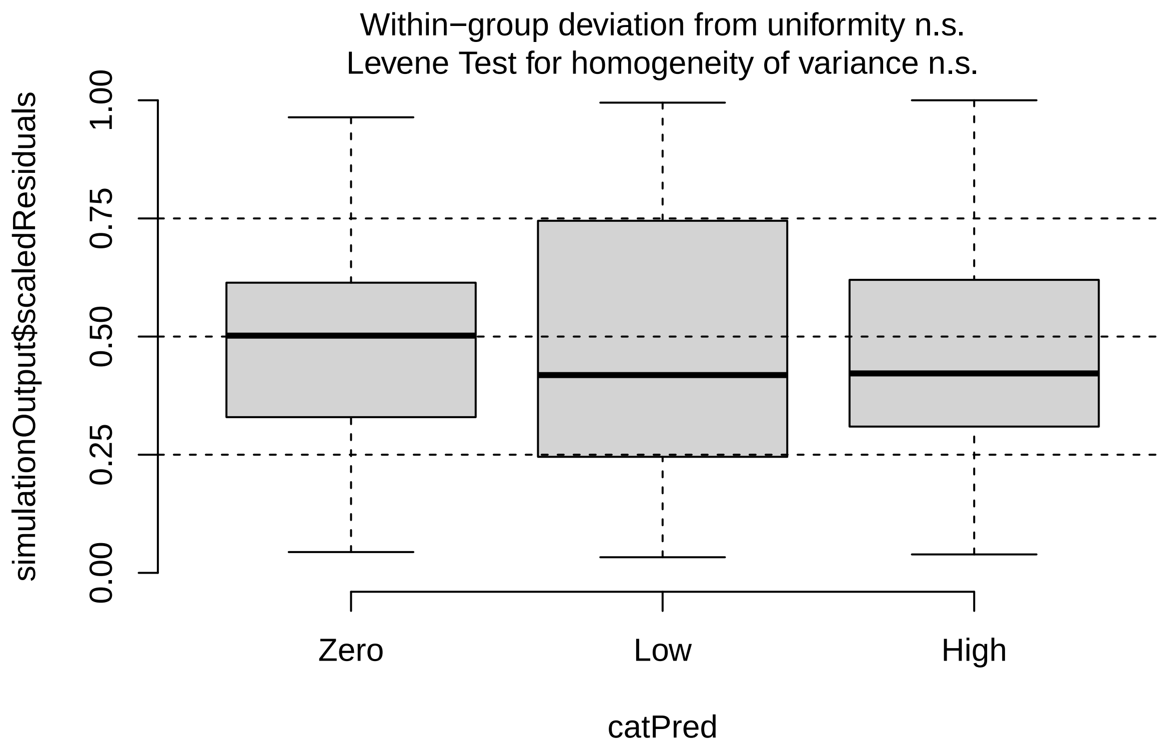

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.95108, p-value = 0.76
## alternative hypothesis: two.sided
```

```
testOutliers(eos_pre_sim)
```

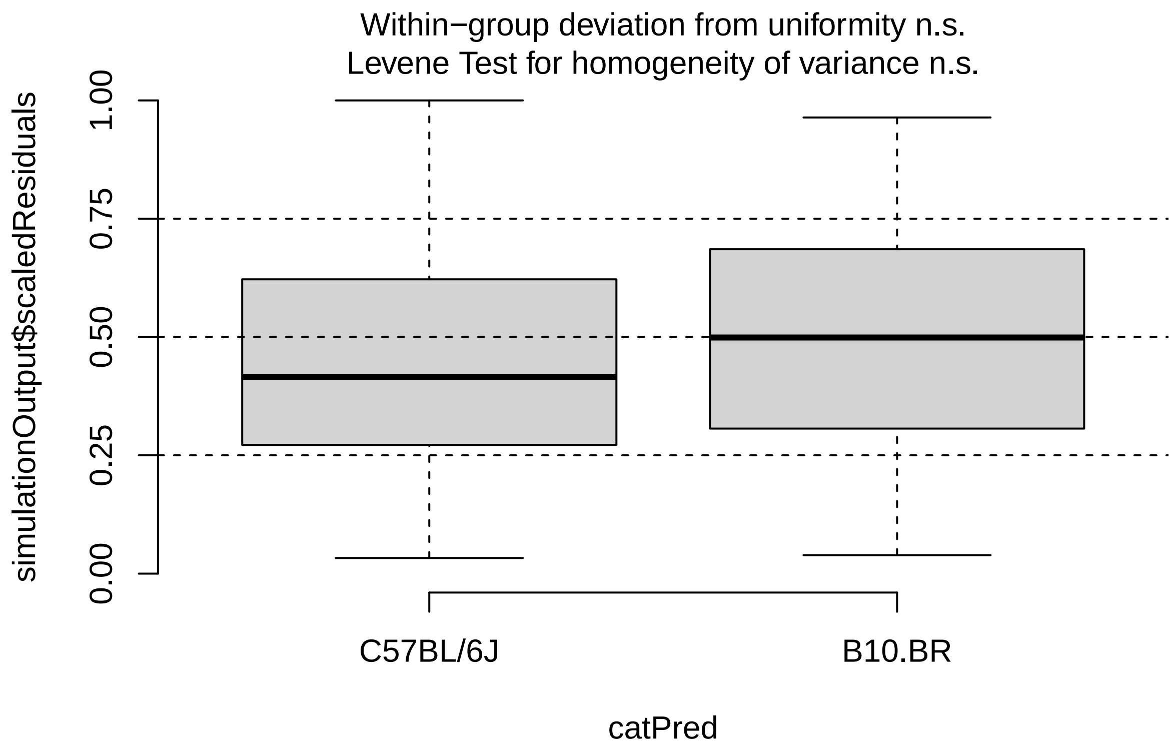



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  eos_pre_sim
## outliers at both margin(s) = 1, observations = 108, p-value = 0.1943
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002343967 0.0505105337
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                        0.009259259
```

```
plotResiduals(eos_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$t_muris_dose)
```




```
plotResiduals(eos_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$strain)
```

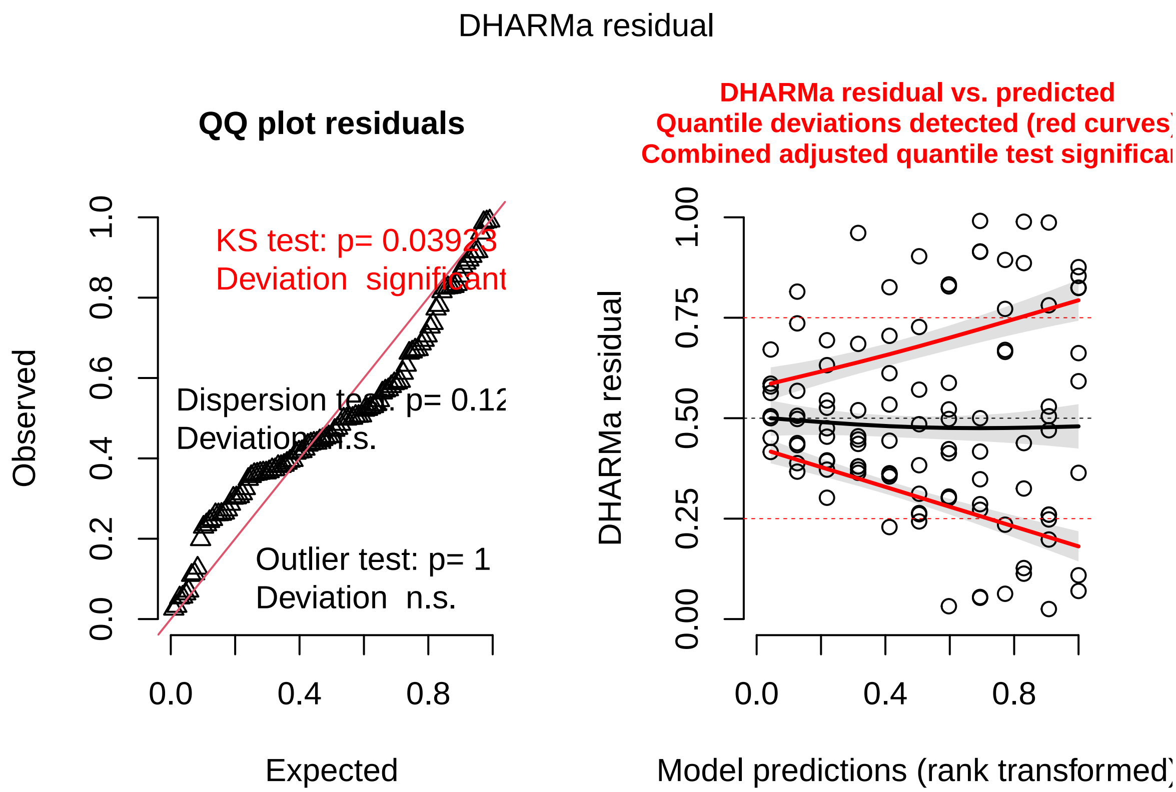



```r
# During RW: environment x dose x strain
eos_during <- lmer(log1p(eos_number) ~ lab_or_rw * t_muris_dose * strain +
                     (1 | cage_wedge) + (1 | cohort),
                   data = filter(cleaned_combined_cbc_data, phase == "During RW"))

print(summary(eos_during), correlation = FALSE)
```

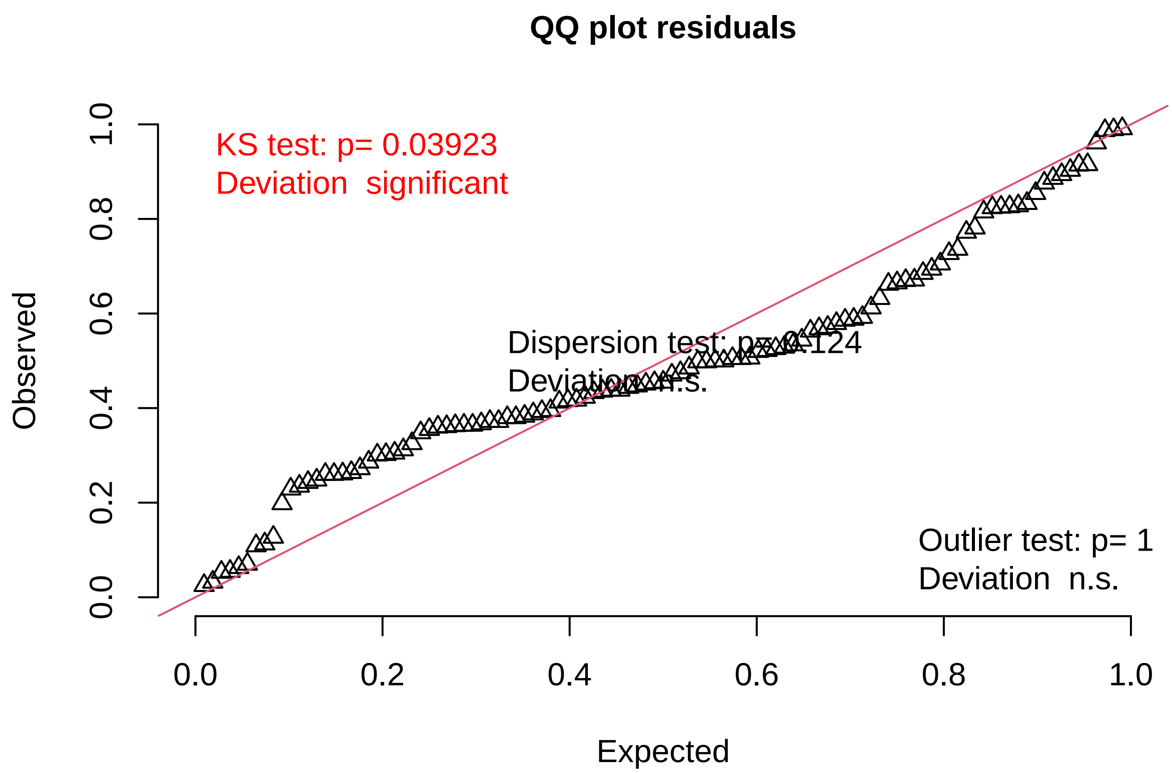
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log1p(eos_number) ~ lab_or_rw * t_muris_dose * strain + (1 |
##     cage_wedge) + (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "During RW")
##
## REML criterion at convergence: -340.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.3873 -0.5639 -0.1116  0.4413  2.7887
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 1.477e-03 3.843e-02
##  cohort     (Intercept) 8.744e-12 2.957e-06
##  Residual               7.839e-04 2.800e-02
## Number of obs: 107, groups:  cage_wedge, 29; cohort, 2
##
```

```
## Fixed effects:
##                                            Estimate Std. Error        df
## (Intercept)                                0.129298   0.014534 47.202234
## lab_or_rwRW                               -0.087023   0.028416 34.993432
## t_muris_doseLow                            0.002758   0.014060 78.664944
## t_muris_doseHigh                           0.008289   0.017617 85.682536
## strainB10.BR                               0.073311   0.020780 46.886287
## lab_or_rwRW:t_muris_doseLow               -0.020270   0.019201 74.910258
## lab_or_rwRW:t_muris_doseHigh              -0.013337   0.021822 80.066671
## lab_or_rwRW:strainB10.BR                  -0.106002   0.034290 76.198369
## t_muris_doseLow:strainB10.BR              -0.017313   0.019560 75.881430
## t_muris_doseHigh:strainB10.BR             -0.072673   0.024413 83.027464
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   0.043429   0.027181 72.808302
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.093202   0.030630 77.901377
##                                           t value Pr(>|t|)
## (Intercept)                                 8.896 1.17e-11 ***
## lab_or_rwRW                                -3.062 0.004203 **
## t_muris_doseLow                             0.196 0.844966
## t_muris_doseHigh                            0.470 0.639210
## strainB10.BR                                3.528 0.000949 ***
## lab_or_rwRW:t_muris_doseLow                -1.056 0.294512
## lab_or_rwRW:t_muris_doseHigh               -0.611 0.542818
## lab_or_rwRW:strainB10.BR                   -3.091 0.002783 **
## t_muris_doseLow:strainB10.BR               -0.885 0.378875
## t_muris_doseHigh:strainB10.BR              -2.977 0.003815 **
## lab_or_rwRW:t_muris_doseLow:strainB10.BR    1.598 0.114420
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR   3.043 0.003194 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
eos_during_sim <- simulateResiduals(fittedModel = eos_during, n = 1000)
plot(eos_during_sim)
```

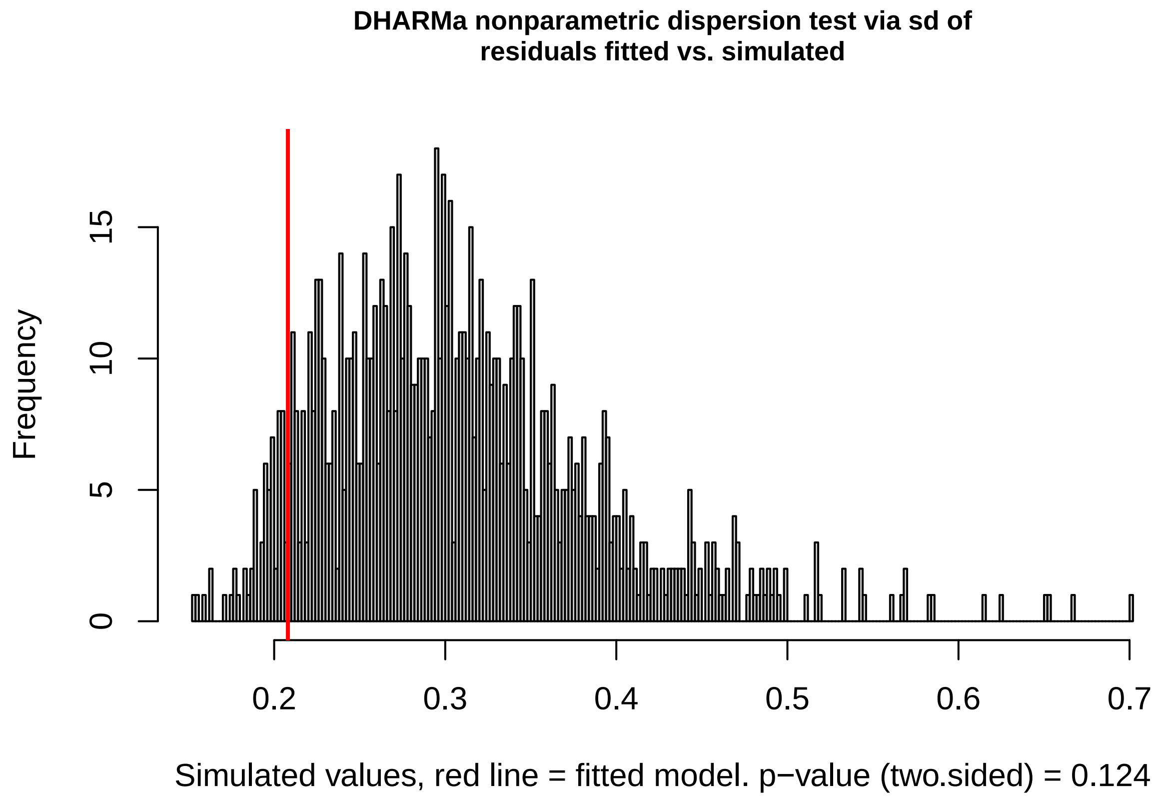



```
testUniformity(eos_during_sim)
```



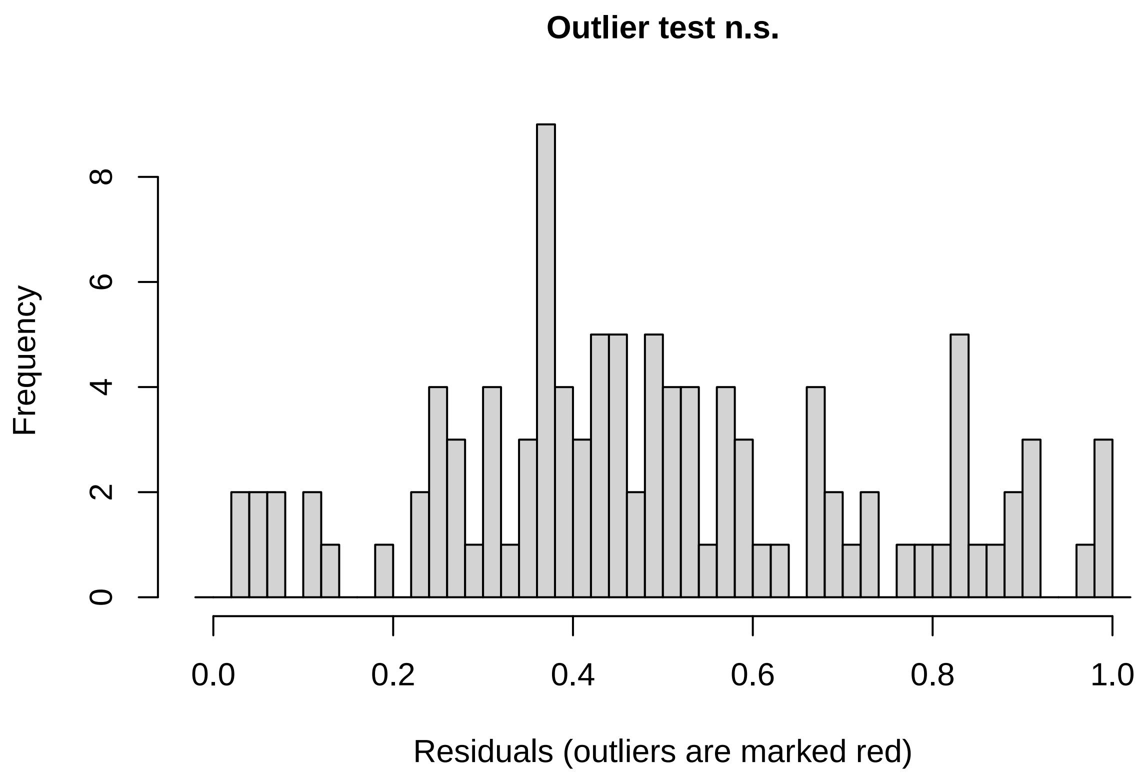

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.13554, p-value = 0.03923
## alternative hypothesis: two-sided
```

```
testDispersion(eos_during_sim)
```



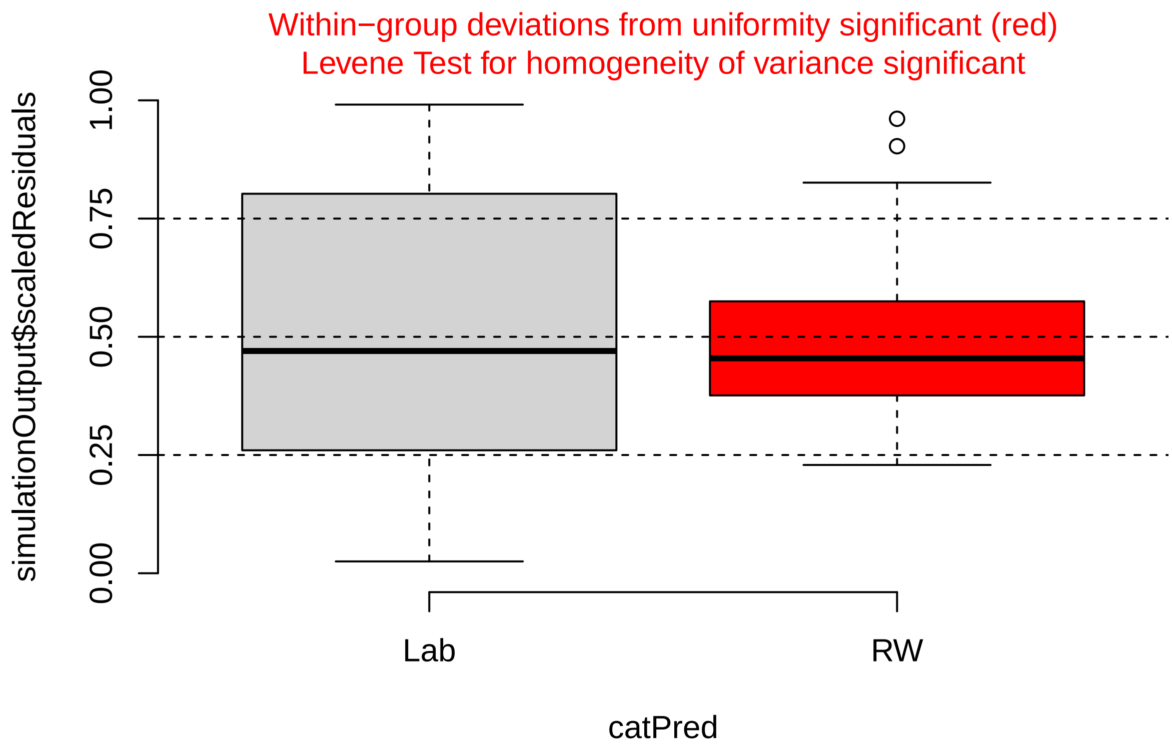

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.67216, p-value = 0.124
## alternative hypothesis: two.sided
```

```r
testOutliers(eos_during_sim)
```

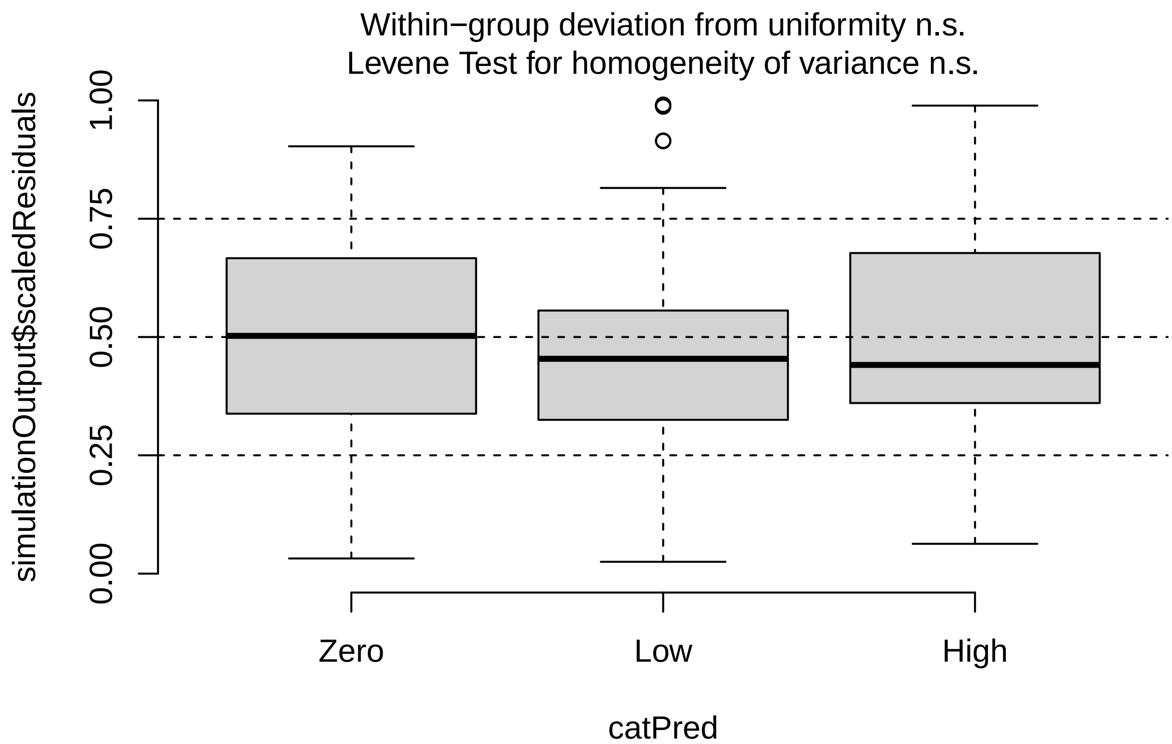



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  eos_during_sim
## outliers at both margin(s) = 0, observations = 107, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.000000 0.033888
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                                    0
```

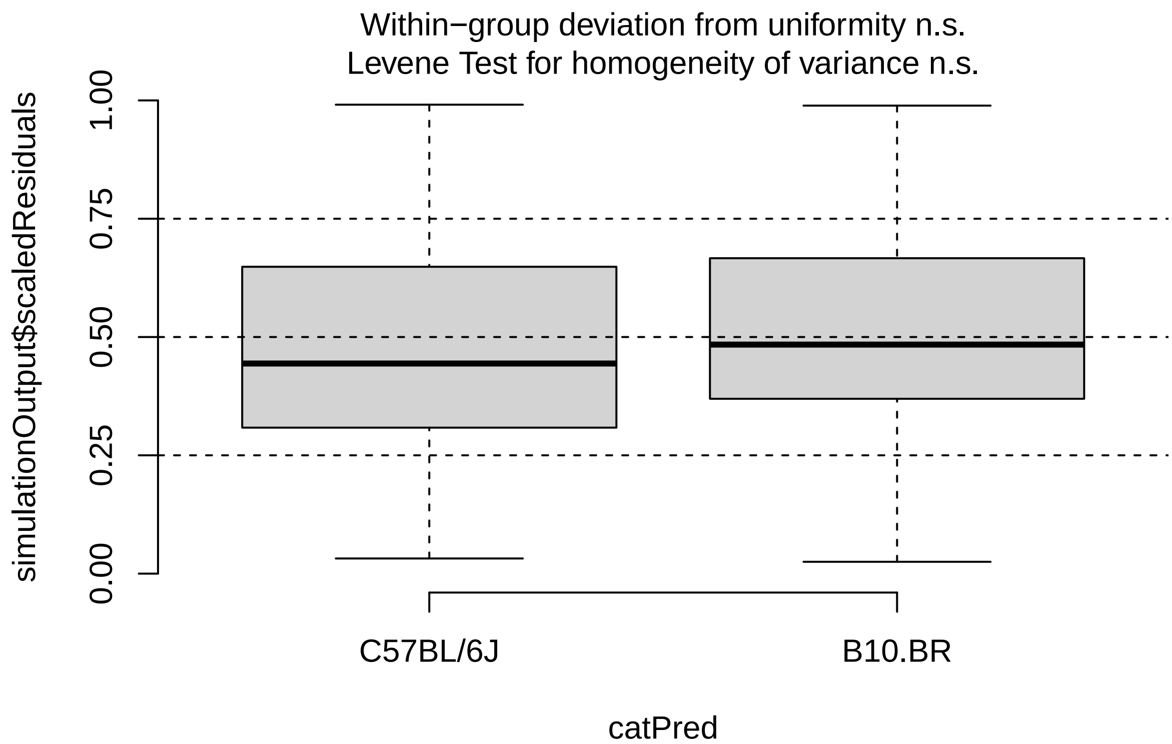
```
plotResiduals(eos_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$lab_or_rw)
```




```
plotResiduals(eos_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$t_muris_dos
```




```
plotResiduals(eos_during_sim, form = filter(cleaned_combined_cbc_data, phase == "During RW")$strain)
```

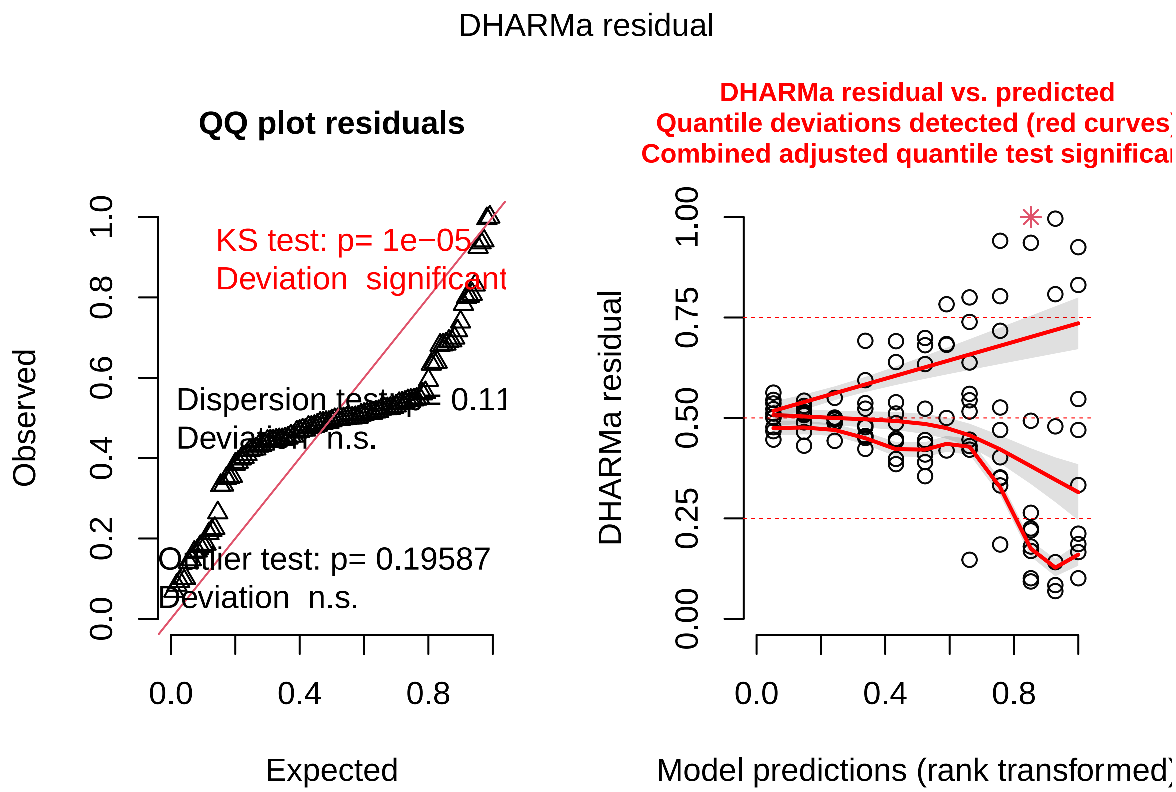



```r
# Post-RW: environment x infection x strain
eos_post <- lmer(log1p(eos_number) ~ lab_or_rw * t_muris_dose * strain +
                   (1 | cage_wedge) + (1 | cohort),
                 data = filter(cleaned_combined_cbc_data, phase == "Post-RW"))

print(summary(eos_post), correlation = FALSE)
```

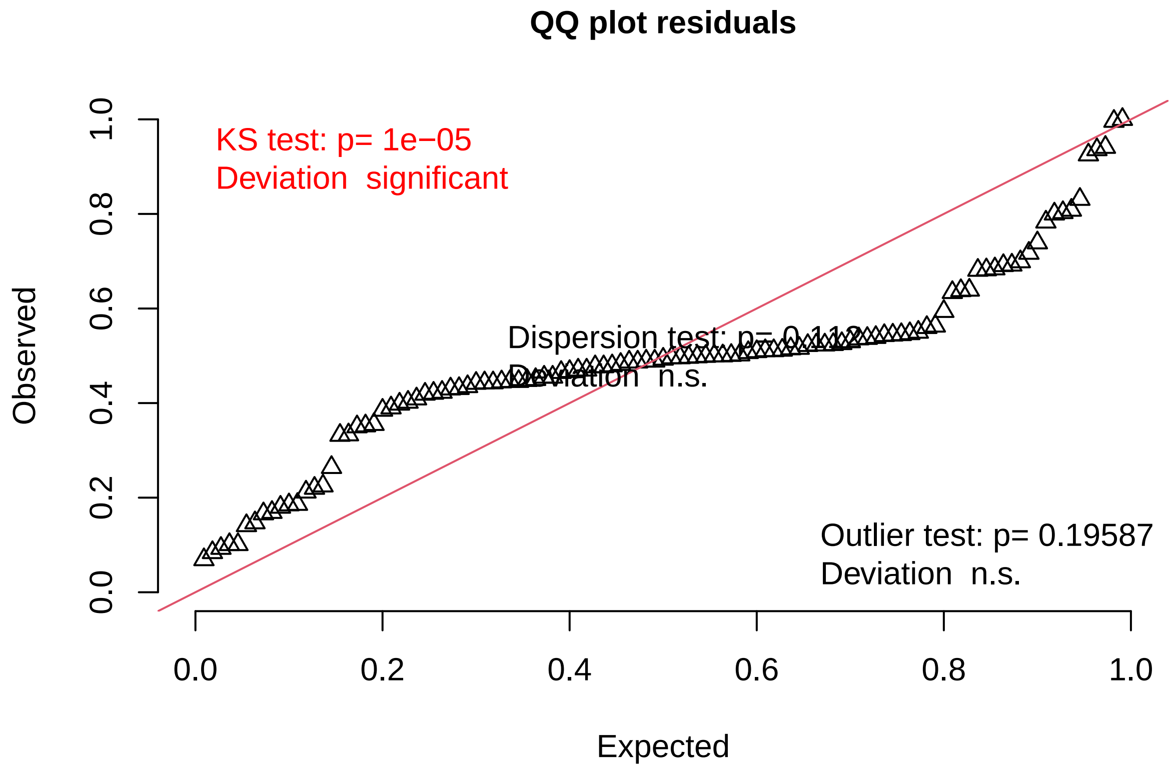
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log1p(eos_number) ~ lab_or_rw * t_muris_dose * strain + (1 |
##     cage_wedge) + (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Post-RW")
##
## REML criterion at convergence: -247.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.8233 -0.2953 -0.0547  0.2539  5.0430
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 3.778e-03 0.061467
##  cohort     (Intercept) 6.705e-05 0.008188
##  Residual               2.349e-03 0.048465
## Number of obs: 109, groups:  cage_wedge, 29; cohort, 2
##
```

```
## Fixed effects:
##                                            Estimate Std. Error       df t value
## (Intercept)                                 0.14620    0.02481  8.67923   5.894
## lab_or_rwRW                                -0.11804    0.05228 20.44104  -2.258
## t_muris_doseLow                            -0.04506    0.02492 86.56250  -1.808
## t_muris_doseHigh                            0.01790    0.03038 91.24733   0.589
## strainB10.BR                                0.04085    0.03448 44.61426   1.185
## lab_or_rwRW:t_muris_doseLow                 0.03068    0.03343 79.86498   0.918
## lab_or_rwRW:t_muris_doseHigh               -0.03813    0.03768 85.07367  -1.012
## lab_or_rwRW:strainB10.BR                   -0.04841    0.07392 20.27463  -0.655
## t_muris_doseLow:strainB10.BR                0.01479    0.03425 81.83809   0.432
## t_muris_doseHigh:strainB10.BR              -0.13021    0.04394 89.34384  -2.963
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   -0.01108    0.04626 76.92383  -0.239
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.13878    0.05383 83.82113   2.578
##                                            Pr(>|t|)
## (Intercept)                                0.000266 ***
## lab_or_rwRW                                0.035030 *
## t_muris_doseLow                            0.074076 .
## t_muris_doseHigh                           0.557097
## strainB10.BR                               0.242413
## lab_or_rwRW:t_muris_doseLow                0.361632
## lab_or_rwRW:t_muris_doseHigh               0.314436
## lab_or_rwRW:strainB10.BR                   0.519887
## t_muris_doseLow:strainB10.BR               0.667027
## t_muris_doseHigh:strainB10.BR              0.003899 **
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   0.811401
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  0.011680 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
eos_post_sim <- simulateResiduals(fittedModel = eos_post, n = 1000)
plot(eos_post_sim)
```

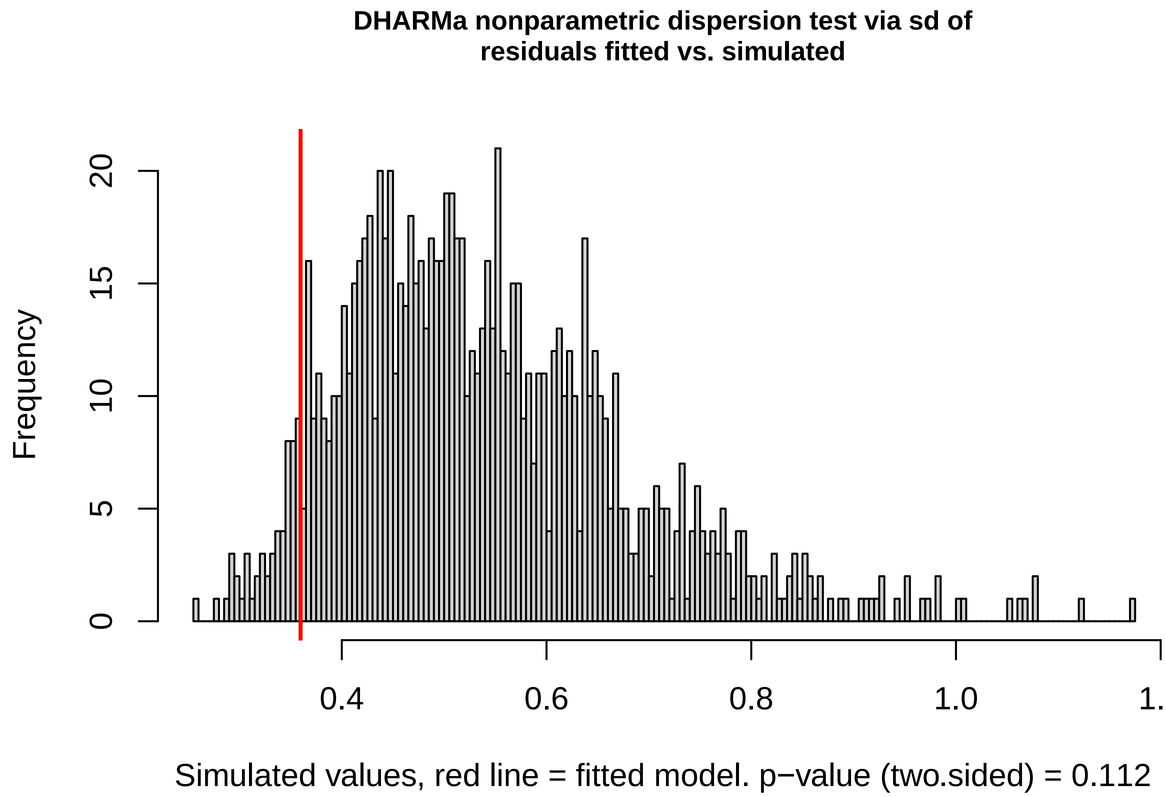



```
testUniformity(eos_post_sim)
```



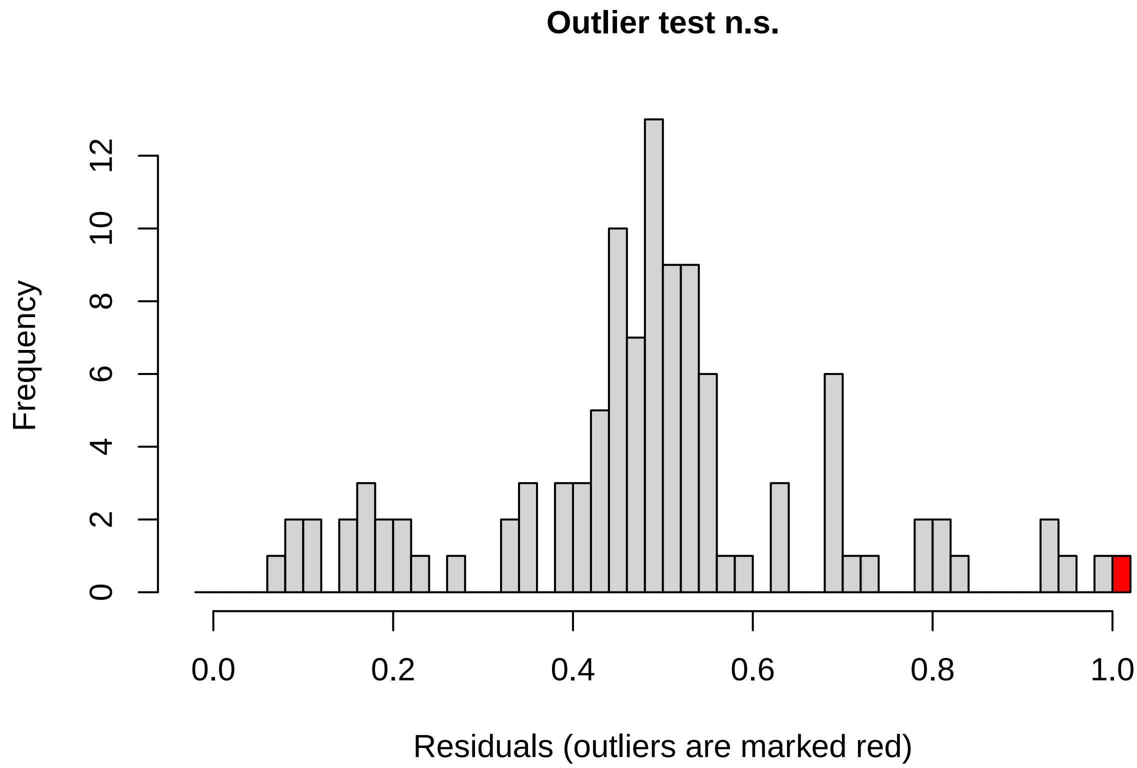

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.23517, p-value = 1.162e-05
## alternative hypothesis: two-sided
```

```
testDispersion(eos_post_sim)
```



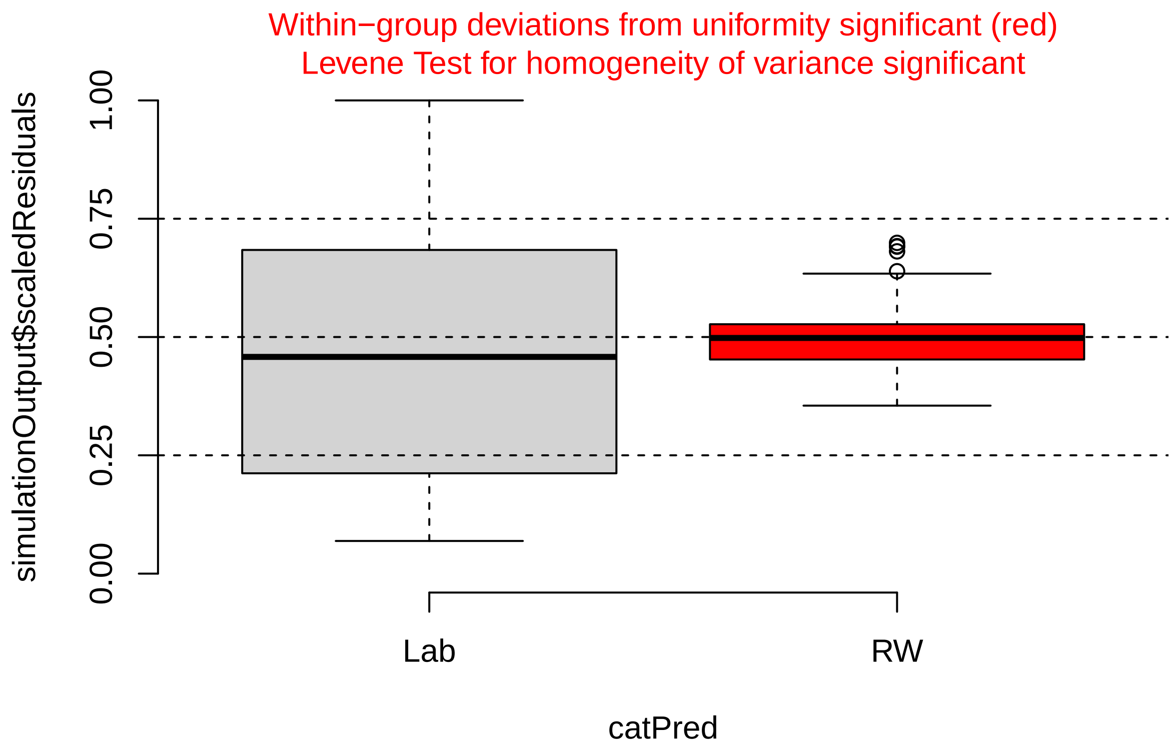

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.66571, p-value = 0.112
## alternative hypothesis: two.sided
```

```r
testOutliers(eos_post_sim)
```

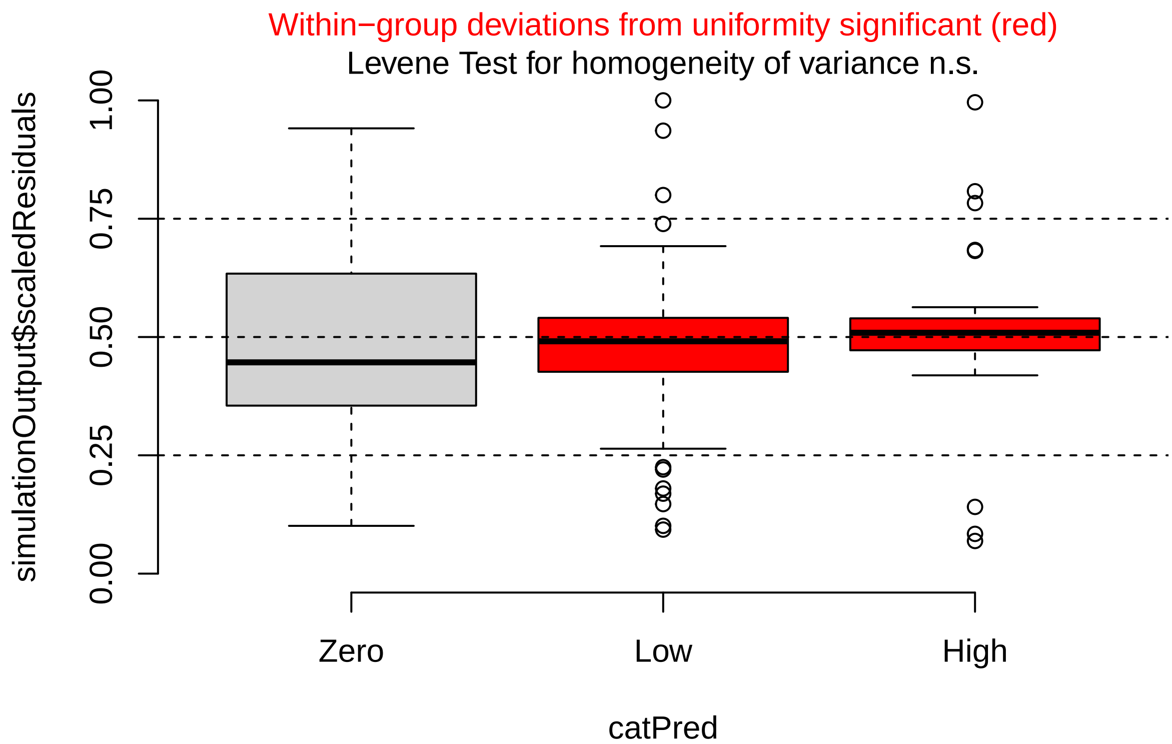



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  eos_post_sim
## outliers at both margin(s) = 1, observations = 109, p-value = 0.1959
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002322465 0.0500568231
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                          0.009174312
```

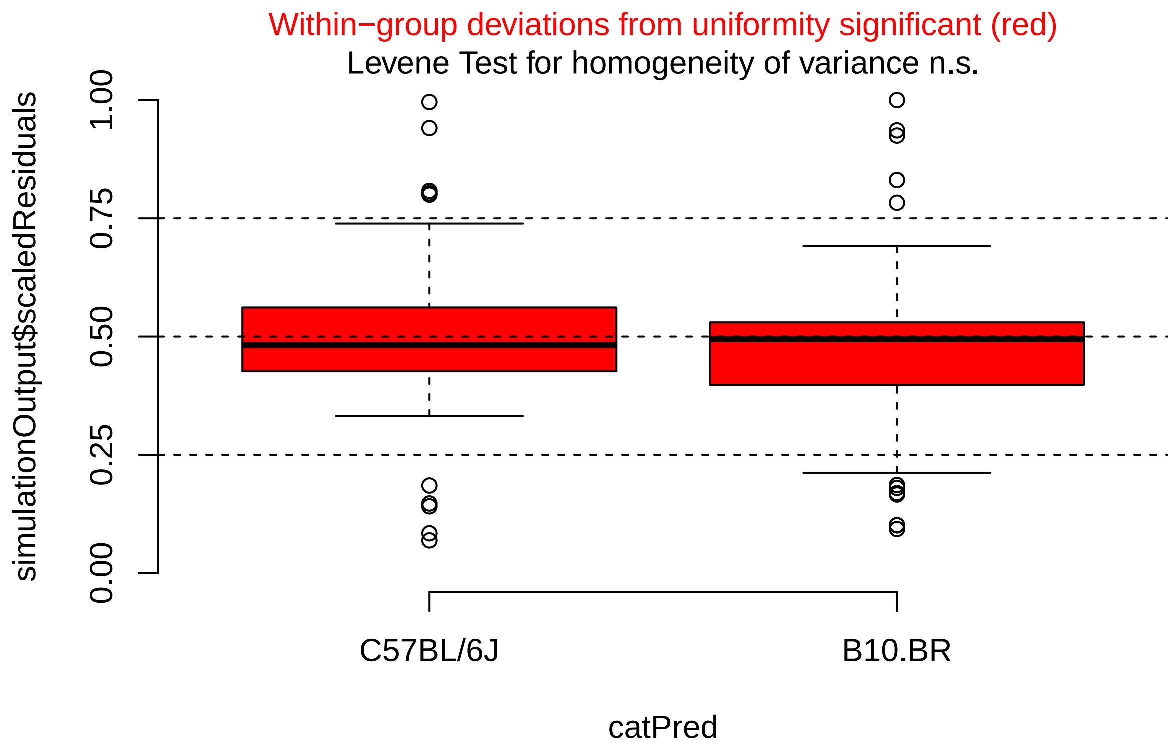
```
plotResiduals(eos_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$lab_or_rw)
```




```
plotResiduals(eos_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$t_muris_dose)
```



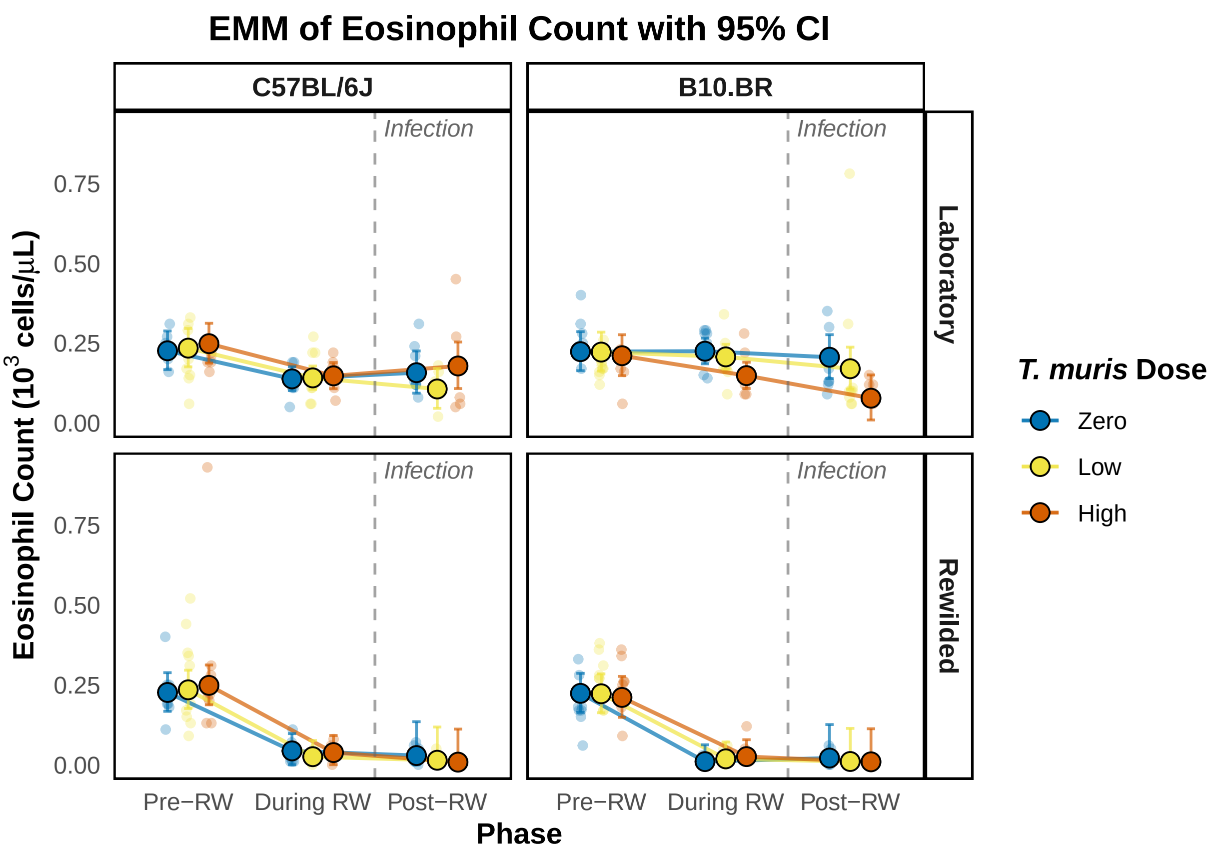

```
plotResiduals(eos_post_sim, form = filter(cleaned_combined_cbc_data, phase == "Post-RW")$strain)
```




```
# log1p back-transform: expm1() reverses log1p()
emm_eos_pre <- as.data.frame(emmeans(eos_pre,
                                      specs = "t_muris_dose",
                                      by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  crossing(lab_or_rw = factor(c("Lab", "RW"),
                              levels = c("Lab", "RW"))) %>%
  mutate(phase = factor("Pre-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_eos_during <- as.data.frame(emmeans(eos_during,
                                         specs = c("t_muris_dose", "lab_or_rw"),
                                         by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  mutate(phase = factor("During RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_eos_post <- as.data.frame(emmeans(eos_post,
                                       specs = c("t_muris_dose", "lab_or_rw"),
                                       by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  mutate(phase = factor("Post-RW", levels = c("Pre-RW", "During RW", "Post-RW")))
```

```r
emm_eos_combined <- bind_rows(emm_eos_pre, emm_eos_during, emm_eos_post) %>%
  mutate(phase = factor(phase, levels = c("Pre-RW", "During RW", "Post-RW")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "Low", "High")),
         strain = factor(strain, levels = c("C57BL/6J", "B10.BR")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

```r
emm_eos_plot <- emm_plotter(
  emm_df = emm_eos_combined,
  cbc_input = cleaned_combined_cbc_data,
  y_col = "eos_number",
  y_label = expression(bold("Eosinophil Count (10"^3~"cells/"*mu*"L)")),
  title = "EMM of Eosinophil Count with 95% CI",
  floor_zero = TRUE
)

emm_eos_plot
```




```r
plot_saver("emm_eos_number.png", emm_eos_plot)
```

Basophil

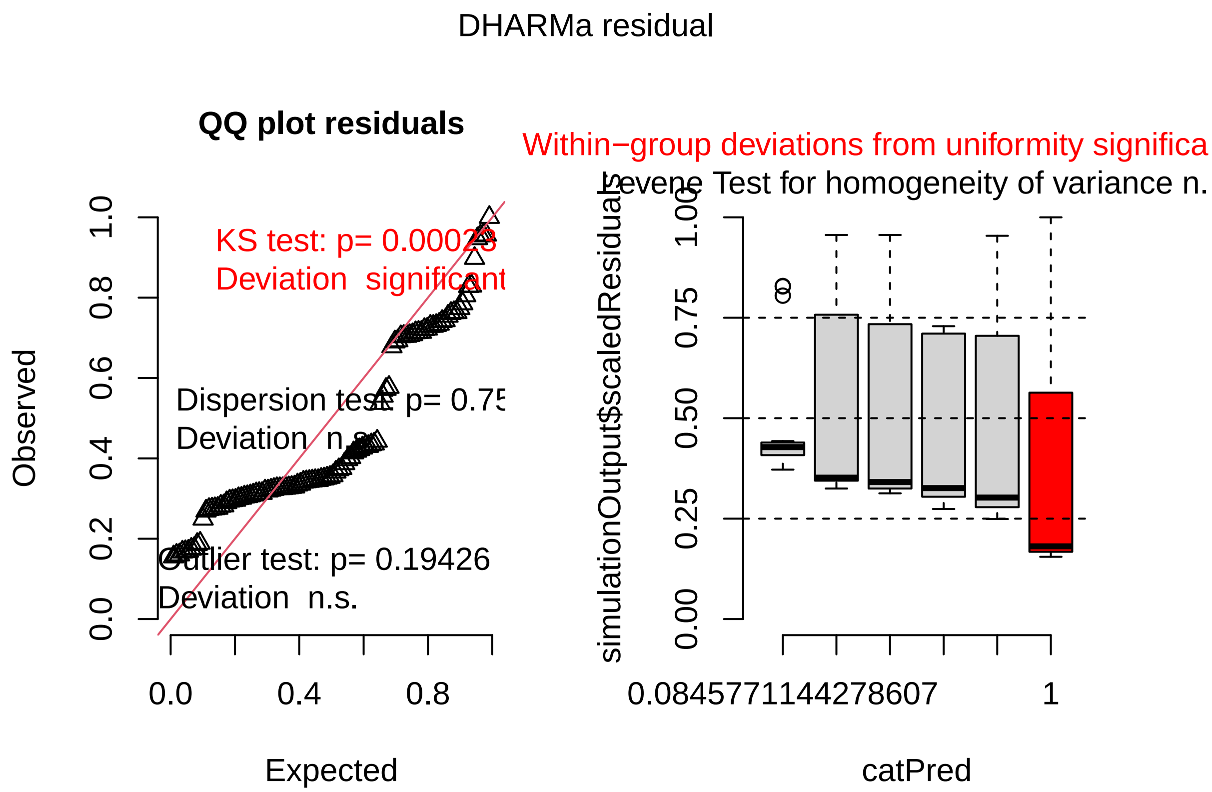
```r
# Pre-RW: dose x strain (log1p for zeros, no environment - all mice are lab-housed)
bas_pre <- lmer(log1p(bas_number) ~ t_muris_dose * strain +
                  (1 | cage_wedge) + (1 | cohort),
                data = filter(cleaned_combined_cbc_data, phase == "Pre-RW"))
```

```
## boundary (singular) fit: see help('isSingular')
```

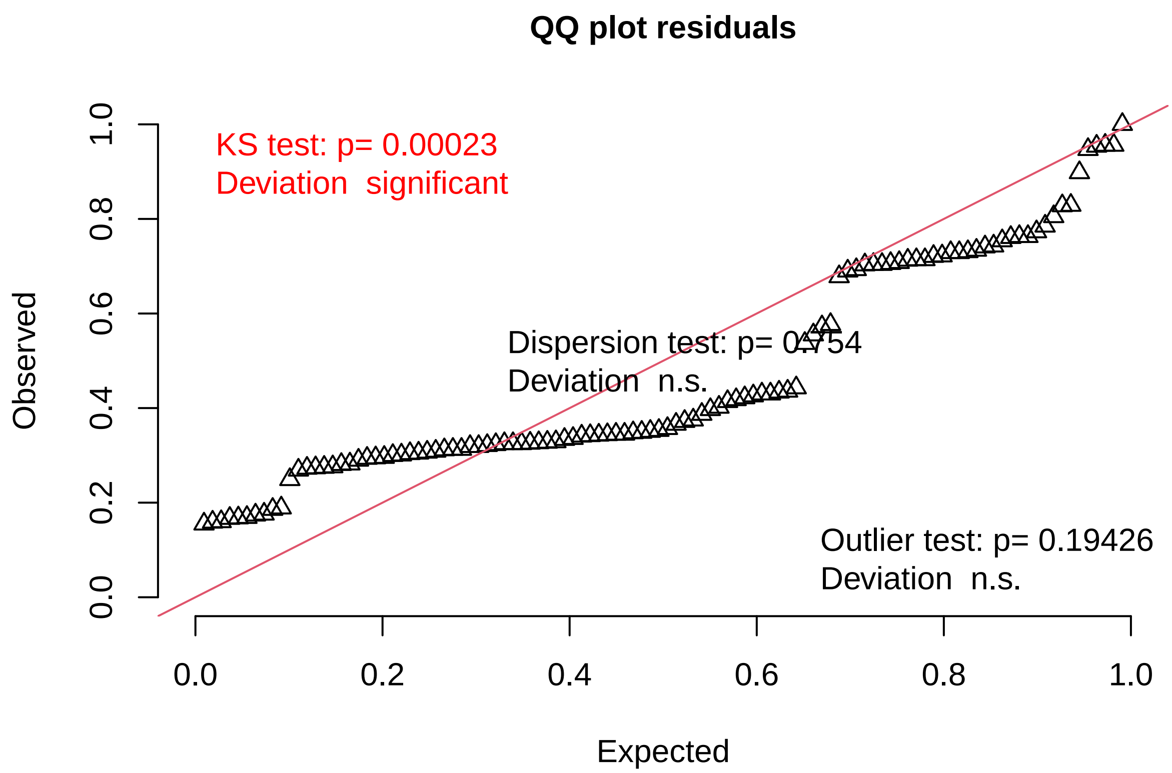
```r
print(summary(bas_pre), correlation = FALSE)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log1p(bas_number) ~ t_muris_dose * strain + (1 | cage_wedge) +
##     (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Pre-RW")
##
## REML criterion at convergence: -652.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.9646 -0.5260 -0.3617  0.5548  7.5324
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 2.938e-14 1.714e-07
##  cohort     (Intercept) 3.945e-07 6.281e-04
##  Residual               8.204e-05 9.057e-03
## Number of obs: 108, groups:  cage_wedge, 21; cohort, 2
##
## Fixed effects:
##                                 Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                    4.103e-03  2.241e-03  2.327e+01   1.831    0.080
## t_muris_doseLow               -6.256e-04  2.988e-03  1.010e+02  -0.209    0.835
## t_muris_doseHigh              -2.238e-03  3.155e-03  1.010e+02  -0.709    0.480
## strainB10.BR                   6.206e-04  3.024e-03  1.010e+02   0.205    0.838
## t_muris_doseLow:strainB10.BR   8.670e-04  4.165e-03  1.010e+02   0.208    0.836
## t_muris_doseHigh:strainB10.BR  6.049e-03  4.404e-03  1.010e+02   1.373    0.173
##
## (Intercept)                   .
## t_muris_doseLow
## t_muris_doseHigh
## strainB10.BR
## t_muris_doseLow:strainB10.BR
## t_muris_doseHigh:strainB10.BR
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
bas_pre_sim <- simulateResiduals(fittedModel = bas_pre, n = 1000)
plot(bas_pre_sim)
```

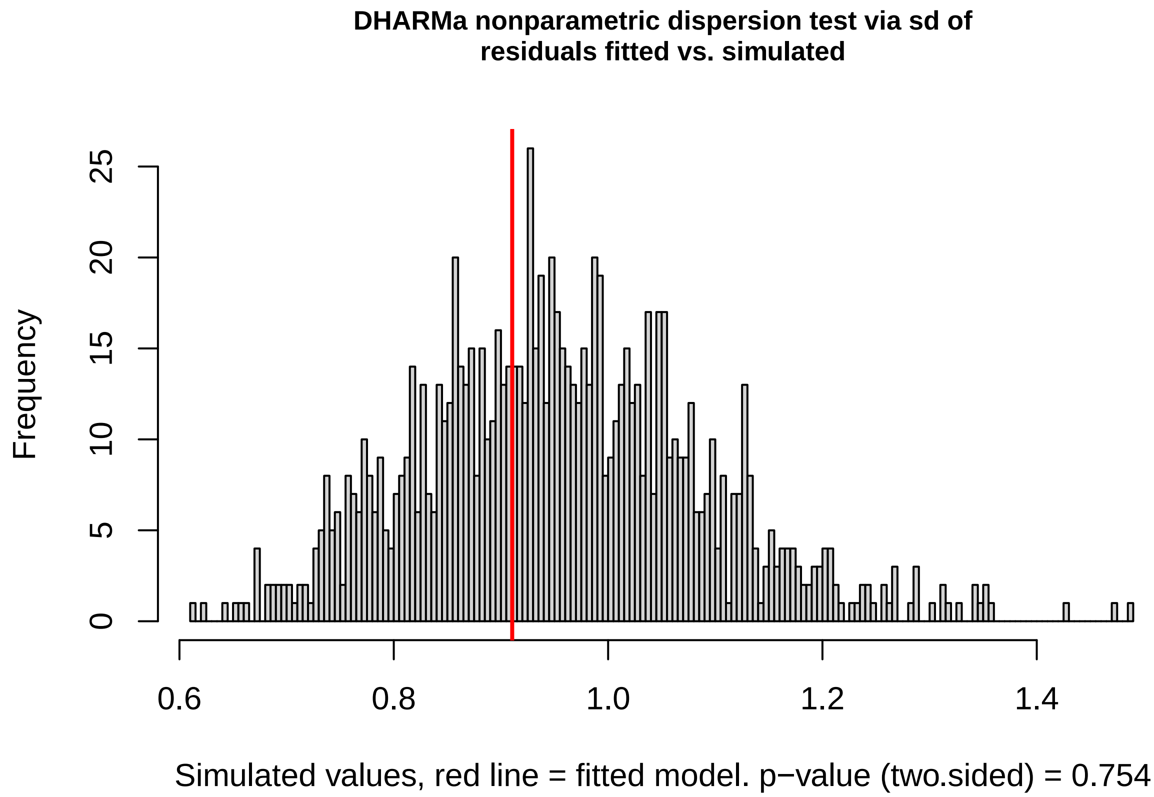



```r
testUniformity(bas_pre_sim)
```



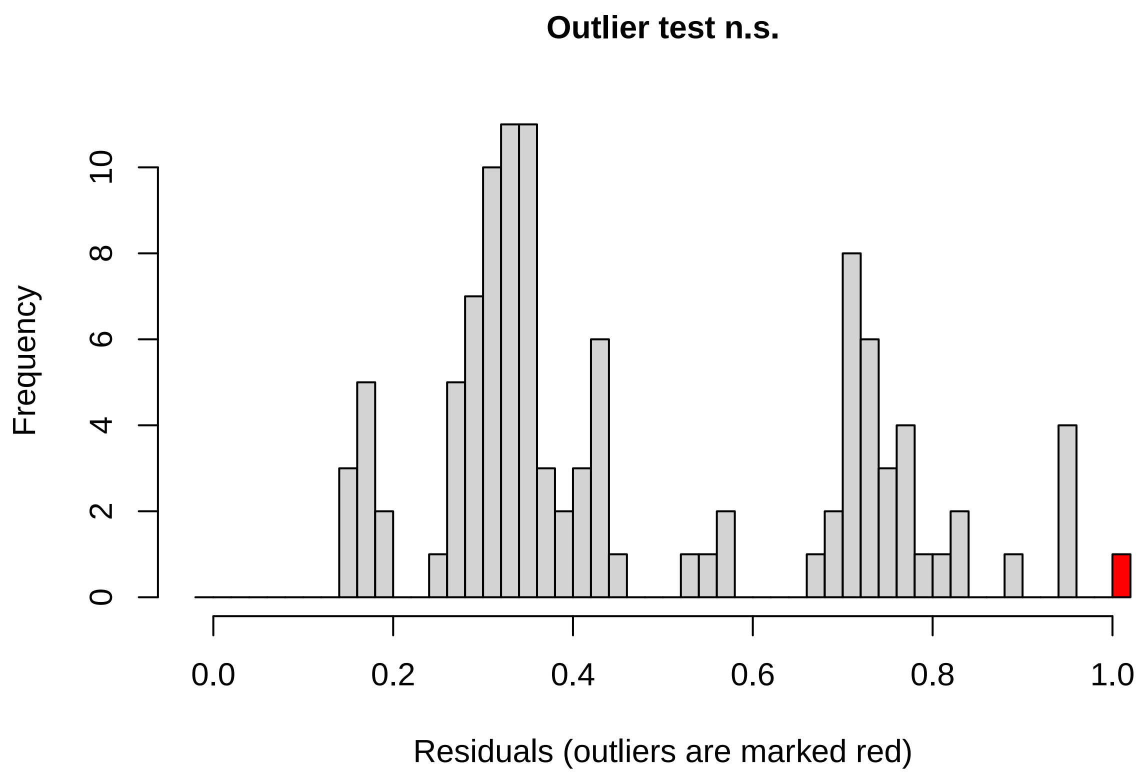

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.20515, p-value = 0.0002255
## alternative hypothesis: two-sided
```

```
testDispersion(bas_pre_sim)
```



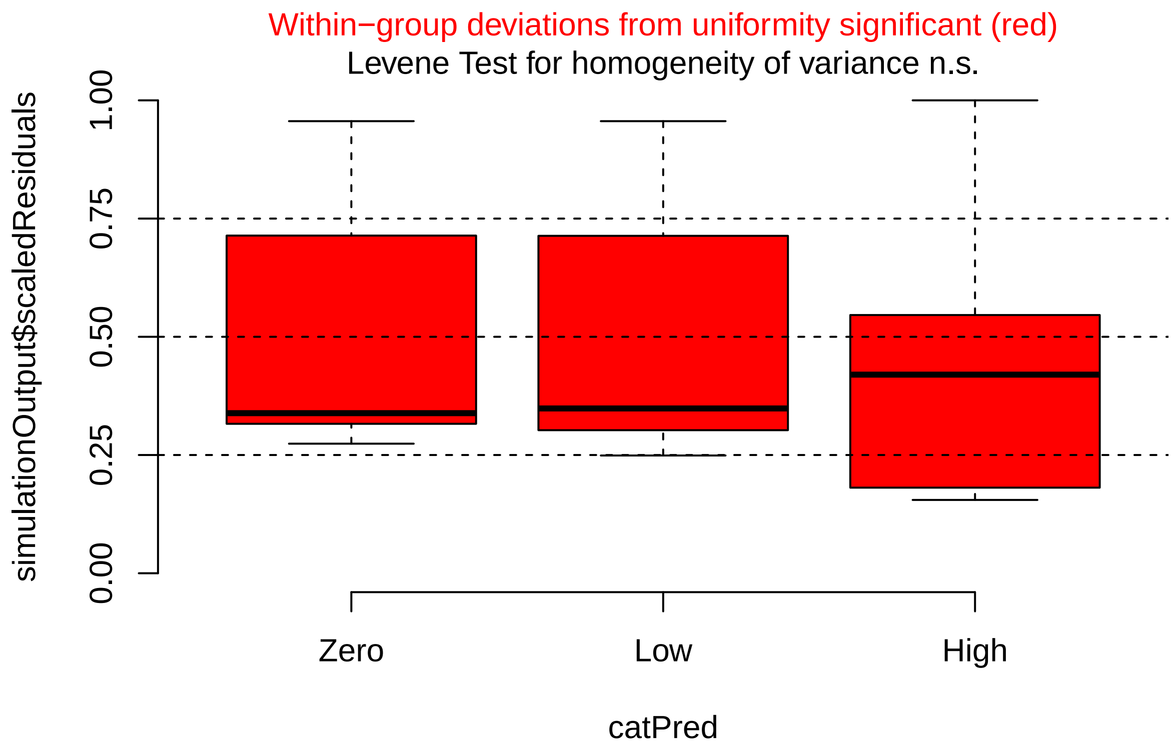

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.95286, p-value = 0.754
## alternative hypothesis: two.sided
```

```
testOutliers(bas_pre_sim)
```

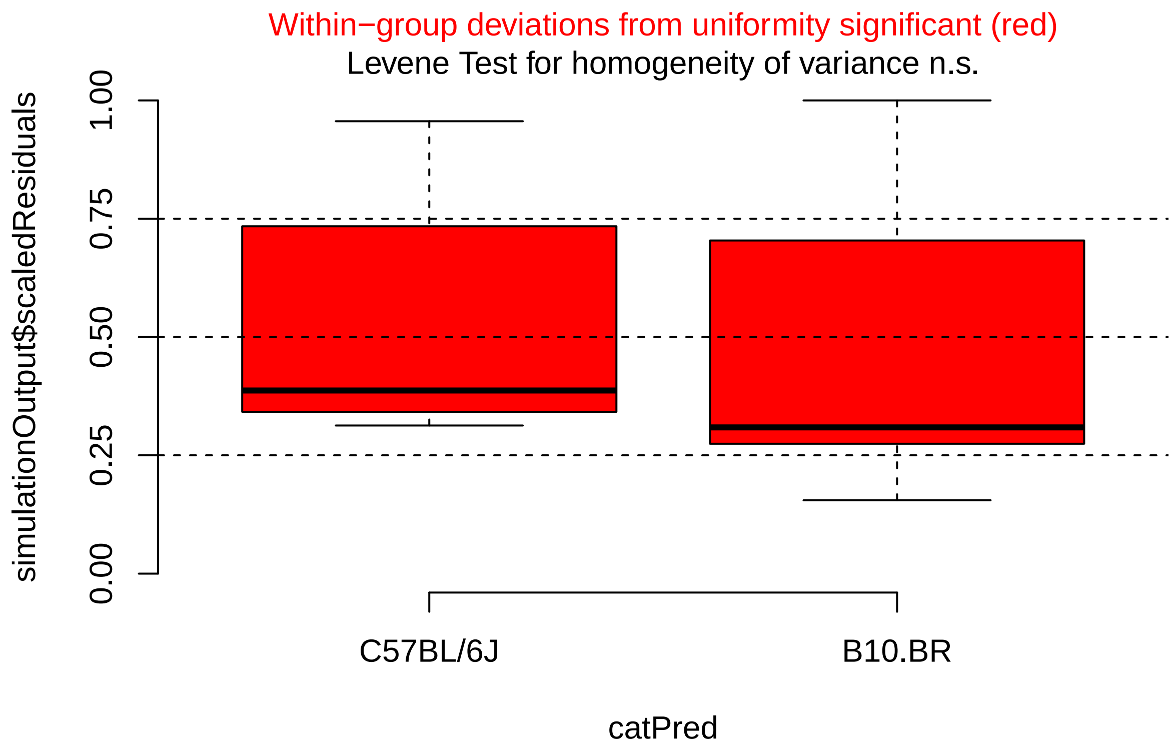



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  bas_pre_sim
## outliers at both margin(s) = 1, observations = 108, p-value = 0.1943
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002343967 0.0505105337
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                            0.009259259
```

```
plotResiduals(bas_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$t_muris_dose)
```




```
plotResiduals(bas_pre_sim, form = filter(cleaned_combined_cbc_data, phase == "Pre-RW")$strain)
```



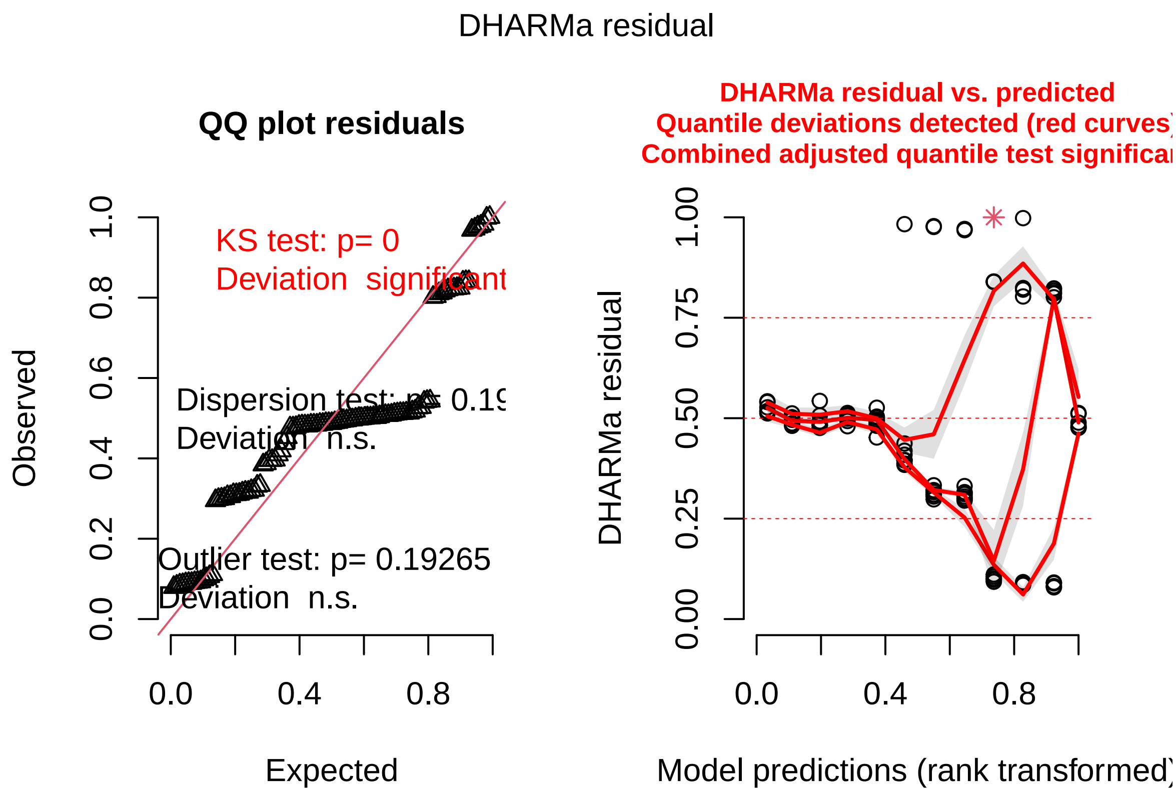

```r
# During RW: environment x dose x strain
bas_during <- lmer(log1p(bas_number) ~ lab_or_rw * t_muris_dose * strain +
                     (1 | cage_wedge) + (1 | cohort),
                   data = filter(cleaned_combined_cbc_data, phase == "During RW"))

print(summary(bas_during), correlation = FALSE)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log1p(bas_number) ~ lab_or_rw * t_muris_dose * strain + (1 |
##     cage_wedge) + (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "During RW")
##
## REML criterion at convergence: -759.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0120 -0.4297 -0.0410  0.1238  3.6882
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 4.331e-06 0.002081
##  cohort     (Intercept) 1.893e-06 0.001376
##  Residual               1.260e-05 0.003550
## Number of obs: 107, groups:  cage_wedge, 29; cohort, 2
##
```

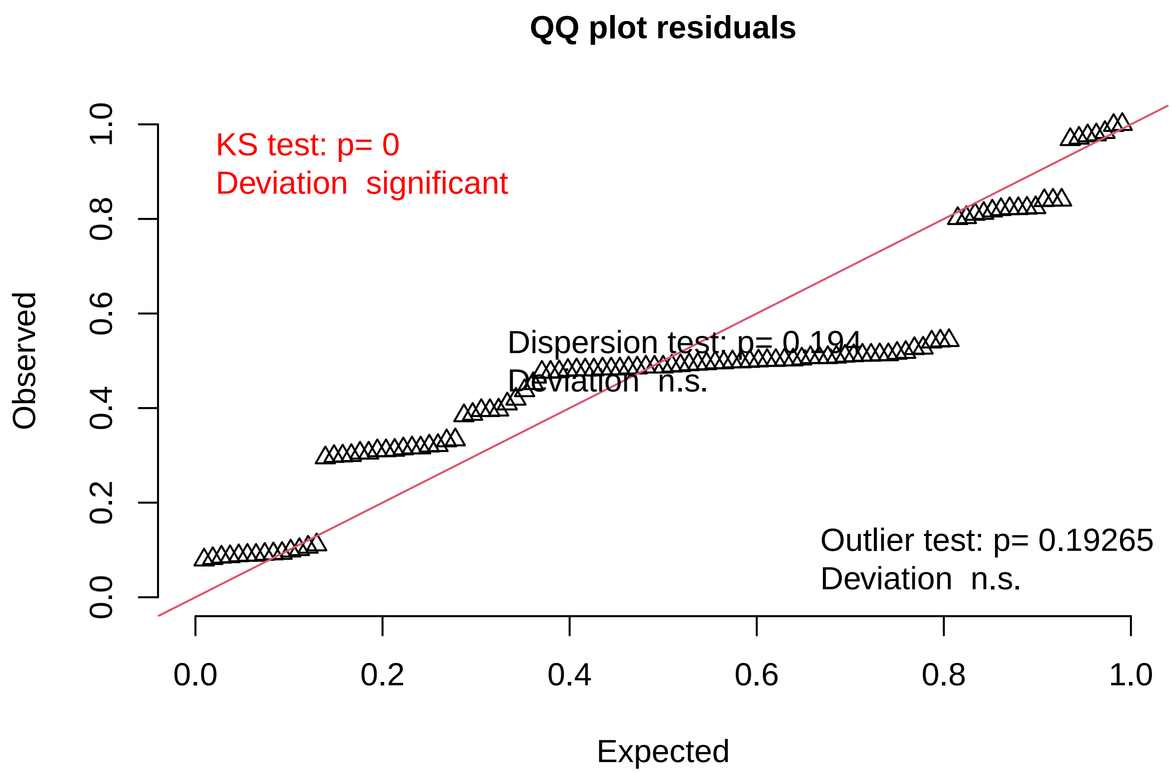
```
## Fixed effects:
##                                              Estimate Std. Error         df
## (Intercept)                                 0.0059408  0.0016362  4.2909103
## lab_or_rwRW                                -0.0048580  0.0022470 22.2812843
## t_muris_doseLow                            -0.0038443  0.0016783 88.3851954
## t_muris_doseHigh                           -0.0063493  0.0020110 93.9817206
## strainB10.BR                               -0.0004336  0.0018881 63.3444569
## lab_or_rwRW:t_muris_doseLow                 0.0027533  0.0023475 79.6878880
## lab_or_rwRW:t_muris_doseHigh                0.0072792  0.0025899 88.9037244
## lab_or_rwRW:strainB10.BR                   -0.0006717  0.0031421 34.0958974
## t_muris_doseLow:strainB10.BR                0.0042748  0.0023716 83.3969595
## t_muris_doseHigh:strainB10.BR               0.0108193  0.0028333 93.5081033
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   -0.0031676  0.0033605 75.5612690
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -0.0117491  0.0036778 85.7693119
##                                            t value Pr(>|t|)
## (Intercept)                                  3.631 0.019619 *
## lab_or_rwRW                                 -2.162 0.041620 *
## t_muris_doseLow                             -2.291 0.024364 *
## t_muris_doseHigh                            -3.157 0.002141 **
## strainB10.BR                                -0.230 0.819096
## lab_or_rwRW:t_muris_doseLow                  1.173 0.244350
## lab_or_rwRW:t_muris_doseHigh                 2.811 0.006081 **
## lab_or_rwRW:strainB10.BR                    -0.214 0.831996
## t_muris_doseLow:strainB10.BR                 1.803 0.075079 .
## t_muris_doseHigh:strainB10.BR                3.819 0.000241 ***
## lab_or_rwRW:t_muris_doseLow:strainB10.BR    -0.943 0.348886
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR   -3.195 0.001960 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
bas_during_sim <- simulateResiduals(fittedModel = bas_during, n = 1000)
plot(bas_during_sim)
```

```
## qu = 0.5, log(sigma) = -4.350275 : outer Newton did not converge fully.
```

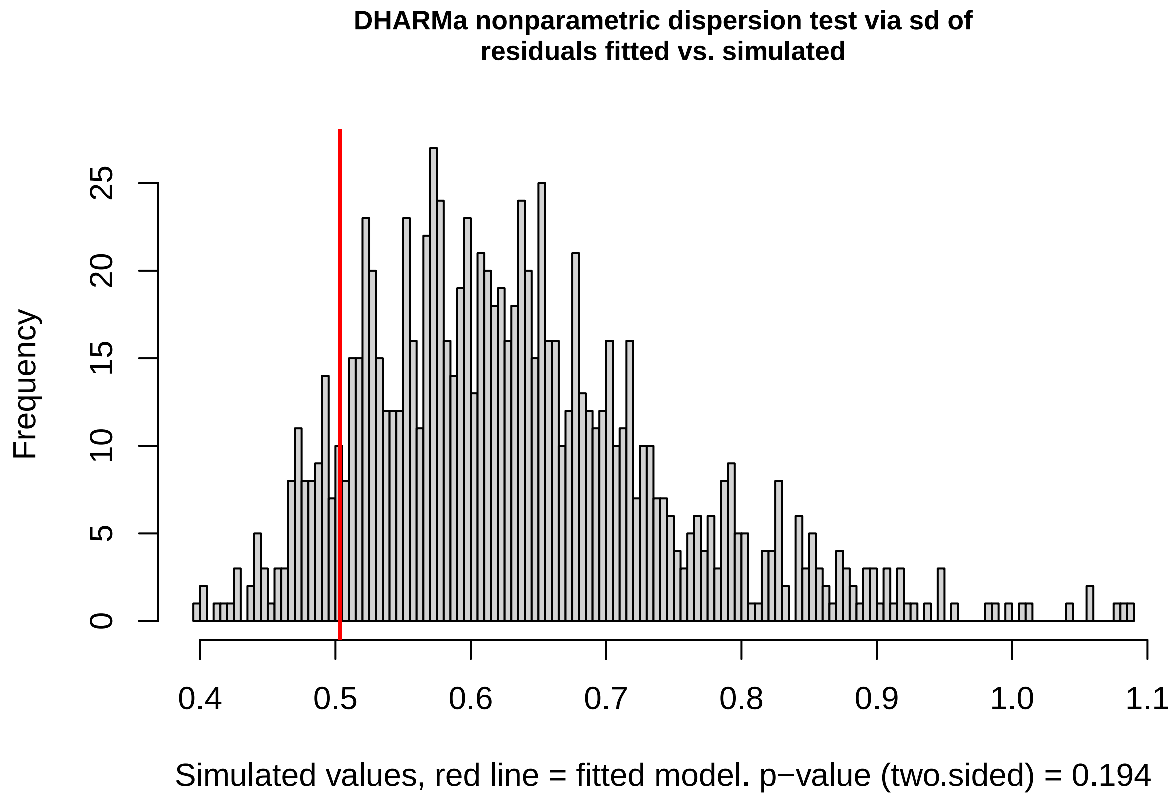



```r
testUniformity(bas_during_sim)
```



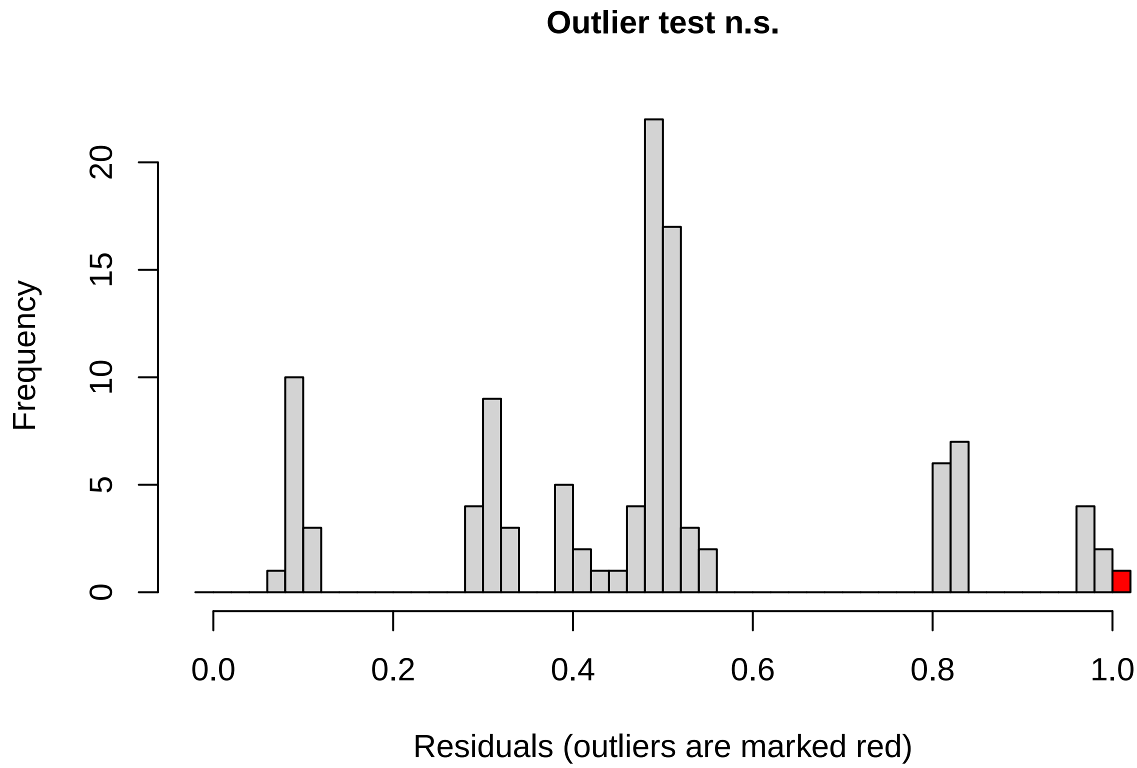

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.27008, p-value = 3.323e-07
## alternative hypothesis: two-sided
```

```
testDispersion(bas_during_sim)
```



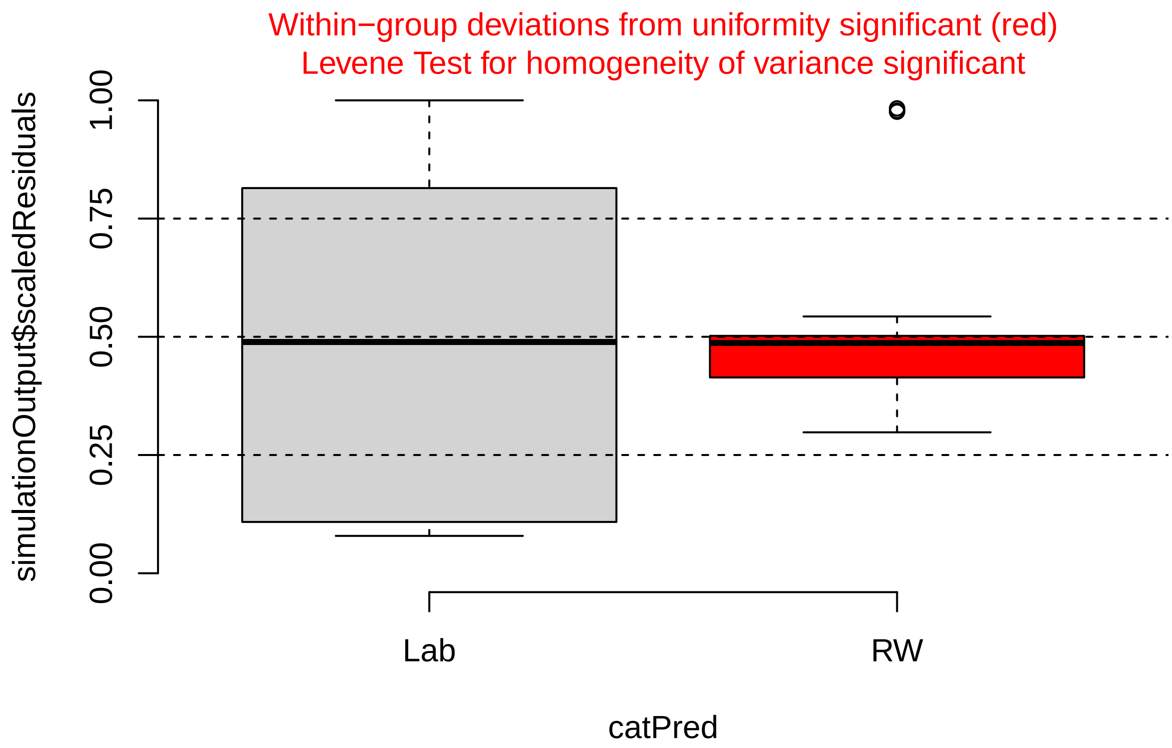

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.79339, p-value = 0.194
## alternative hypothesis: two.sided
```

```
testOutliers(bas_during_sim)
```

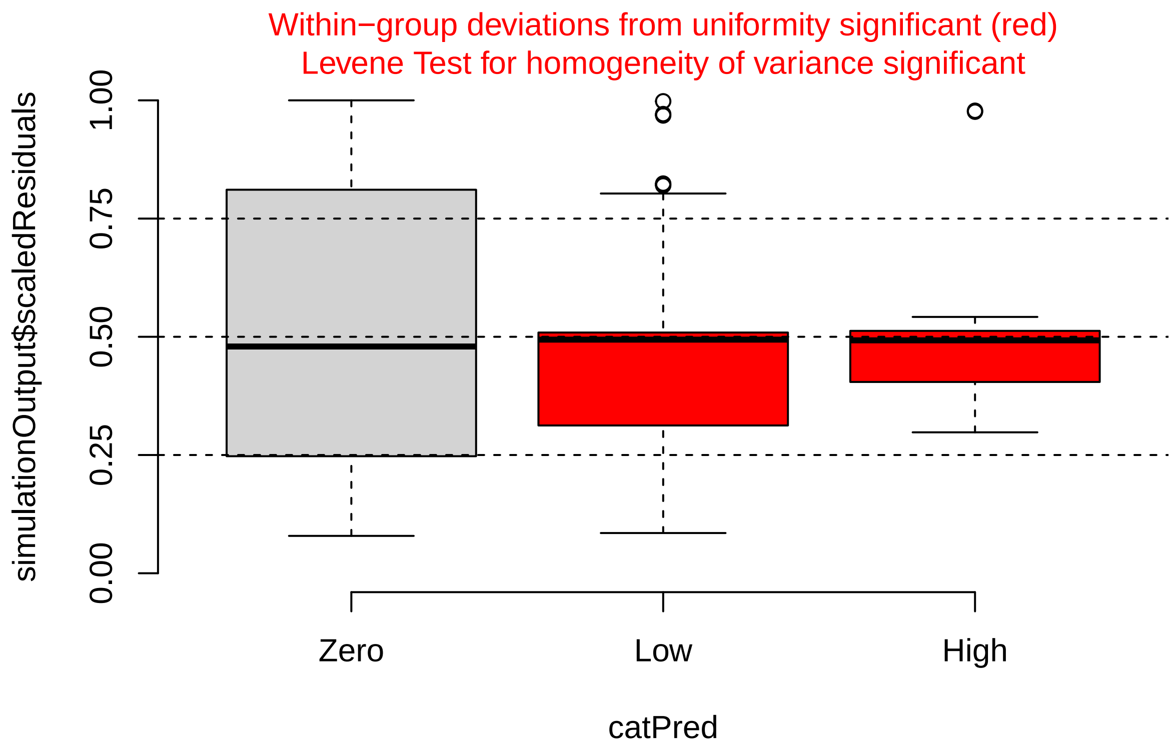



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  bas_during_sim
## outliers at both margin(s) = 1, observations = 107, p-value = 0.1927
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.000236587 0.050972543
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                         0.009345794
```

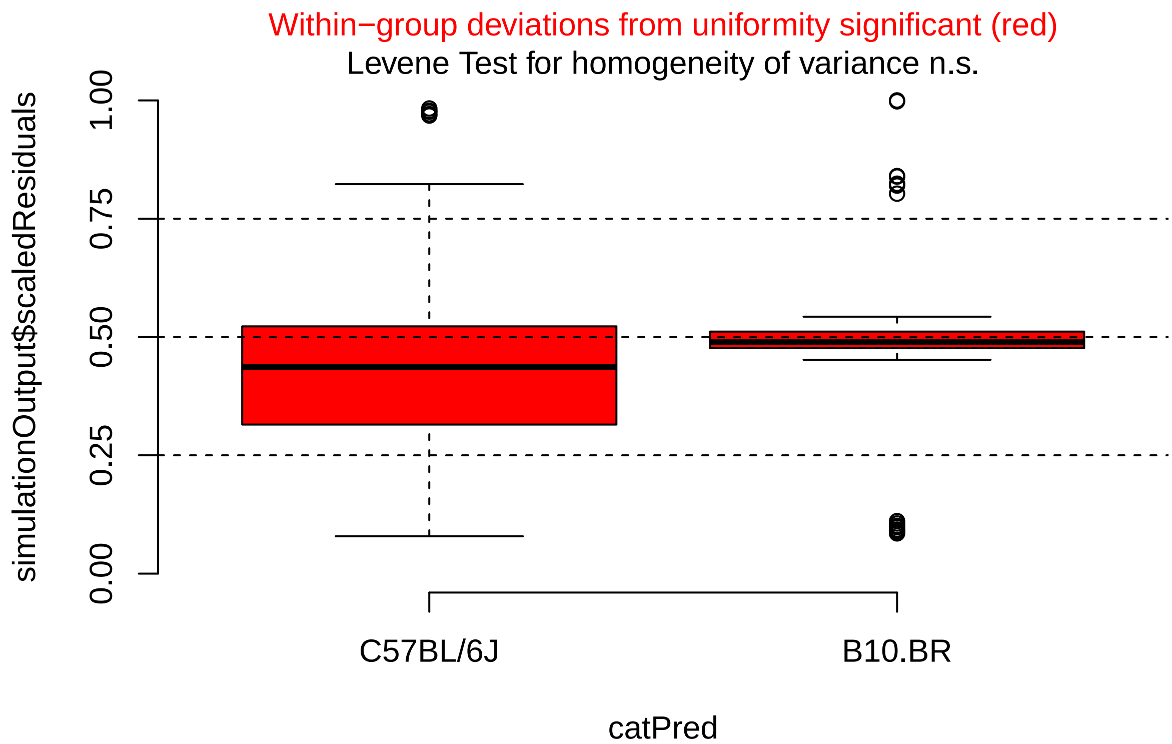
```
plotResiduals(bas_during_sim, form = filter(cleaned_combined_cbc_data,
                                            phase == "During RW")$lab_or_rw)
```




```
plotResiduals(bas_during_sim, form = filter(cleaned_combined_cbc_data,
                                            phase == "During RW")$t_muris_dose)
```




```r
plotResiduals(bas_during_sim, form = filter(cleaned_combined_cbc_data,
                                            phase == "During RW")$strain)
```

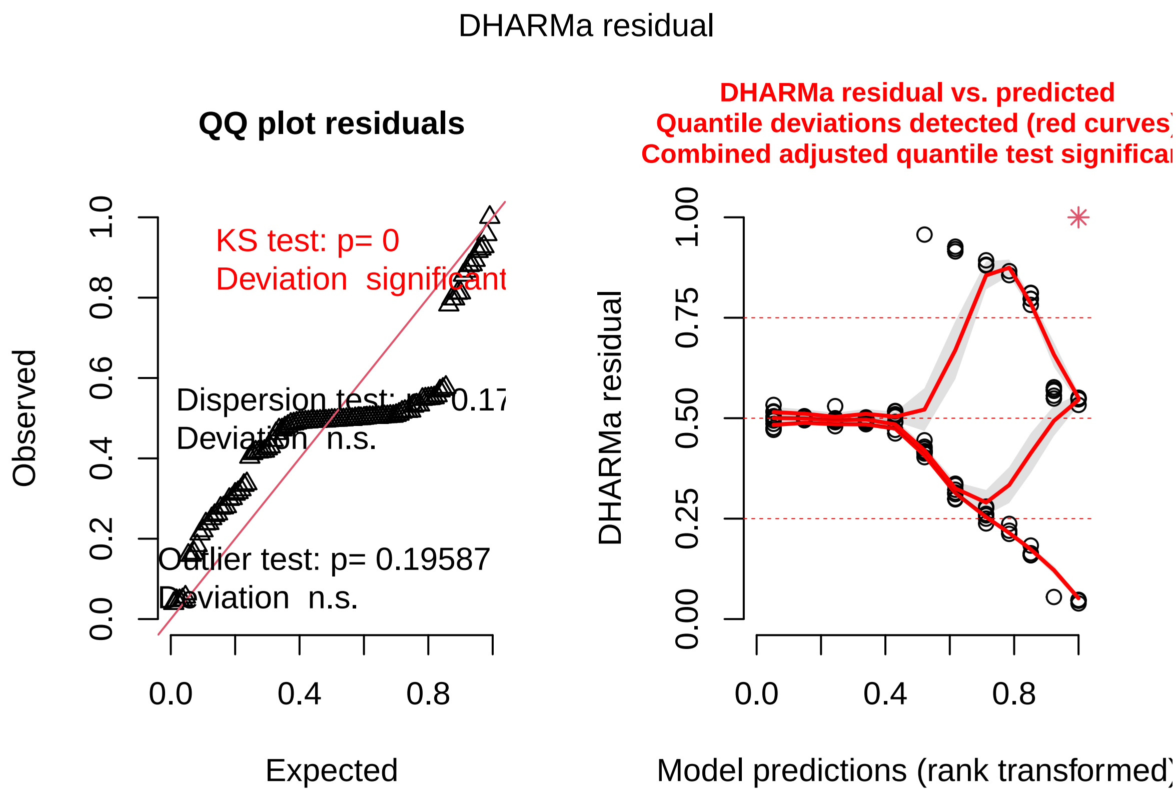



```r
# Post-RW: environment x infection x strain
bas_post <- lmer(log1p(bas_number) ~ lab_or_rw * t_muris_dose * strain +
                   (1 | cage_wedge) + (1 | cohort),
                 data = filter(cleaned_combined_cbc_data, phase == "Post-RW"))

print(summary(bas_post), correlation = FALSE)
```

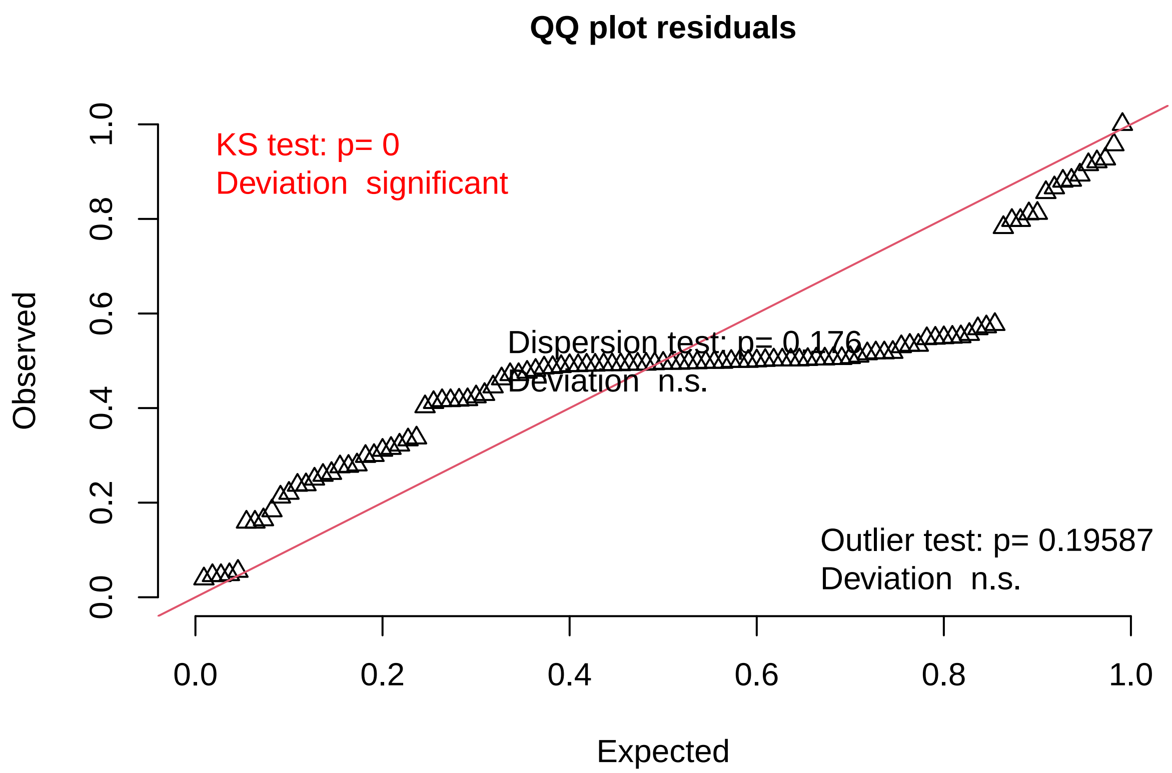
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log1p(bas_number) ~ lab_or_rw * t_muris_dose * strain + (1 |
##     cage_wedge) + (1 | cohort)
##    Data: filter(cleaned_combined_cbc_data, phase == "Post-RW")
##
## REML criterion at convergence: -747.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.8999 -0.1915 -0.0201  0.0979  5.4155
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cage_wedge (Intercept) 1.198e-05 0.003462
##  cohort     (Intercept) 2.753e-06 0.001659
##  Residual               1.519e-05 0.003897
## Number of obs: 109, groups:  cage_wedge, 29; cohort, 2
##
```

```
## Fixed effects:
##                                             Estimate Std. Error         df
## (Intercept)                                0.0094090  0.0020228  4.0272387
## lab_or_rwRW                               -0.0094003  0.0032249 19.0806279
## t_muris_doseLow                           -0.0068116  0.0019401 91.6545416
## t_muris_doseHigh                          -0.0004330  0.0023340 95.5874352
## strainB10.BR                              -0.0042411  0.0023562 54.1463710
## lab_or_rwRW:t_muris_doseLow                0.0068029  0.0026412 83.2042810
## lab_or_rwRW:t_muris_doseHigh               0.0004243  0.0029427 89.6867361
## lab_or_rwRW:strainB10.BR                   0.0052274  0.0045550 18.9504495
## t_muris_doseLow:strainB10.BR               0.0049865  0.0026927 86.2835479
## t_muris_doseHigh:strainB10.BR             -0.0008223  0.0033918 94.5488790
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  -0.0059728  0.0036744 79.4800893
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -0.0001640  0.0042137 88.5495896
##                                           t value Pr(>|t|)
## (Intercept)                                 4.652 0.009494 **
## lab_or_rwRW                                -2.915 0.008855 **
## t_muris_doseLow                            -3.511 0.000695 ***
## t_muris_doseHigh                           -0.186 0.853223
## strainB10.BR                               -1.800 0.077439 .
## lab_or_rwRW:t_muris_doseLow                 2.576 0.011772 *
## lab_or_rwRW:t_muris_doseHigh                0.144 0.885679
## lab_or_rwRW:strainB10.BR                    1.148 0.265410
## t_muris_doseLow:strainB10.BR                1.852 0.067469 .
## t_muris_doseHigh:strainB10.BR              -0.242 0.808958
## lab_or_rwRW:t_muris_doseLow:strainB10.BR   -1.626 0.108008
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR  -0.039 0.969040
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
bas_post_sim <- simulateResiduals(fittedModel = bas_post, n = 1000)
plot(bas_post_sim)
```

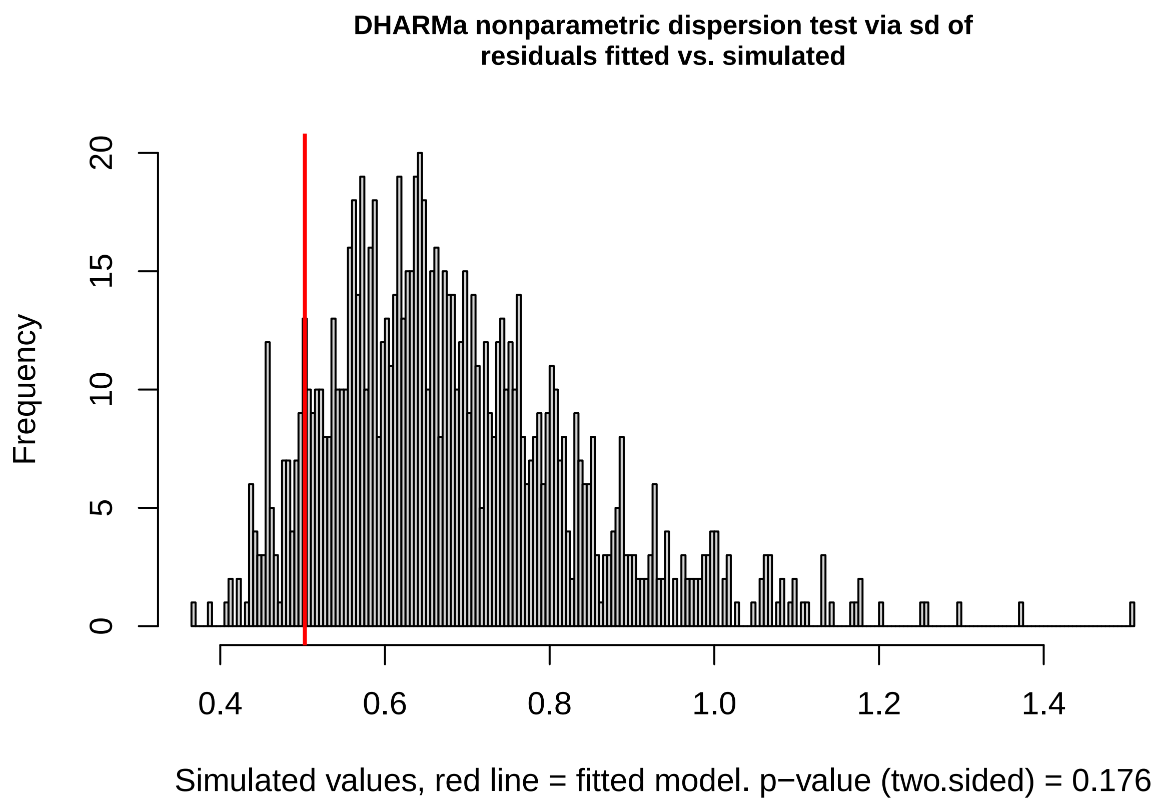



```r
testUniformity(bas_post_sim)
```



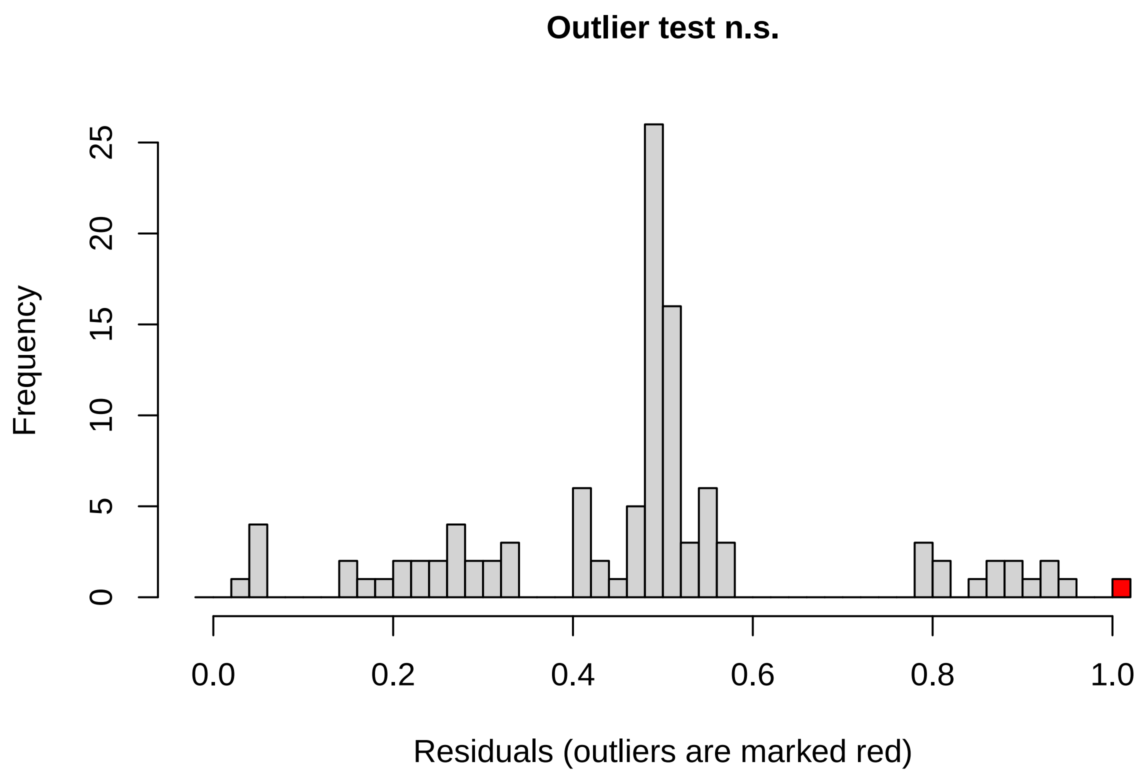

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.28539, p-value = 3.892e-08
## alternative hypothesis: two-sided
```

```
testDispersion(bas_post_sim)
```



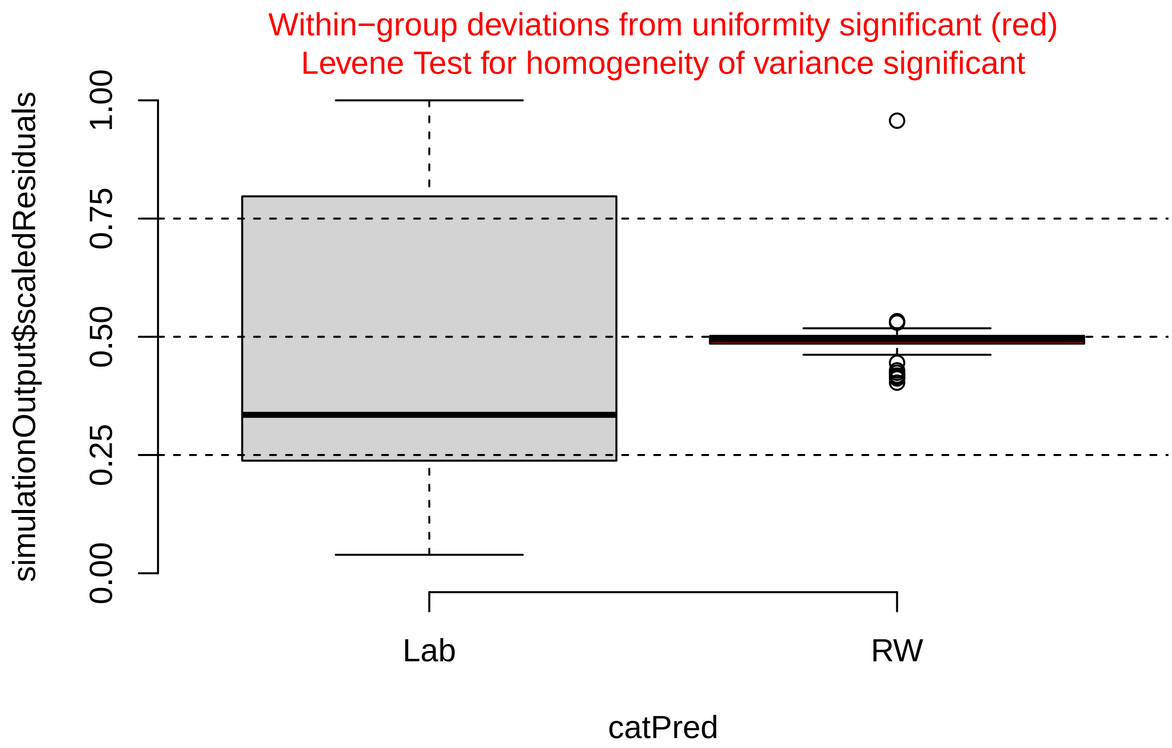

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.7338, p-value = 0.176
## alternative hypothesis: two.sided
```

```
testOutliers(bas_post_sim)
```

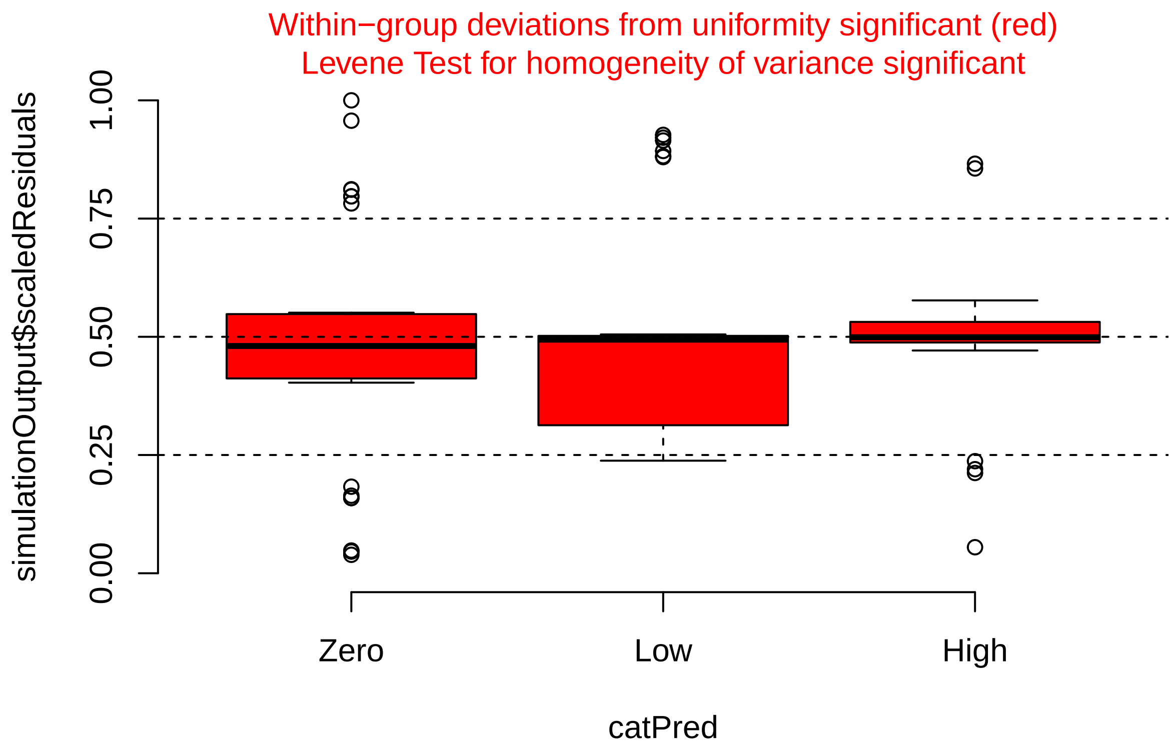



```
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  bas_post_sim
## outliers at both margin(s) = 1, observations = 109, p-value = 0.1959
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002322465 0.0500568231
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                          0.009174312
```

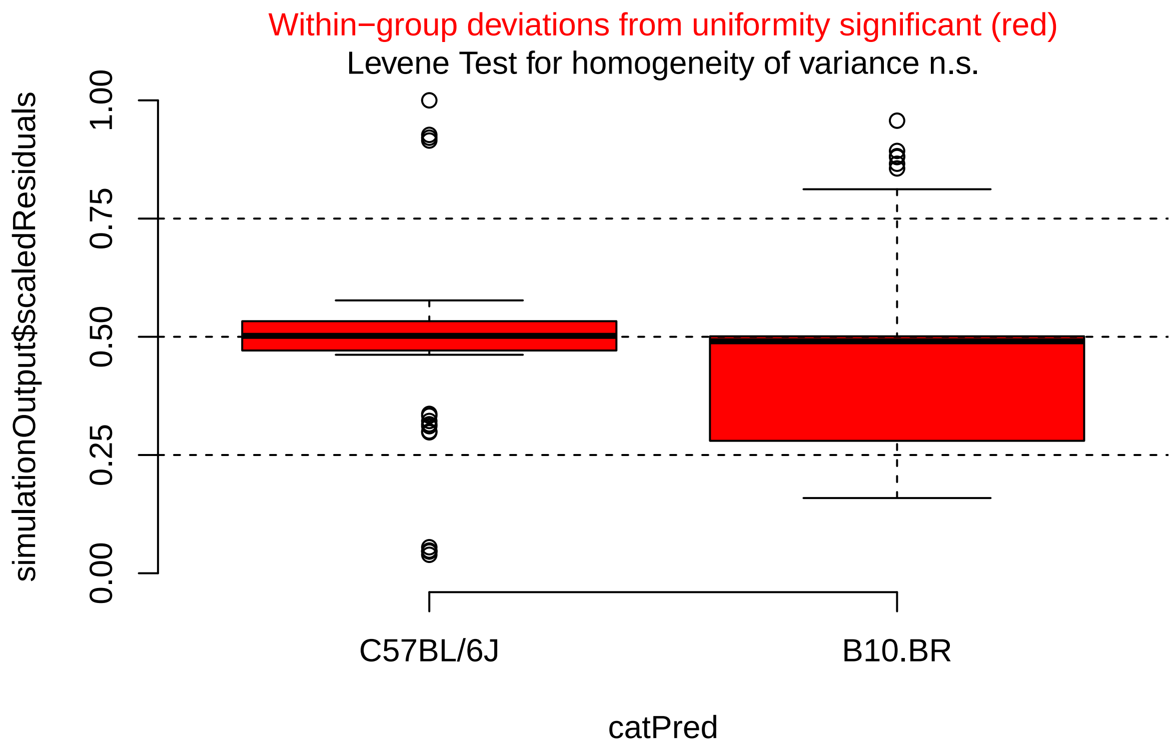
```
plotResiduals(bas_post_sim, form = filter(cleaned_combined_cbc_data,
                                          phase == "Post-RW")$lab_or_rw)
```




```
plotResiduals(bas_post_sim, form = filter(cleaned_combined_cbc_data,
                                          phase == "Post-RW")$t_muris_dose)
```



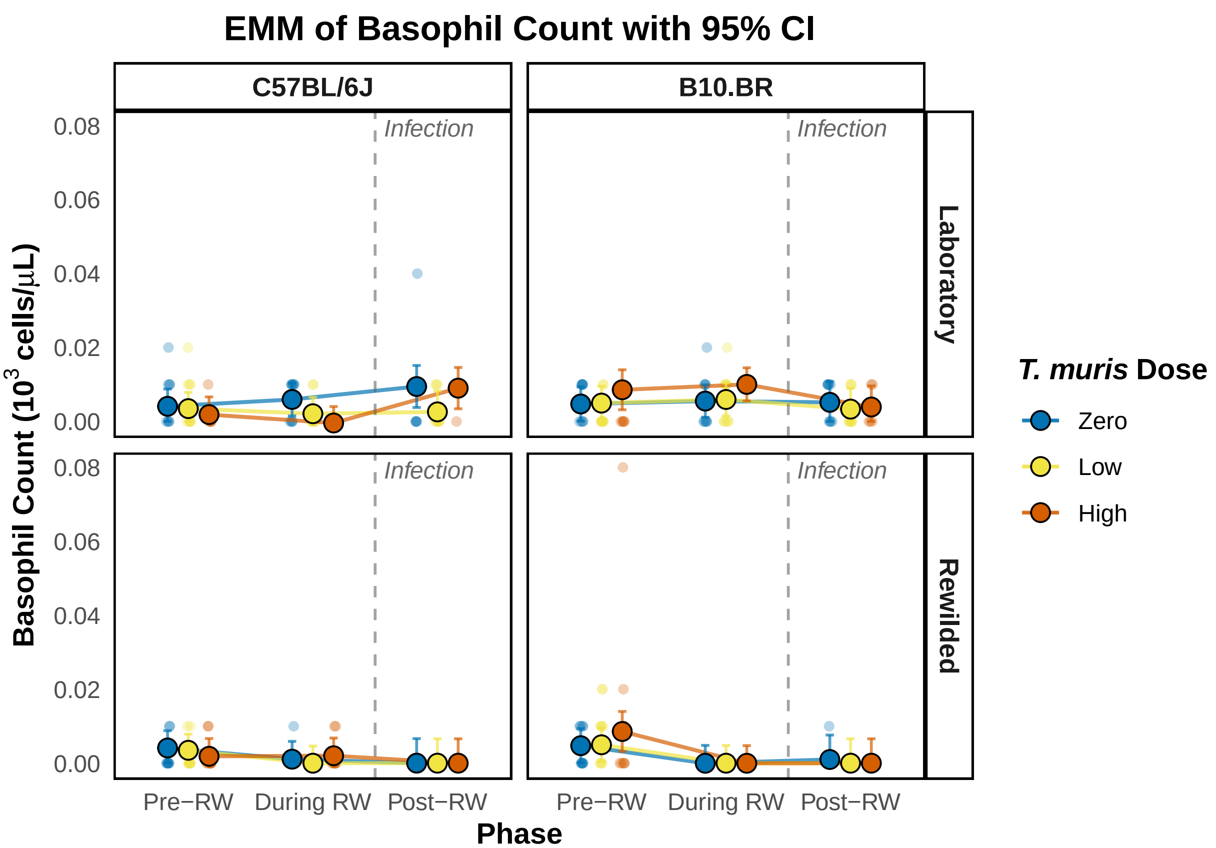

```
plotResiduals(bas_post_sim, form = filter(cleaned_combined_cbc_data,
                                          phase == "Post-RW")$strain)
```




```
# log1p back-transform: expm1() reverses log1p()
emm_bas_pre <- as.data.frame(emmeans(bas_pre,
                                     specs = "t_muris_dose",
                                     by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  crossing(lab_or_rw = factor(c("Lab", "RW"),
                              levels = c("Lab", "RW"))) %>%
  mutate(phase = factor("Pre-RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_bas_during <- as.data.frame(emmeans(bas_during,
                                        specs = c("t_muris_dose", "lab_or_rw"),
                                        by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  mutate(phase = factor("During RW", levels = c("Pre-RW", "During RW", "Post-RW")))

emm_bas_post <- as.data.frame(emmeans(bas_post,
                                      specs = c("t_muris_dose", "lab_or_rw"),
                                      by = "strain")) %>%
  mutate(response = expm1(emmean),
         lower.CL = expm1(lower.CL),
         upper.CL = expm1(upper.CL)) %>%
  mutate(phase = factor("Post-RW", levels = c("Pre-RW", "During RW", "Post-RW")))
```

```r
emm_bas_combined <- bind_rows(emm_bas_pre, emm_bas_during, emm_bas_post) %>%
  mutate(phase = factor(phase, levels = c("Pre-RW", "During RW", "Post-RW")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "Low", "High")),
         strain = factor(strain, levels = c("C57BL/6J", "B10.BR")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

```r
emm_bas_plot <- emm_plotter(
  emm_df = emm_bas_combined,
  cbc_input = cleaned_combined_cbc_data,
  y_col = "bas_number",
  y_label = expression(bold("Basophil Count (10"^3~"cells/"*mu*"L)")),
  title = "EMM of Basophil Count with 95% CI",
  floor_zero = TRUE
)

emm_bas_plot
```




```r
plot_saver("emm_bas_number.png", emm_bas_plot)
```